

QFT Notes

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Part I

Classical Field Theory

Chapter 1

The Lorentz group and its Irreducible Representations

Quantum mechanics is broken - it does not respect Lorentz invariance. We want to try to fix this.

To do this we will start by reviewing some fundamental aspects of physics and math. In particular we're going to quickly revisit the concept of vectors, tensors, and rotations. These should be quite familiar by the time someone is taking this course, but modern academia is unfortunately focused on rushing teaching so there is more time for research and grant writing. (Also for some reason General Relativity has been de-emphasized in curricula).

Having revisited these fundamental concepts we will take a look at group theory, in particular the “spin group.” We will briefly review the standard quantum mechanics problem of adding the spin of two spin- $\frac{1}{2}$ particles, but we'll put particular emphasis on the idea of irreducible representations of a group.

These irreducible representations are going to be the starting point of our QFT. We will see that they allow us to define fields that have well-defined transformation properties under the Lorentz group and therefore will allow us to, in the next chapter, form Lorentz-invariant Lagrangian densities – what will be our four-dimensional analogue of Lagrangians.

1.1 Back to the basics

What is a vector?

In introductory physics we would most likely have defined a vector as “a quantity with magnitude and direction” or something along these lines. Hopefully somewhere along the way we learned¹,

A **vector** is a quantity whose components transform in a specific way under rotations, in particular:

$$v'_i = R_{ij} v_j. \quad (1.1)$$

So far, we most likely are only familiar with R as a rotation in 3-dimensional space, and maybe from electrodynamics we’ve seen:

$$\bar{x}^\mu = \Lambda^\mu_\nu x^\nu. \quad (1.2)$$

Where instead of the usual “rotation” R , we have a Lorentz transformation Λ which represents boosts and rotations.

The reason we are taking so much time on something as simple as the definition of a vector is the intimate connection between our more formal definition above a symmetry. Symmetry will become the foundation of pretty much everything we do in QFT so we need to know the “language” of symmetry. Let’s look at the square of the norm of v and v' from Eq. 1.1.

$$|v'|^2 = (R_{ij} v_j) (R_{ik} v_k) = v_j R_{ji}^T R_{ik} v_k = v_j \delta_{jk} v_k = |v|^2. \quad (1.3)$$

To get here, we used the orthogonal property of rotations which in practice means $R^T R = \mathbb{1}$. Rotations in N -dimensional space are referred to as $O(N)$ when using group theory language. Unsurprisingly, rotations preserve the norm, or length, of a vector.

For quantum field theory this will be of profound importance! We will use this idea of vectors and transformations to ensure our theory is Lorentz invariant

¹It is a good time to note our default will be to use Einstein summation notation unless otherwise specified. That is, repeated indices are summed. For three spatial dimensions we use latin characters, for the four spacetime dimensions we use greek characters. Unfortunately, eventually we will have many other indices we need to assign characters to. So ultimately we will need to use context to remember the details and cannot rely on the specific alphabet to indicate the range of the sum.

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as well as invariant under other symmetries. For example, for quantum electrodynamics (QED) we will impose a “ $U(1)$ ” symmetry on our theory. This will correspond to “gauge symmetry” in the electrodynamics sense².

Now, this isn’t the whole story. We can also take two different vectors, like momentum and velocity, and show that their product is preserved under rotations:

$$p'_i x'_i = p_i R_{ij}^T R_{jk} x_k = p_i x_i . \quad (1.4)$$

Again, this is a very simple statement, but the implications are profound. As long as we are taking the product of quantities which transform appropriately under some symmetry, we can form a scalar under the corresponding transformation. Consider two objects with two components v_1 and v_2 , we will define them as:

$$v_1 = \begin{pmatrix} x \\ y \end{pmatrix} , \quad v_2 = \begin{pmatrix} r \\ \theta \end{pmatrix} . \quad (1.5)$$

v_1 is familiar, and it’s probably already clear it will transform appropriately. However, v_2 is strange as it explicitly depends nonlinearly on x and y , that is $r = \sqrt{x^2 + y^2}$ and $\theta = \arctan(y/x)$. We’re going to *choose* v_1 and v_2 to be the same vector. We’ve just chosen to label v_2 in a really asinine manner.

If we calculate the norms of v_1 and v_2 we find:

$$|v_1|^2 = r^2 = x^2 + y^2 , \quad |v_2|^2 = \theta^2 + r^2 \neq r^2 . \quad (1.6)$$

Rotation by ϕ corresponds to a rotation matrix,

$$R = \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix} . \quad (1.7)$$

Applying this matrix to our first vector gives,

$$v'_1 = \begin{pmatrix} x \cos \phi - y \sin \phi \\ x \sin \phi + y \cos \phi \end{pmatrix} \quad |v'_1|^2 = (x^2 + y^2)(\sin^2 \phi + \cos^2 \phi) = r^2 , \quad (1.8)$$

the length is preserved. Now, we also know that a simple rotation of v_2 will simply add ϕ to θ giving the rotated v_2 :

$$v'_2 = \begin{pmatrix} r \\ \theta + \phi \end{pmatrix} , \quad (1.9)$$

²Recall from Classical Mechanics and Noether’s theorem that every symmetry corresponds to a conserved charge. For this “ $U(1)$ ” symmetry that will be electric charge, for rotations we have conservation of angular momentum.

If we do this, but preserve our usual concept of the squared norm as the sum of the squares of the components we find,

$$v_2 = \begin{pmatrix} r \\ \theta + \phi \end{pmatrix} \quad |v_2|^2 = (\theta + \phi)^2 + r^2 \neq \theta^2 + r^2, \quad (1.10)$$

notice that $v'_2 \neq Rv_2$ this was due to our choice of how we apply the rotation. By maintaining the usual definition of the norm, we can see it is not preserved:

$$|v'_2|^2 = r^2 + (\theta')^2 \neq r^2 + \theta^2. \quad (1.11)$$

If this feels like a very contrived example, it is. We could make a new definition of the norm that would behave properly under rotations for vectors defined like v_2 , but then we may as well return to our definition in Eq. 1.1. Why do we prefer the usual way of writing vectors as defined in Eq. 1.1? This definition is manifestly linear. We don't know how to calculate much beyond the simplest examples in math and physics, but linear equations generally are something we know how to solve.

We will deal with a lot of new vectors in quantum field theory, but the ones that should be familiar at this stage are four-momenta,

$$p^\mu = (E, \vec{p}) \quad p^2 = p^\mu p_\mu = p^\mu \eta_{\mu\nu} p^\nu = E^2 - \vec{p}^2 = m^2, \quad (1.12)$$

as well as the four-position vector,

$$x^\mu = (t, \vec{x}), x^2 = -t^2 + \vec{x}^2 \equiv s^2, \quad (1.13)$$

where s^2 is the “space-time interval.”

What is a Tensor?

Now that we have a fundamental understanding of what a vector is, we can extrapolate this not only to a “tensor” but an N^{th} rank tensor.

An N^{th} **rank tensor** is a quantity whose components transform in a specific way under rotations, in particular for N indices we have N rotation matrices:

$$T'_{i'j'k'...} = R_{i'i} R_{j'j} R_{k'k} ... T_{ijk...} \quad (1.14)$$

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That's just to say an N^{th} -rank tensor transforms under the application of N rotations. So a scalar is a 0^{th} -rank tensor, a vector a 1^{st} -rank tensor, and so on.

It may not be obvious, but as early as classical mechanics we became familiar with a 2nd rank tensor. The moment of inertia relates angular momentum and angular velocity as:

$$L_i = I_{ij}\omega_j \quad (1.15)$$

We know that $\vec{L}$ and $\vec{\omega}$ are well defined vectors. So let's have a look at how the norm of Eq. 1.15 transforms under rotations:

$$L'_i = R_{ij}L_j = R_{ij}(I_{jk}\omega_k) = R_{ij}I_{jk} \underbrace{R_{kl}^T R_{lm}}_{\delta_{km}} \omega_m \quad (1.16)$$

We explicitly take the transpose of R^T to arrive at,

$$L'_i = R_{ij}L_j = [R_{ij}R_{lk}I_{jk}][R_{lm}\omega_m] \equiv I'_{il}\omega'_l \quad (1.17)$$

If we want an example from special relativity and classical field theory, we have for a general field-strength tensor,

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{pmatrix}. \quad (1.18)$$

As a cute example of Lorentz invariance, let's consider F in some reference frame where there is only a z component to the electric field and the magnetic field exactly vanishes:

$$F^{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & -E_z \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ E_z & 0 & 0 & 0 \end{pmatrix}. \quad (1.19)$$

Applying a boost only in the x direction with a velocity v corresponds to the Lorentz transformation,

$$\Lambda^\mu{}_\nu = \begin{pmatrix} \gamma & -\beta\gamma & 0 & 0 \\ -\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (1.20)$$

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We already asserted that F is a (second-rank) tensor, so each of its indices transforms with its own Lorentz transformation matrix,

$$F'^{\mu\nu} = \Lambda^\mu{}_\sigma \Lambda^\nu{}_\rho F^{\sigma\rho} \quad (1.21)$$

$$= \begin{pmatrix} 0 & 0 & 0 & -\gamma E_z \\ 0 & 0 & 0 & \beta\gamma E_z \\ 0 & 0 & 0 & 0 \\ \gamma E_z & -\beta\gamma E_z & 0 & 0 \end{pmatrix} \quad (1.22)$$

$$\equiv \begin{pmatrix} 0 & 0 & 0 & -E'_z \\ 0 & 0 & 0 & B'_y \\ 0 & 0 & 0 & 0 \\ E'_z & -B'_y & 0 & 0 \end{pmatrix}. \quad (1.23)$$

We see after a Lorentz boost, we've changed from a reference frame where there is only a z component of the E field, to one where there are x and z components of the E field as well as a y component for the B field. Notice that $F_{\mu\nu}F^{\mu\nu}$ is a Lorentz scalar (all indices are contracted) so $F'_{\mu\nu}F'^{\mu\nu} = F_{\mu\nu}F^{\mu\nu}$, so the physics remains unchanged.

Using just classical electrodynamics we see the unification of electricity and magnetism as was first demonstrated in the early 20th century. This unification is seen as a fundamental achievement of classical physics and is consistently sought after in modern fundamental physics. We will eventually see how electrodynamics and the “weak nuclear force” are unified in the electroweak Standard Model. Physicists have further tried to unify the entire Standard Model of particle physics in Grand Unified theories, though any such realization by nature has yet to be observed.

So far the math and physics topics shouldn't be too unfamiliar. That's the point. My goal is to show you that you already understand how symmetries and invariances are built into physics. Then we're going to take these concepts and extend them little by little until using tools we already know from Mechanics, Electrodynamics, and Quantum Mechanics we accidentally stumble on QFT and get a slightly more technical vocabulary along the way.

1.2 Adding Spin

Spin addition is a notoriously confusing topic in Quantum Mechanics courses. Frequently it is introduced in a second semester QM class which should be covered in undergraduate studies. Unfortunately it is often pushed off to a Master's course in QM so it may not be entirely familiar to someone who is pushing ahead in their studies to get to the “good stuff” (particle physics and QFT) faster.

As such we will review the basic premise of what is often referred to as “adding spin.”

Adding two spin- $\frac{1}{2}$ particles

In quantum mechanics we usually work with quanta in a Hilbert space. For example, when you first solve the Hydrogen atom you have kets like $|n, l, m\rangle$. Where n is the “principle quantum number” which is related to the energy of the state, l is the orbital angular momentum, and m the magnetic quantum number. Typically in the second semester of quantum mechanics we learn that we actually need some extra label for the electron “spin” so that we can incorporate the spin-orbit coupling into our Hamiltonian and calculate corrections to the spectrum.

Spin is thrown in randomly, and we're just told that the electron has spin $\frac{1}{2}$. Soon we'll see how spin naturally comes out of representation theory and choosing that our quantum field correspond to an “irreducible representation” of the Lorentz group. For now let's follow the QM example and simply stipulate the spin operator for spin $\frac{1}{2}$ particles,

$$\vec{S} = \frac{1}{2}\vec{\sigma}, \quad (1.24)$$

where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z) = \sigma_i$ and the σ_i are the Pauli matrices. Further, we have the commutation relation between the spin operator components:

$$[S_i, S_j] = i\epsilon_{ijk}S_k. \quad (1.25)$$

Since these don't commute we can't have simultaneous eigenvalues, *meaning* that we can't prepare a state with definite S_i and $S_{j \neq i}$ as well as that operators have an uncertainty principle relating them. We can, however, define the operator,

$$S^2 = \vec{S} \cdot \vec{S}, \quad (1.26)$$

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which separately commutes with all S_i :

$$[S^2, S_i] = 0. \quad (1.27)$$

From here we generally choose a particular spin operator and use its eigenvalue to label the spin of a state. The common choice is S_z with labels $|\pm\rangle$ corresponding to spin projection $\pm\frac{1}{2}$:

$$S_z |\pm\rangle = \pm\frac{1}{2} |\pm\rangle \quad (1.28)$$

Recalling the Pauli matrices,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (1.29)$$

and defining,

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (1.30)$$

we can work out the following relations:

$$S_x |\pm\rangle = \frac{1}{2} |\mp\rangle, \quad (1.31)$$

$$S_y |\pm\rangle = \pm\frac{i}{2} |\mp\rangle, \quad (1.32)$$

$$S^2 |\pm\rangle = \frac{1}{2} \left(1 + \frac{1}{2}\right) |\pm\rangle \equiv s(s+1) |\pm\rangle. \quad (1.33)$$

Where we have rewritten Eq. 1.33 so that the total spin s is manifest. From this we can label a given spin state in terms of its total spin and its S_z projection:

$$|s, m\rangle \xrightarrow{\text{spin } 1/2} \left|\frac{1}{2}, \pm\right\rangle \quad (1.34)$$

Now we're in a position to "add two spin $\frac{1}{2}$ " particles. More precisely, we can now explore how to take the tensor product of two spin- $\frac{1}{2}$ Hilbert spaces and decompose them into irreducible representations of definite total spin. Right now that more formal part isn't really clear, but let's start by going through what we already know about spin to understand "definite total spin" as a concept then in the next section we'll tackle "irreducible representation."

Let's call a spin- $\frac{1}{2}$ Hilbert space $\mathcal{H}_{\frac{1}{2}}$, we can write this problem as:

$$\mathcal{H}_{\frac{1}{2} \oplus \frac{1}{2}} = \mathcal{H}_{\frac{1}{2}} \otimes \mathcal{H}_{\frac{1}{2}} \quad (1.35)$$

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Now, we're certainly welcome to invent our own two-spin states. For example:

$$\left. \begin{aligned} |+\rangle \otimes |+\rangle &= |++\rangle \\ |+\rangle \otimes |-\rangle &= |+-\rangle \\ |-\rangle \otimes |+\rangle &= |-+\rangle \\ |-\rangle \otimes |-\rangle &= |--\rangle \end{aligned} \right\} \text{Order Matters!} \quad (1.36)$$

Now we want to do the same analysis as above. We define a new S^2 and S_i that act on the new Hilbert space (notice the use of boldface for operators in $\mathcal{H}_{\frac{1}{2}} \otimes \mathcal{H}_{\frac{1}{2}}$):

$$\vec{\mathbf{S}} = \mathbf{S}_i = \frac{1}{2}\sigma_i \otimes \mathbb{1} + \mathbb{1} \otimes \frac{1}{2}\sigma_i \equiv S_i^{(1)} + S_i^{(2)}. \quad (1.37)$$

Taking the z component of this operator and acting on one of our new eigenkets we have:

$$\mathbf{S}_z |++\rangle = (S_z^{(1)} + S_z^{(2)}) |++\rangle \quad (1.38)$$

$$= (S_z |+\rangle) \otimes |+\rangle + |+\rangle \otimes (S_z |+\rangle) \quad (1.39)$$

$$= \left(\frac{1}{2} + \frac{1}{2}\right) |++\rangle \quad (1.40)$$

For the total spin operator, directly in analogy with S^2 from $\mathcal{H}_{\frac{1}{2}}$ we have:

$$\mathbf{S}^2 = (S_i \otimes \mathbb{1} + \mathbb{1} \otimes S_i)^2 \quad (1.41)$$

$$= \frac{1}{4} [6\mathbb{1} \otimes \mathbb{1} + 2(\sigma_x \otimes \sigma_x + \sigma_y \otimes \sigma_y + \sigma_z \otimes \sigma_z)] \quad (1.42)$$

To arrive here we need to know how to expand the product of two “ $\otimes$,”

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD), \quad (1.43)$$

if this is not intuitive, remember that for $(C \otimes D)$, C acts on the first state, and D on the second. So:

$$(C \otimes D) |\psi\rangle \otimes |\phi\rangle = (C|\psi\rangle) \otimes (D|\phi\rangle) \quad (1.44)$$

$$(A \otimes B)(C \otimes D) |\psi\rangle \otimes |\phi\rangle = (A \otimes B)(C|\psi\rangle) \otimes (D|\phi\rangle) = (AC|\psi\rangle) \otimes (BD|\phi\rangle) \quad (1.45)$$

we also made use of the Pauli matrix identity:

$$\sigma_i \sigma_j = \delta_{ij} \mathbb{1} + i\epsilon_{ijk} \sigma_k. \quad (1.46)$$

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Applying $\mathbf{S}^2$ on our $|++\rangle$ state gives:

$$\mathbf{S}^2 |++\rangle = \frac{1}{4} [6 |++\rangle + 2(|--\rangle - |--\rangle + |++\rangle)] = 2 |++\rangle = s(s+1) |++\rangle \quad (1.47)$$

Giving total spin for the combined state of $s = 1$. This is pretty intuitive, in essence the both spin projections have added to 1. Don't be fooled, however, this is still quantum mechanics. Let's take a look at the action $\mathbf{S}_z$ and $\mathbf{S}^2$ on other eigenkets from Eq. 1.36:

$$\mathbf{S}_z |\pm\pm\rangle = \pm |\pm\pm\rangle \quad (1.48)$$

$$\mathbf{S}_z |\pm\mp\rangle = 0 |\pm\mp\rangle \quad (1.49)$$

$$\mathbf{S}^2 |\pm\pm\rangle = 2 |\pm\pm\rangle = 1(1+1) |\pm\pm\rangle \quad (1.50)$$

$$\mathbf{S}^2 |\pm\mp\rangle = (|+-\rangle + |-+\rangle) \quad (1.51)$$

Consider the first and third lines, we have simultaneous eigenvalues for $\mathbf{S}_z$ and $\mathbf{S}^2$ for both eigenkets $|\pm\pm\rangle$. We run into a problem for $|\pm\mp\rangle$: *the state does not have a definite total spin*. Is this a problem in a laboratory? No. Of course we can work with mixed states and experimentalists do. We will find this topic useful later, however, when talking about fields and their transformations under the Lorentz group.

Remember, we said we want to “add two spin $\frac{1}{2}$ ” or take the tensor product of two spin- $\frac{1}{2}$ Hilbert spaces and decompose them into **irreducible representations** of *definite total spin*. Now I've made good on my promise to explain what I mean by “definite total spin.” But how do we achieve this?

Let's explore another tool we have from looking at single spin- $\frac{1}{2}$ eigenket. We have the ladder (or raising/lowering) operators. Recall:

$$S_{\pm} = S_x \pm iS_y \quad (1.52)$$

$$S_{\mp} |\pm\rangle = |\mp\rangle \quad (1.53)$$

$$S_{\mp} |\mp\rangle = 0 \quad (1.54)$$

$$[S^2, S_{\pm}] = 0 \quad (1.55)$$

That last equation will be key. For the single spin- $\frac{1}{2}$ state, $S_{\pm}$ will preserve the eigenspace of S^2 . So $S_{\pm}$ moves among states of different m while preserving s in $|s, m\rangle$. If we can achieve the same for our two-particle state we

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will not change the total spin but will find a new state. Let's extend this to the case of $\mathcal{H}_{\frac{1}{2}} \otimes \mathcal{H}_{\frac{1}{2}}$:

$$\mathbf{S}_{\pm} \equiv S_{\pm} \otimes \mathbb{1} + \mathbb{1} \otimes S_{\pm} \equiv \mathbf{S}_{\pm}^{(1)} + \mathbf{S}_{\pm}^{(2)}. \quad (1.56)$$

From this definition and a bit of algebra we can show,

$$[\mathbf{S}^2, \mathbf{S}_{\pm}] = 0, \quad (1.57)$$

the only thing that may not be total obvious is that operators acting on different Hilbert spaces for example:

$$[\mathbf{S}_i^{(1)}, \mathbf{S}_{\pm}^{(2)}] = [\mathbf{S}_i^{(2)}, \mathbf{S}_{\pm}^{(1)}] = 0. \quad (1.58)$$

We won't take the space to do the full derivation here, but it is made substantially easier by rewriting the total spin operator in terms of the ladder operators:

$$\mathbf{S}^2 = \mathbf{S}_z^2 + \frac{1}{2} (\mathbf{S}_+ \mathbf{S}_- + \mathbf{S}_- \mathbf{S}_+) \quad (1.59)$$

So we can see that the ladder operators *preserve total spin* while changing the state. We also saw that $|++\rangle$ had a well-defined quantum number under $\mathbf{S}^2$. So what we do now is use $\mathbf{S}_-$ to find new eigenvalues until we run out. That is, “we start at the top of the ladder and move our way down.” Each new state will have definite total spin. At the bottom of the ladder $\mathbf{S}_-$ will annihilate the lowest “rung on the ladder” telling us we've reached the end of our set of states linked by their total spin quantum number. We'll also see that while operators like $\mathbf{S}_i$ may mix between states with differing $\mathbf{S}_z$ projection, we'll never lose the meaning of this total spin quantum number.

Applying $\mathbf{S}_-$ to $|++\rangle$ we find:

$$\mathbf{S}_- |++\rangle = (\mathbf{S}_{\pm}^{(1)} + \mathbf{S}_{\pm}^{(2)}) |++\rangle, \quad (1.60)$$

$$= |-+\rangle + |+-\rangle. \quad (1.61)$$

Applying the ladder operator again gives,

$$\mathbf{S}_- (|-+\rangle + |+-\rangle) = (\mathbf{S}_{\pm}^{(1)} + \mathbf{S}_{\pm}^{(2)}) (|-+\rangle + |+-\rangle), \quad (1.62)$$

$$= (0 + |--\rangle + |--\rangle + 0) = 2 |--\rangle, \quad (1.63)$$

Where the 0's are the terms where $\mathbf{S}_-$ acts on $|- \rangle$ giving zero. Applying $\mathbf{S}_-$ again gives 0, we have reached the bottom of our ladder. Let's quickly

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rewrite these states so they have norm “1” and we’ll give them nicknames whose origin will become clear shortly:

$$|1, 1\rangle = |++\rangle , \quad (1.64)$$

$$|1, 0\rangle = \frac{1}{\sqrt{2}} (|+-\rangle + |-+\rangle) , \quad (1.65)$$

$$|1, -1\rangle = |--\rangle . \quad (1.66)$$

Applying $\mathbf{S}^2$ and $\mathbf{S}_z$ to these eigenstates and performing a fair amount of algebra gives:

$$\mathbf{S}^2 |1, 1\rangle = 1(1+1) |1, 1\rangle = s(s+1) |1, 1\rangle , \quad (1.67)$$

$$\mathbf{S}_z |1, 1\rangle = 1 |1, 1\rangle , \quad (1.68)$$

$$\mathbf{S}^2 |1, 0\rangle = 1(1+1) |1, 0\rangle = s(s+1) |1, 0\rangle , \quad (1.69)$$

$$\mathbf{S}_z |1, 0\rangle = 0 |1, 0\rangle , \quad (1.70)$$

$$\mathbf{S}^2 |1, -1\rangle = 1(1+1) |1, -1\rangle = s(s+1) |1, -1\rangle , \quad (1.71)$$

$$\mathbf{S}_z |1, -1\rangle = -1 |1, -1\rangle . \quad (1.72)$$

Something is missing though. Remember we had four example states in Eq. 1.36, a little scribbling on paper shows this is all possible pairs of $\pm$ where order matters. Shouldn’t we have one more state? The answer is yes! We can guess where it went, although I’m going to abuse language here: The spin-one combination came from aligning the $s = \frac{1}{2}$ quantum number of both particles, what if their spins were anti-aligned? Wouldn’t we get a spin 0 particle?

We can take inspiration from $|1, 0\rangle$, the $\mathbf{S}_z$ projection there is 0. We just need to also work out total spin $s = 0$ as well. We can make that work with a relative minus sign:

$$|0, 0\rangle = \frac{1}{\sqrt{2}} (|+-\rangle - |-+\rangle) \quad (1.73)$$

Really we didn’t need to guess to start from a form similar to $|1, 0\rangle$. We can arrive at this state by requiring it be orthogonal to the other three and we’ll conveniently arrive to the same $|0, 0\rangle$ (up to a sign which is irrelevant in QM). In this more systematic method we get the relative sign for free. Notice the state $|0, 0\rangle$ is antisymmetric. We’re going to come back to this once we better understand what we’ve really accomplished here.

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All of these states have well defined total spin quantum numbers³. What we want to get to next is that the $s = 1$ states mix among themselves under rotations, but they don't "mix out" of $s = 1$. Similarly for the $s = 0$ case (in this case it can only mix into itself).

For example, let's look at the action of $\mathbf{S}_x$ on each of our states:

$$\mathbf{S}_x |1, 1\rangle = \frac{1}{\sqrt{2}} |1, 0\rangle \quad (1.74)$$

$$\mathbf{S}_x |1, 0\rangle = \frac{1}{\sqrt{2}} (|1, 1\rangle + |1, -1\rangle) \quad (1.75)$$

$$\mathbf{S}_x |1, -1\rangle = \frac{1}{\sqrt{2}} |1, 0\rangle \quad (1.76)$$

$$\mathbf{S}_x |0, 0\rangle = 0 |0, 0\rangle \quad (1.77)$$

If we wanted to dig really deeply into the math behind this, we could show that an arbitrary rotation by the angle theta along the axis $\hat{n}$, $R(\theta, \hat{n})$, acting on a state $|s, m\rangle$ is given by,

$$R(\theta, \hat{n}) |s, m\rangle = \exp\left(-i\frac{\theta}{2}\hat{n} \cdot \mathbf{S}\right) |s, m\rangle = \sum_{m'=-s}^s D_{m'm}^s(\theta, \hat{n}) |s, m'\rangle, \quad (1.78)$$

where the $D_{m'm}^s$ are the "Wigner D-matrices." Their form isn't relevant here, but what's to their right is: The decomposition is into eigenstates of *definite* s . Also bear in mind that the $\mathbf{S}_i$ are those as defined in the total s space, so they are $(2s + 1) \times (2s + 1)$ dimensional as in our example above.

To recap, we reviewed spin $\frac{1}{2}$ in the context of QM. We then explored what it means to take the tensor product of two spin- $\frac{1}{2}$ Hilbert spaces. We're free to work with any state we like, such as the ones in Eq. 1.36. We can choose a certain way to write these states which exhibits a particularly attractive property: under rotations they preserve their total spin quantum number. Our $s = 1$ and $s = 0$ states are what we will learn to refer to as "irreducible representations." We will define this (a little) more carefully in the next section. These irreducible representations (irreps) will have direct analogues to the irreps of the Lorentz group. We'll use these to construct Lorentz invariant Lagrangians, which form the fundamental building blocks of our QFTs.

³They also have well-defined quantum numbers under $\mathbf{S}_z$, but this isn't super important for our purposes.

1.3 Lie Groups, Lie Algebras, and Representation Theory

In the last section we mentioned that $SU(2)$ was the “spin group” without really explaining what we meant. $SU(2)$ is a Lie Group, which can be described locally through the $su(2)$ Lie Algebra. The Lorentz group is another, more complex, Lie Group. We’ll use its properties to impose Lorentz invariance on our QFT.

Lie Algebras and Groups

Here we will flesh out the concepts of the last section by reviewing the definition of a Lie Algebra, Lie Group, and introducing the formal idea of a “representation” and “irreducible representation.” This will finally give us the equipment to define vectors in different representations of the Lorentz group which we will eventually upgrade to the fields from which we form our QFTs.

We’ll start quickly with our slightly simplified definition of a Lie Algebra:

A **Lie Algebra** is a mathematical structure consisting of a set of elements equipped with two operations:

1. Addition - we can add any two elements together to get another element. Addition here is commutative, associative, we have an additive identity, and we have additive inverses.
2. The Lie Bracket - Denoted $[x, y]$, the Lie bracket takes two elements of the Lie Algebra and produces another element in the set. This operation satisfies the following properties:

- Bilinearity: It is linear in both arguments, for scalars a and b :

$$[ax + by, z] = a[x, z] + b[y, z] \quad (1.79)$$

- Antisymmetry: $[x, y] = -[y, x]$ for all x and y .
- Jacobi Identity: For any three elements x , y , and z , the bracket satisfies:

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0. \quad (1.80)$$

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A Lie Algebra consists of a set of elements that form a vector space. So we can add any two elements together or multiply them by scalars, and the result is still an element of the Lie Algebra. A minimal subset of these elements (not unique!), called the “generators,” can be chosen such that, through scalar multiplication and addition, they span the entire vector space of the Lie Algebra.

For example, in our familiar case of $SU(2)$ the Pauli matrices can be chosen as the generators. This choice is not unique, I could for example choose any linear combination of the Pauli matrices that still spans the vector space. A caveat we’ll understand better soon, is that this is also “representation” dependent. The Pauli matrices are 2×2 and a single spin $\frac{1}{2}$ particle lives in a two-dimensional Hilbert space. For our $s = 1$ (three-dimensional) case resulting from adding the spin of two spin $\frac{1}{2}$ particles we would instead need generators that are 3×3 .

With Lie Algebras better understood we want to take a look at Lie Groups. The Lorentz group isn’t the only Lie Group we’ll see, we’ll be dealing with them regularly in the context of “gauge symmetry.” We should already have some familiarity with gauge invariance from electrodynamics, but we’ll also see some new structures arise as we assemble the Standard Model.

A **Lie Group** is a group in the usual mathematical sense, there’s a group operation satisfying the usual group properties - closure, associativity, inverses for all elements, and the existence an identity element. What makes Lie groups unique is *they also form a smooth, continuous space (manifold), and the group operation is smooth (i.e. differentiable).*

Think about rotating an object. This is a smooth operation, and if we add up a bunch of infinitesimal rotations we can achieve a finite rotation. In fact, rotations in N -dimensional space constitute the group $O(N)$ the “orthogonal group of order N .” While seemingly simple, this has the very important implication that we can perform calculus on these operations. This makes Lie Groups particularly useful in physics.

If the Lie Algebra describes the infinitesimal structure of the group around the identity element and everything is smooth, then we can generate any group element by compounding these infinitesimal structures through exponentiation:

$$g = e^{i\theta_a T_a}, \quad (1.81)$$

where the T_a are the generators of the Lie Algebra, θ_a are real parameters, and g is an element of the Lie Group. *We can create a finite rotation (more*

generally, transformation) by summing up a series of infinitesimal rotations.

This is profound, the entire Lie Group can be understood in terms of its Lie Algebra. This will allow us to study complex symmetries and their associated transformations by focusing on their simpler infinitesimal counterparts.

Now, we’ve swept a subtlety under the rug. We’re only constructing the Lie Group locally. For “simply connected” Lie Groups this is sufficient to fully describe the group. For groups that are not simply connected, however, this approach may not capture the full global structure of the group.

Let’s consider the example of rotations in two dimensions. We can write any rotation R as:

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \quad (1.82)$$

These rotations have $\det R = 1$. This group is called $SO(2)$, it’s the *special orthogonal group of order 2*. That is, it’s the group describing *proper* rotations with determinant 1 in two dimensions. “Special” indicates determinant 1, “O” tells us orthogonal (rotations in this example), and the “2” indicates the dimension of the space on which the group acts. So $SO(N)$ corresponds to proper rotations in N -dimensional space.

We can toss in “improper” rotations by simply including reflections:

$$F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (1.83)$$

Qualitatively, proper rotations preserve the “handedness” of a rotation. This reflection reverses the orientation of the system.

Orthogonal Groups are defined by the structure $R^T R = \mathbb{1}$. Similarly, **Unitary Groups** are defined by $U^\dagger U = \mathbb{1}$. The “special unitary group of order N ” is denoted $SU(N)$ and we can immediately see that the Spin group falls into this category. The simplest unitary group is $U(1)$ which corresponds only to a phase:

$$U = e^{i\theta} \quad (1.84)$$

Since this is a scalar and has norm 1, we neglect to call it “special” as it is implicit that it has “determinant 1” (which doesn’t make much sense to talk about for a scalar quantity anyway). If you think about it $U(1)$ and $SO(2)$ look quite similar: both are one parameter (θ) groups and while $SO(2)$ is composed of real 2×2 matrices $U(1)$ represents complex phases. They are in fact isomorphic, although we won’t go into these details here.

So Lie Algebras describe locally the Lie Groups we’ll be using throughout these notes. For gauge symmetries, by a happy coincidence, we’ll be dealing

strictly with the “special unitary groups.” For the most part they’ll describe the full structure of the Groups we’ll discuss. Later we’ll see the gauge symmetry of the Standard Model is $U(1) \times SU(2) \times SU(3)$. In the next section we’ll see how that’s not the case for the Lorentz group.

Representation Theory

Finally we’re in a position to discuss *representation theory*. For spin $\frac{1}{2}$, i.e. $SU(2)$, we can represent any element of the group as:

$$U(\theta) = \exp\left(i\theta^a \frac{\sigma^a}{2}\right). \quad (1.85)$$

This is a two-dimensional representation of $SU(2)$. Why is it two dimensional? Because it’s represented by matrices of dimension two! That simple.

We could also form a four-dimensional representation of $SU(2)$ by exponentiating the $\mathbf{S}_i$ of Eq 1.37 from Chapter 1.2. After all the work we did in that section, it would be really nice if we could form a three-dimensional representation. We won’t go through the details of actually achieving this, but the answer is after a similarity transformation acting on the $\mathbf{S}_i$ we can put them in “block diagonal” form⁴. This results in one block of 3×3 matrices:

$$S_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad S_y = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad S_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (1.86)$$

Exponentiating these matrices generates all elements of $SU(2)$ in a three-dimensional representation. We also had the one-dimensional, $s = 0$, or trivial representation. Trivial in the sense that all elements of the group act as the identity operator on the vector space.

So what’s the difference between the two-, three-, and four-dimensional representations (besides their dimensionality)? As we saw in Chapter 1.2 the elements of the two- and three-dimensional representations don’t mix into other representations under the group action – “nothing leaks out”. In the four-dimensional representation we have combinations of the three-dimensional and one-dimensional representations. These combinations rotate within the

⁴This is guaranteed by “Maschke’s theorem” (for finite or compact groups). In the context of $SU(2)$, it is likely also familiar as the Clebsch-Gordon decomposition of tensor product representations.

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full four-dimensional space, but their components belonging to the one- and three-dimensional representations do not.

To understand this concept of representation better, we can look at a more formal definition of representation:

A **representation** of a group G on a vector space V is a way to assign a matrix to each group element such that multiplying group elements corresponds to multiplying the matrices (respecting the group multiplication):

$$\underbrace{g_1, g_2 \in G}_{\text{abstract group elements}} \Rightarrow \underbrace{\rho(g_1 g_2) = \rho(g_1) \rho(g_2)}_{\rho(g) \text{ are matrices}} \quad (1.87)$$

This definition fits very well with what we've done so far. For $SU(2)$ we had group elements that we mapped to exponentiated Pauli matrices for the two-dimensional representation, similarly we saw how this works for the one-, three-, and four-dimensional representations.

Then we can quite naturally arrive at irreducible representations:

A subspace $W \subset V$ that is invariant under the group action is called a **subrepresentation**.

If V has two subrepresentations which are V itself and the trivial subspace $\{0\}$, we refer to the representation as an **irreducible representation**.

We shouldn't confuse $\{0\}$ with the trivial representation. It is the 0 vector which is always invariant under any linear operation.

Looking at our four dimensional-representation we can see it can be reduced to the three-dimensional and one-dimensional representations. It is *not irreducible*. We haven't necessarily shown that the one-, two-, and three-dimensional representations are irreducible. This could be proven by showing that the decomposition in Eq. 1.78 holds for these representations.

We showed that adding the spin of two spin one-half particles results in a spin-one and spin-zero irreducible representation (frequently referred to as an "irrep"). We can write this mathematically as:

$$\frac{1}{2} \otimes \frac{1}{2} = 1_S \oplus 0_A, \quad (1.88)$$

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where we've labeled the representations by their spin and the subscript S refers to the fact the spin-one representation is symmetric, while A refers to the fact that the spin-0 irrep is antisymmetric (under exchange of $|+-\rangle$ and $| - + \rangle$, see e.g. Eq. 1.73). A mathematician would prefer,

$$2 \otimes 2 = 3_S \oplus 1_A, \quad (1.89)$$

which instead refers to the dimension of the irreps. Interestingly enough, we can actually infer this directly from the product of two Pauli matrices:

$$\sigma_i \sigma_j = \delta_{ij} \mathbb{1} + i \epsilon_{ijk} \sigma_k. \quad (1.90)$$

The Kronecker delta is symmetric in its indices, while the Levi-Civita tensor is antisymmetric.

For $SU(2)$, the two-dimensional representation is referred to as the Fundamental (or doublet), while the three-dimensional representation is referred to as the Adjoint (or triplet). “**Doublet**” and “**Triplet**” refer to two- or three-dimensional representations. “**Fundamental**” refers to the N -dimensional representation of $SU(N)$ and “**Adjoint**” to the $(N^2 - 1)$ -dimensional representation. Fundamental and Adjoint have more profound implications we'll explore much later when we look at gauge symmetries. Don't worry about remembering all this jargon, we'll keep repeating the definitions as necessary.

For Mathematicians a **representation** of a group is literally these matrix representations we've discussed. Particle physicists will frequently refer to, for example “a field/vector/tensor in the Adjoint representation.” This is lazy language that physicists love and irritates mathematicians. Formally speaking, when we refer to these fields etc., we should be saying “a field transforming according to the Adjoint representation.”

We've now had our first look at how Lie Algebras and Groups have a deep connection to symmetries that underpin much of physics, with their differentiability demonstrating their utility as indispensable tools. Along the way, we also discussed the role of the Unitary Group ($U(N)$) in particle physics. The example of spin $\frac{1}{2}$ bridged these topics with representation theory, providing a tangible application of the underlying mathematics. Finally, we have a solid enough foundation in Lie Groups and representation theory to delve into the Lorentz Group.

1.4 The Lorentz Group

In constructing our QFT we want to achieve what QM fails to do. We want to respect the tenets of Special Relativity, i.e. that the laws of physics are the same in all inertial frames. To achieve this we want to write the theory in terms of Lorentz invariants.

The Lorentz Algebra and Group

We'll start by taking the idea of a Lorentz Transformation which preserves the spacetime interval as an analogue of rotations preserving a vector's length. We want to show, just for concreteness, that Lorentz transformations do indeed form a group.

Analogous to rotational invariants we have:

$$\bar{x}^\mu = \Lambda^\mu{}_\nu x^\nu \quad (1.91)$$

$$\bar{x}^2 = \eta_{\mu\nu} \bar{x}^\mu \bar{x}^\nu \quad (1.92)$$

$$= \eta_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma x^\rho x^\sigma \quad (1.93)$$

$$= x^2 \quad (1.94)$$

The last equivalence follows from our knowledge that x^μ is a well-defined Lorentz vector. As such we can infer:

$$\eta_{\rho\sigma} = \eta_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma \quad (1.95)$$

$$= \Lambda_{\nu\rho} \Lambda^\nu{}_\sigma \quad (1.96)$$

$$\eta^{\rho\alpha} \eta_{\alpha\sigma} = \eta^{\rho\alpha} \Lambda_{\nu\alpha} \Lambda^\nu{}_\sigma \quad (1.97)$$

$$\delta^\rho{}_\sigma = \Lambda_{\nu}{}^\rho \Lambda^\nu{}_\sigma \quad (1.98)$$

That is, the inverse of Λ is given by:

$$(\Lambda^{-1})^\rho{}_\nu = \Lambda_\nu{}^\rho \quad (1.99)$$

With this we can establish that Lorentz transformations form a group:

- The product of two Lorentz Transformations is a Lorentz Transformation

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- The product is associative
- There is an identity ($\delta^\mu{}_\nu$)
- There exists an inverse for all $\Lambda^\mu{}_\nu$

Now we've swept the first two points under the rug, this is to reduce tedious calculations. We will need the existence of an inverse later, justifying the added effort with respect to that proof.

We learned that because of the “smoothness” of a Lie Group, its infinitesimal structure allows us to get a bigger picture of the Group as a whole through exponentiating the generators. Remember, the generators form the Lie Algebra of the group and this exponential map connects the Lie Algebra and the Lie Group. So our first step will be to look at the infinitesimal structure of Lorentz transformations, this will let us obtain the generators of the Lorentz group – analogues of the Pauli matrices for the two-dimensional irrep of the Spin group.

We can express a given infinitesimal Lorentz Transformation⁵ as:

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \delta\Lambda^\mu{}_\nu. \quad (1.100)$$

This is really just expanding an exponential to its first order, we should therefore associate $\delta\Lambda$ with the generators multiplied by real parameters (analogous to $i/2\theta^a\sigma^a$ for $SU(2)$, see Eq. 1.85). Also notice that we've chosen to do this in the representation that acts on Lorentz vectors. Why? Without much QFT background this is likely the most familiar vector space that Lorentz transformations act on. Don't worry: we'll dig into other representations of the Lie group in great detail in the next chapter.

Next we'll take Eq. 1.96 and expand it infinitesimally:

$$\eta_{\rho\sigma} = \eta_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma, \quad (1.101)$$

$$= \eta_{\mu\nu}(\delta^\mu{}_\rho + \delta\Lambda^\mu{}_\rho)(\delta^\nu{}_\sigma + \delta\Lambda^\nu{}_\sigma), \quad (1.102)$$

$$= \eta_{\rho\sigma} + \underbrace{\delta\Lambda_{\sigma\rho} + \delta\Lambda_{\rho\sigma}}_{=0 \rightarrow \delta\Lambda_{\sigma\rho} = -\delta\Lambda_{\rho\sigma}} + \mathcal{O}(\delta\Lambda^2). \quad (1.103)$$

We find that $\delta\Lambda_{\rho\sigma}$ is a 4×4 antisymmetric matrix. It is real, as it takes e.g. x^μ to $\bar{x}^\mu$. A real antisymmetric matrix is composed of 6 real parameters. Why?

⁵Up to pesky complications analogous to the our discussion of improper rotations, $SO(2)$ and $O(2)$ around Eq. 1.83

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- A real 4×4 matrix has 16 elements.
- Antisymmetric means the diagonal four elements must vanish: $12 = 16 - 4$.
- The upper right elements of the matrix are minus the lower left: $6 = 12/2$.

These constitute three rotations and three boosts. The rotations can be expressed as $\delta\Lambda_{ij} = -\epsilon_{ijk}\hat{n}_k\delta\theta$, this is a rotation in the ij -plane. The boosts can be written $\delta\Lambda_{i0} = \hat{n}_i\delta\eta$, corresponding to a boost in the i direction. Here and below, we've split Lorentz indices μ into their spatial components i and temporal 0.

We want to transition to speaking about the Lorentz Group more generally. To do so we'll discuss the Lorentz Transformations $U(\Lambda)$ as the abstract mathematical elements of the group. Then our discussion becomes representation independent. This will be necessary for us to learn information about the various irreps of the Lorentz group we'll want to work with. We can write an infinitesimal Lorentz Transformation as,

$$U(\Lambda) = \mathbb{1} + \frac{i}{2}\delta\omega_{\mu\nu}M^{\mu\nu}, \quad (1.104)$$

where the ω s are the free parameters (analogues of θ s for $SU(2)$ in Eq. 1.85) and the M s are the generators of the Lorentz Group. The Lie Algebra of the Lorentz group, $SO(1,3)$, is given by:

$$[M^{\mu\nu}, M^{\rho\sigma}] = i(\eta^{\mu\rho}M^{\nu\sigma} - \eta^{\nu\rho}M^{\mu\sigma}) - i(\eta^{\mu\sigma}M^{\nu\rho} - \eta^{\nu\sigma}M^{\mu\rho}). \quad (1.105)$$

If we rewrite the generators M in terms of spatial rotations J and boosts K (mixing space and time coordinates) we find:

$$M^{ij} = \epsilon^{ijk}J_k, \quad (1.106)$$

$$M^{0i} = K_i. \quad (1.107)$$

notice there are three Js and three Ks for a sum of 6, corresponding to the six Lorentz transformations we found in the vector representation. After a fair amount of algebra we find that the Lie Algebra of the Lorentz group reduces to:

$$[J_i, J_j] = i\epsilon_{ijk}J_k, \quad (1.108)$$

$$[K_i, K_j] = -i\epsilon_{ijk}J_k, \quad (1.109)$$

$$[J_i, K_j] = i\epsilon_{ijk}K_k. \quad (1.110)$$

The commutators give us an intuitive view of these operations: rotations mix to form rotations (they have a closed algebra!), boosts in different directions generate rotations, and mixing rotations and boosts yield new boosts.

1. The Lorentz group and its Irreducible Representations

We’re going to take this rewriting of generators one step further. This abstracts the physical intuition of the generators, but will help us understand the irreps of the Lorentz group in terms of the irreps of $SU(2)$ that are now very familiar. We start by rewriting:

$$J_i = (N_i + N_i^\dagger) , \quad (1.111)$$

$$K_i = i (N_i - N_i^\dagger) . \quad (1.112)$$

N_i and $N_i^\dagger$ are our new generators. Plugging this into Eqs. 1.108–1.110 gives us the Lie Algebra in terms of these new generators:

$$[N_i, N_j] = i\epsilon_{ijk}N_k , \quad (1.113)$$

$$[N_i^\dagger, N_j^\dagger] = i\epsilon_{ijk}N_k^\dagger , \quad (1.114)$$

$$[N_i, N_j^\dagger] = 0 . \quad (1.115)$$

In essence we’ve rewritten the Lie Algebra of the Lorentz group in terms of two $SU(2)$ Lie Algebras that “don’t talk to each other” ($[N_i, N_j^\dagger] = 0$).

Representations of the Lorentz Group

We’ve reduced the problem of finding irreps of the Lorentz group to the addition of spin problem from Chapter 1.2. In fact, for most purposes in QFT it’s not even necessary to go beyond the addition of spin $\frac{1}{2}$!

Let’s start by labelling our irreps by $(2n + 1, 2n' + 1)$, where the first label corresponds to the irrep under $SU(2)$ formed by the generators N_i , and the second corresponding to $N_i^\dagger$. Such an object has $(2n + 1) \times (2n' + 1)$ components with n and n' integers or half integers as with spin.

Then by making choices of n and n' we can find different irreps of the Lorentz Group. Let’s embrace the particle physicist language of referring to “vectors transforming according to an irrep” just as fields in an irrep (even though we haven’t established the idea of a field yet). We list some of these irreps Table 1.1. A **scalar field** corresponds to the trivial representation. Left- and right-handed Weyl fields look like spin $\frac{1}{2}$ particles in $SU(2)$, but transforming separately under the two different $SU(2)$ algebras. Finally a vector, perhaps counter-intuitively, transforms like a spin $\frac{1}{2}$ particle under both $SU(2)$ algebras. (Why isn’t the vector $(4, 1)$ or $(1, 4)$? More on this later.)

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Name	n	n'	$(2n + 1, 2n' + 1)$	Components
Scalar (trivial irrep)	0	0	(1, 1)	1 component
Left handed spinor	$\frac{1}{2}$	0	(2, 1)	2 components
Right handed spinor	0	$\frac{1}{2}$	(1, 2)	2 components
Vector	$\frac{1}{2}$	$\frac{1}{2}$	(2, 2)	4 components

Table 1.1: List of the lowest dimension irreps of the Lorentz Group.

Now we see the similarities between spin and the Lorentz group. While in QM spin is just thrown in by hand, seemingly ad hoc, it actually naturally comes out of these irreps and will be a consequence of forming fields out of irreps of the Lorentz Group. We'll do this by touching on the relevant topics without working out the details.

We start with one of the “Casimir” operators of the Lorentz Group. For context, S^2 was the (only) Casimir for our discussion of the Spin Group. The Pauli-Lubanski vector is defined as:

$$W^\mu = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} M_{\nu\rho} P_\sigma \quad (1.116)$$

For a particle with mass, we can take $P_\sigma = (m, \vec{0})$ reducing W to,

$$W^i = \frac{m}{2} \epsilon^{ijk} M_{jk}, \quad (1.117)$$

where we've included that requiring $\sigma = 0$ limits all other Lorentz indices to spatial indices and that the four-dimensional Levi Civita tensor is reduced to three dimensions. Employing Eq. 1.106 gives:

$$W^i = \frac{m}{2} \epsilon^{ijk} \epsilon^{jkl} J_l, \quad (1.118)$$

$$= \frac{m}{2} (\delta^{il} \delta^{jj} - \delta^{ij} \delta^{jl}) J_l, \quad (1.119)$$

$$= m J_i. \quad (1.120)$$

When this operator acts on a Field in a specific irrep its eigenvalues therefore are dependent on some quanta of angular momentum. This angular momentum is intrinsic to the field in the sense that it comes from the Field's intrinsic property of being in an irreducible representation.

We've achieved something quite profound: we've found a fundamental understanding of spin as an *intrinsic angular momentum quantum number* associated with a specific particle. Why intrinsic? Well W^2 is a Lorentz invariant

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quantity! So spin is reference frame invariant. This also clarifies why we do non-obvious things in QM like adding normal angular momentum and spin and why they are conserved together as a part of total angular momentum, rather than as unrelated quantum numbers.

We'll state without proof: the eigenvalues of W^2 are proportional to $s(s+1)$, with s the spin corresponding to the field. Indeed W^2 labels irreps of the Lorentz group in a similar manner to S^2 for the Spin group. This comes with a subtlety for massless particles that we will neglect in this text. Acting with W^2 on a scalar field returns $s = 0$, acting on either spinor will give $s = \frac{1}{2}$, what happens when it acts on the vector, $(2, 2)$? Well the vector is analogous to adding the spin of two spin- $\frac{1}{2}$ particles, so it contains a spin 1 and spin 0. This will lead to some problems later on and require us to use gauge fixing to remove the spin 0 component to properly describe spin 1 particles like the photon and gluons.

Let's rewind to the vector irrep $(2, 2)$, why isn't it the four-dimensional irrep $(4, 1)$? If we apply W^2 to a field in the $(4, 1)$ irrep we'll find $s = 3/2$. This actually corresponds to adding the spin of three spin $\frac{1}{2}$ particles:

$$\frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} = \frac{3}{2} \oplus 1 \oplus \frac{1}{2}. \quad (1.121)$$

So the $(4, 1)$ irrep of the Lorentz group corresponds to the $3/2$ irrep of $SU(2)$ in the above equation. An example spin $3/2$ particle is the Δ hadron. The reason we aren't emphasizing this or higher spin irreps of the Lorentz group is we won't need them for our discussions. In fact we'll eventually see that, for a theory with a finite number of free parameters⁶ we need only consider the scalar, spinor, and vector representations.

Parity, Time reversal, and the Lorentz group

We've taken a general Lorentz Transformation and expanded it infinitesimally, this allowed us to identify the infinitesimal action of the Lorentz Group (boosts and rotations). Remember though, if we exponentiate just this we don't get back the full Lorentz Group!

We're missing things analogous to improper rotations. What we would get through exponentiation is the “**Proper Orthochronous Lorentz group**” (in fancy language: $SO^+(1, 3)$). What are the missing ingredients? The parity transformation, $\vec{x} \rightarrow -\vec{x}$, and time reversal transformation $t \rightarrow -t$.

⁶This has to do with something called “renormalizability.”

1. The Lorentz group and its Irreducible Representations

Generally when we discuss “Lorentz Invariant QFTs” we mean under the Proper Orthochronous Lorentz Group. There are theories we’ll consider, like Quantum Electrodynamics, that are invariant under the full Lorentz Group $O(1,3)$.

The irreducible representations of the full Lorentz Group differ from those of the Proper Orthochronous Lorentz Group. It won’t be clear why until the next chapter, but parity swaps between the left-handed spinor $(2,1)$ irrep and the right-handed spinor irrep $(1,2)$. This demonstrates that for the full Lorentz group these aren’t the irreps. Instead we have something we’ll later define as a Dirac spinor. The Dirac spinor combines the $(1,2)$ and $(2,1)$ irreps into a single representation.

Dirac spinors are the fundamental fermionionic building blocks of Quantum Electrodynamics (QED) and Quantum Chromodynamics (QCD). For both of these theories parity is a good symmetry. In the Standard Model, in contrast, we’ll have both parity and time reversal violation. In this case the left- and right-handed spinors will be our building blocks.

1.5 Big Picture

The way a physicist thinks of vectors and tensors is in terms of the way transformations act on them. This is going to be fundamental to our construction of not only Lorentz invariant QFTs, but also QFTs that are invariant under gauge and other symmetries. Familiar examples from introductory physics courses include rotations acting on vectors in three dimensions and the field-strength tensor $F_{\mu\nu}$ which is acted on by Lorentz transformations.

We also revisited the Spin group and the classic QM problem of “adding spin.” This gave us a visceral sense of representations and irreducible representations of the Spin group. We saw that the defining feature of these irreducible representations is that under rotations they don’t “mix out” of their representation.

The interplay between transformations and the vector spaces they act on is at its core described by Lie Algebras and their corresponding Lie Groups. The most familiar cases being rotations in N spatial dimensions, the Orthogonal group of order N $O(N)$ and the spin group $SU(2)$. The example of adding spin gave us the context in which to understand representation theory.

By looking at the infinitesimal structure of the Lorentz group, $O(1,3)$, we were able to see how infinitesimal transformations correspond to the familiar rotations and boosts of Lorentz transformations. By manipulating the Algebra of the Lorentz group in its abstract form (in terms of generators $M^{\mu\nu}$) allowed us to see how the generators of the group are in fact rotations and boost, but also allowed us to reduce the algebra to two copies of the $SU(2)$ algebra. Since we know how to add spin, we know how to use these two copies to form the most important irreps of the Lorentz group.

Now we are in a position to study the transformation properties of the irreps under the Lorentz group and write down our first Lorentz invariant Lagrangians.

Chapter 2

Lorentz Invariant Lagrangians

In this chapter we will work toward the goal of understanding how to put together Lorentz invariant Lagrangians. A natural question at this point for many physics students is, why Lagrangians? Aren't Hamiltonians the fundamental building blocks of Quantum Mechanics?

We can, in fact, construct and perform calculations using Lagrangians in the path integral formulation. Our focus in these notes will be on the path integral formulation of QFT. The alternate formulation, referred to as “canonical quantization” is beautiful and has a nice physical interpretation. So we will discuss it qualitatively later. However, the path integral formulation can do things the canonical approach cannot and it often makes proofs far simpler than in the canonical formulation. We'll get to path integrals in the next chapter.

When we reference “Lagrangians” we'll really be referring “Lagrangian densities” which is just a generalization of the Lagrangian, recall the relation between the (usual) Lagrangian L and the action S :

$$S = \int dt L. \quad (2.1)$$

This equation doesn't “treat t and $\vec{x}$ on the same footing. That is, the measure of the integral only depends on t . This won't work for a special relativistic theory (think about how boosts mix spatial and temporal components). Instead we want to work with,

$$S = \int d^4x \mathcal{L}, \quad (2.2)$$

where $\mathcal{L}$ is the **Lagrangian density**. This is hopefully somewhat familiar

from electrodynamics courses where we first see classical field theory. We'll learn in later chapters how to get physical observables from the Lagrangian density.

What we want to achieve is to write Lagrangians that are Lorentz invariant and are in terms of the different irreducible representations of the Lorentz group discussed in Chapter 1. So naturally we'll start by writing Lagrangians in terms of scalars (out simplest irrep), next we'll consider spinors which require quite a bit more effort to construct, and finally we'll look at vectors.

2.1 Lagrangians for Scalars

We'll start with the simplest example, a Lorentz scalar. We need to go from just the abstract concept of an object transforming under the action of the Lorentz group to a field we use in a Lagrangian. This is just a matter of promoting the Lorentz tensor (remember a scalar is a 0th-rank tensor) to a function of a given spacetime point x^μ :

$$\phi \rightarrow \phi(x). \quad (2.3)$$

It's a bit messy to always write x^μ so we'll suppress the Lorentz index in the argument of ϕ . What we've done here is a bit tricky, we now have the abstract Lorentz tensor that transforms one way under the Lorentz group (trivially) and its argument x^μ which transforms in the (2, 2) irrep.

We start by recalling,

$$x^\mu \rightarrow \bar{x}^\mu = \Lambda^\mu{}_\nu x^\nu. \quad (2.4)$$

Since ϕ as a Lorentz tensor transforms trivially, we have for the action of ϕ under the Lorentz group:

$$\phi(x) \rightarrow \phi(\bar{x}) = \phi(\Lambda x). \quad (2.5)$$

That is, after a Lorentz transformation $\bar{x}$ is our new coordinate, which can be written in terms of our old coordinate as $\bar{x} = \Lambda x$.

So if we consider polynomials in $\phi(x)$ with scalar **classical numbers (often referred to as c-numbers)**, c_i , we could have a general Lagrangian:

$$\mathcal{L} = c_0 + c_1 \phi(x) + c_2 \phi(x)^2 + \dots, \quad (2.6)$$

$$\rightarrow c_0 + c_1 \phi(\bar{x}) + c_2 \phi(\bar{x})^2 + \dots. \quad (2.7)$$

I promised Lorentz invariant *Lagrangians*, but this doesn't look Lorentz Invariant! The arguments have just changed from one space-time point to another. It isn't actually our Lagrangians that are invariant, it's the action. To see this we'll need the transformation of the measure,

$$d^4x \rightarrow \left| \det \frac{\partial x^\mu}{\partial \bar{x}^\nu} \right| d^4\bar{x} = d^4\bar{x}, \quad (2.8)$$

where the two measures are related by the Jacobian. For proper Lorentz transformations the determinant is 1, for improper it's -1, either way the absolute value of the determinant gives Jacobian 1. Later we'll see that an

overall normalization of the action doesn't change the physics at the quantum level either¹.

Finally we can see the action transforms as:

$$S[\phi] = \int d^4x \mathcal{L}[\phi(x)] \rightarrow \int d^4\bar{x} \mathcal{L}[\phi(\bar{x})] \quad (2.9)$$

Since x and $\bar{x}$ are just labels², we can see that the action remains invariant, $S[\phi] \rightarrow S[\bar{\phi}] = S[\phi]$. When we talk about Lorentz invariant Lagrangians, we're going to let this be implicit in the conversation. If after a Lorentz transformation the Lagrangian is invariant up to a relabelling of the space-time coordinate, we'll just call it a Lorentz invariant Lagrangian (because we know the action will be invariant).

Adding derivatives

Without derivatives, there are no dynamics in our theory, so we'll need the transformation of a four-derivative, ∂_μ :

$$\bar{\partial}_\mu \phi(\bar{x}) = \frac{\partial}{\partial \bar{x}^\mu} \phi(\bar{x}) = \frac{\partial \phi(x)}{\partial x^\nu} \frac{\partial x^\nu}{\partial \bar{x}^\mu}, \quad (2.10)$$

$$= \frac{\partial x^\nu}{\partial \bar{x}^\mu} \partial_\nu \phi(x), \quad (2.11)$$

$$= (\Lambda^{-1})^\nu{}_\mu \partial_\nu \phi(x). \quad (2.12)$$

The first line is the chain rule, in the last line we invert Eq. 2.4 and take the derivative with respect to $\bar{x}$. We've also employed $\bar{\partial}$ to indicate a derivative acting on the barred coordinate $\bar{x}^\mu$. A Lorentz vector like x^μ transforms under Λ , while ∂_μ transforms under Λ^{-1} . We say x transforms **covariantly** and the derivative **contravariantly**.

Recalling from Chapter 1.4 Eq. 1.99,

$$(\Lambda^{-1})^\rho{}_\nu = \Lambda_\nu{}^\rho, \quad (2.13)$$

and swapping the role of $x \leftrightarrow \bar{x}$ we have the derivative of our field transforms

¹In fact this is going to be critical to our ability to formulate QM from path integrals.

²Remember, under integration the integration variables are just dummy indices.

2. Lorentz Invariant Lagrangians

as:

$$\partial_\mu \phi(x) \rightarrow (\Lambda^{-1})^\nu{}_\mu \bar{\partial}_\nu \phi(\bar{x}), \quad (2.14)$$

$$= \Lambda_\mu{}^\nu \bar{\partial}_\nu \phi(\bar{x}). \quad (2.15)$$

The bar over ∂ indicates the derivative is with respect to the transformed spacetime coordinate $\bar{x}$.

For Lorentz invariance we'll need all indices contracted. So our simplest Lagrangian that makes use of Eq. 2.15 is:

$$\mathcal{L} = \partial^\mu \phi(x) \partial_\mu \phi(x), \quad (2.16)$$

$$\rightarrow (\eta^{\mu\nu}) \Lambda_\nu{}^\sigma \bar{\partial}_\sigma \phi(\bar{x}) (\Lambda^{-1})^\rho{}_\mu \bar{\partial}_\rho \phi(\bar{x}), \quad (2.17)$$

$$= \Lambda^{\mu\sigma} (\Lambda^{-1})^\rho{}_\mu \bar{\partial}_\sigma \phi(\bar{x}) \bar{\partial}_\rho \phi(\bar{x}), \quad (2.18)$$

$$= \bar{\partial}^\sigma \phi(\bar{x}) \bar{\partial}_\sigma \phi(\bar{x}). \quad (2.19)$$

This Lagrangian is invariant (up to relabelling of x allowed by the definition of the action) under Lorentz transformations and contains derivatives. This term will actually end up describing how a noninteracting field goes from one spacetime point to another in QFTs without interactions. It's closely related to what's called the “propagator.”

This proof of invariance was a bit of overkill. The derivative of ϕ is a well-defined Lorentz vector so really all we need to do is contract the Lorentz indices and we know we have a Lorentz scalar (recall our discussion in Chapter 1.1) So we can write a whole slew of derivative based Lagrangian terms, as long as all Lorentz indices come in pairs. For example,

$$\mathcal{L} = \kappa_1 \partial^\mu \phi(x) \partial_\mu \phi(x) + \kappa_2 \phi(x) \square \phi(x) + \kappa_3 \partial_\mu \phi(x) \partial^\mu \square \phi(x) + \dots, \quad (2.20)$$

where $\square = \partial^\mu \partial_\mu$. We can write infinitely many terms as polynomials in ∂_μ and ϕ as long as they are Lorentz invariant. It turns out that many of these operators are redundant under **Integration By Parts identities (IBP identities)**.

To see why IBP identities work, we start with Gauss's theorem,

$$S = \int d^4x \partial_\mu A^\mu(x) = \int d\Sigma_\mu A^\mu(x), \quad (2.21)$$

where $A_\mu(x)$ can be any product of fields and their derivatives. We've related the integral over the full spacetime volume to one over $d\Sigma$ the surface element.

For all applications discussed in these notes³ we will assume that the fields fall off sufficiently fast at infinity that we can take $A_\mu(x)$ to zero. Then we are just integrating “0” over the surface, and the integral is 0. That is, the integrand vanishes on the boundary. In this way we can see the two terms in Eq. 2.20 labeled by κ_1 and κ_2 are related to each other:

$$\partial^\mu [\phi(x) \partial_\mu \phi(x)] = \partial^\mu \phi(x) \partial_\mu \phi(x) + \phi(x) \square \phi(x). \quad (2.22)$$

So we can safely rewrite Eq. 2.20 as,

$$\mathcal{L} = (\kappa_1 - \kappa_2) \partial^\mu \phi(x) \partial_\mu \phi(x) + \kappa_3 \partial_\mu \phi(x) \partial^\mu \square \phi(x) + \dots, \quad (2.23)$$

effectively reducing the number of terms in the Lagrangian.

Some additional notes

We can also introduce a complex field Φ and write Lagrangians like (suppressing the spacetime argument x^μ),

$$\mathcal{L} = \Phi^\dagger \square \Phi + \Phi^\dagger \Phi + \dots, \quad (2.24)$$

or some real or complex ϕ_i with extra components:

$$\mathcal{L} = \phi_i \square \phi_i + \phi_i \phi_i + \dots. \quad (2.25)$$

All of these will produce viable and interesting QFTs. The reason we always write our terms Hermitian is that a QFT without a Hermitian Lagrangian will violate unitarity (i.e. that probabilities add to one). We won’t get into the details of this here, but this will place a restriction on the Lagrangians we consider.

We have one last topic to address before moving on to fields in other representations. Why are all of our fields evaluated at the same spacetime point x^μ ? We could consider a Lagrangian like,

$$\mathcal{L}(x_1) = \int d^4 x_2 \phi(x_1) \phi(x_2) K(x_1, x_2), \quad (2.26)$$

but this Lagrangian is nonlocal. Nonlocal meaning it involves fields evaluated at two different spacetime points. Considering a Lagrangian like this would lead to a theory that violates causality. Again, demonstrating this is beyond the scope of these notes.

³This argument assumes that fields vanish at infinity or that spacetime has no boundary. In finite-volume systems, or for example in lattice field theory with periodic boundary conditions, surface terms need not vanish.

2.2 Weyl Spinors

In Chapter 1.4 we rewrote the Lorentz group in terms of two $SU(2)$ subalgebras:

$$[N_i, N_j] = i\epsilon_{ijk}N_k, \quad (2.27)$$

$$[N_i^\dagger, N_j^\dagger] = i\epsilon_{ijk}N_k^\dagger, \quad (2.28)$$

$$[N_i, N_j^\dagger] = 0. \quad (2.29)$$

From the spin analogy we then were able to identify the irreducible representations of the Lorentz groups as $(2n+1, 2n'+1)$ with n and n' either integer or half integer. We then named the $(2, 1)$ irrep as a “left-handed spinor” and the $(1, 2)$ irrep as a right-handed spinor, both of which have two components. In general, **spinors** will refer to fields in the $(1, 2)$ or $(2, 1)$ irrep, or the Dirac spinor which will combine the two.

We then want to label these spinors according to their vector space with indices. We'll write them as,

$$\psi_a(x), \quad \psi_{\dot{a}}^\dagger(x), \quad (2.30)$$

where we've chosen a to be the index corresponding to left-handed spinors and $\dot{a}$ to be that corresponding to right-handed spinors. That is, a ($\dot{a}$) transforms according to the subalgebra spanned by N_i ($N_i^\dagger$). These spinors are usually referred to as **Weyl spinors**, we'll most commonly refer to them as left- or right-handed spinors instead of e.g. “left-handed Weyl spinor” for brevity. Later we'll work with Dirac spinors which are composed of two Weyl spinors.

Next we want to focus on the left-handed spinor and its transformations under the Lorentz group. After that we get the transformation for the right-handed spinor for free since it's related to the left-handed spinor through hermitian conjugations (i.e. the difference between N and $N^\dagger$).

Left-handed spinors

Under an infinitesimal Lorentz transformation, $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \delta\Lambda^\mu{}_\nu$, a left-handed spinor transforms according to:

$$L_a{}^b(1 + \delta\Lambda) = \delta_a{}^b + \frac{i}{2}\delta\omega_{\mu\nu}(S_L^{\mu\nu})_a{}^b. \quad (2.31)$$

If you're wondering where this equation comes from, it's nothing more than the infinitesimal expansion of our definition of a vector (and in particular, transforming under the Lorentz group), see Eq. 1.1. So the transformation of our left-handed spinor field is given by,

$$\psi_a(x) \rightarrow [L(\Lambda)]_a^b \psi_b. \quad (2.32)$$

Starting from our knowledge of rotations in the Spin $\frac{1}{2}$ case (Eq. 1.24), we can immediately write the generators of rotations in terms of Pauli matrices as:

$$J_i = \frac{1}{2} \sigma_i. \quad (2.33)$$

Then, we know the relationship between N_i , rotations (J_i), and boosts (K_i) from Chapter 1.4 Eq. 1.111 and 1.112,

$$J_i = (N_i + N_i^\dagger), \quad (2.34)$$

$$K_i = i(N_i - N_i^\dagger), \quad (2.35)$$

which allows us to determine the generators of boosts to be:

$$K_i = \frac{i}{2} \sigma_i. \quad (2.36)$$

This requires us to remember that $N^\dagger$ acting on the $(2, 1)$ representation vanishes (i.e. the 1 in $(2, 1)$ means singlet of $N^\dagger$). Next we need to recall the relationship between the generators in the general group sense (i.e. abstracted from specific representations) $M^{\mu\nu}$ and rotations and boosts. These were (see Eqs. 1.106 and 1.107):

$$M^{ij} = \epsilon^{ijk} J_k, \quad (2.37)$$

$$M^{0i} = K_i. \quad (2.38)$$

In the specific representation for left-handed spinors, we call the generators $S_L^{\mu\nu}$, which gives us the form for the generators of a general infinitesimal Lorentz transformation acting on a left-handed spinor:

$$S_L^{ij} = \epsilon^{ijk} J_k = \frac{1}{2} \epsilon^{ijk} \sigma_k \quad (2.39)$$

$$S_L^{0i} = K_i = \frac{i}{2} \sigma_i \quad (2.40)$$

Right-handed spinors

Remember, the generators acting on our right-handed spinor representation ($N^\dagger$) are just the Hermitian conjugate of those acting on the left-handed representation (N). So we can relate left- and right-handed spinors through Hermitian conjugation:

$$[\psi_a(x)]^\dagger = \psi_{\dot{a}}^\dagger(x). \quad (2.41)$$

Take careful note, the hermitian conjugate of an undotted index a is a dotted index $\dot{a}$.

Taking Eq. 2.32 and Hermitian conjugating we find:

$$\psi_{\dot{a}}^\dagger \rightarrow \psi_b^\dagger [L(\Lambda)^\dagger]_{\dot{a}}^b \equiv \psi_b^\dagger [R(\Lambda)]_{\dot{a}}^b. \quad (2.42)$$

Again, note that $a \rightarrow \dot{a}$, as well as that the transpose aspect of the Hermitian conjugate swaps the order. We then take the hermitian conjugate of L from Eq. 2.31,

$$R_{\dot{a}}^b = [L_a^b(1 + \delta\omega)]^\dagger = \delta_{\dot{a}}^b - \frac{i}{2}\delta\omega_{\mu\nu} [(S_L^{\mu\nu})^\dagger]_{\dot{a}}^b, \quad (2.43)$$

$$= \delta_{\dot{a}}^b + \frac{i}{2}\delta\omega_{\mu\nu} [-(S_L^{\mu\nu})^\dagger]_{\dot{a}}^b. \quad (2.44)$$

Notice, the Hermitian conjugation operation acts only on the representation space, so it doesn't swap the μ and ν which are external labels referring to the (μ, ν) plane. That is, they label which generator in the abstract group theory sense we're referring to. (a, b) or $(\dot{a}, \dot{b})$ refer to the generator in a specific representation and therefore are acted on by Hermitian conjugation.

From Eq. 2.44 we can see that the right-handed generators are fully determined by the left-handed ones:

$$(S_R^{\mu\nu})_{\dot{a}}^b = -[(S_L^{\mu\nu})^\dagger]_{\dot{a}}^b. \quad (2.45)$$

Indeed, this is no surprise as the two subalgebras are related by hermitian conjugation.

Invariant symbols

In order to produce Lorentz invariant Lagrangians in terms of spinors we'll need the idea of something called an **invariant symbol**. As we'll see, these

correspond to tensors with indices that transform under a certain representation, but they transform into themselves.

Equation 1.96 expresses the invariance of the Minkowski metric under Lorentz transformations, which ensures that spacetime intervals are preserved:

$$\eta_{\rho\sigma} = \eta_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma \quad (2.46)$$

But notice: the Minkowski metric is a Lorentz tensor, it has two open Lorentz indices. After a Lorentz transformation it transforms into itself (i.e. is invariant)! The Minkowski metric is an invariant symbol.

Consider our relation resulting from the tensor product of two spin- $\frac{1}{2}$ Hilbert spaces:

$$\frac{1}{2} \otimes \frac{1}{2} = 1_S \oplus 0_A . \quad (2.47)$$

So if we took some two-index object, C_{ab} which has two left-handed spinor indices a and b , we should be able to decompose it into two distinct irreps as,

$$C_{ab} = G\delta_{ab} + A\epsilon_{ab} , \quad (2.48)$$

where ϵ_{ab} is the Levi Civita tensor in two dimensions, and G and A label the symmetric and antisymmetric parts respectively. That we can decompose C_{ab} is guaranteed by representation theory. Think about Chapter 1.2: The tensor product of two spin- $\frac{1}{2}$ Hilbert spaces forms a four-dimensional space (e.g. the reducible representation from Eq. 1.36). However, this space could also be decomposed in terms of the irreducible symmetric triplet (Eqs. 1.64–1.66) and antisymmetric singlet (Eq. 1.73).

A in Eq. 2.48 is a scalar. As we know that under $SU(2)$ transformations the terms labeled G and A only transform within their own irreps, we can conclude that ϵ is an invariant symbol:

$$L_a{}^c(\Lambda)L_b{}^d(\Lambda)\epsilon_{cd} = \epsilon_{ab} . \quad (2.49)$$

That is, because 0_A doesn't mix into the symmetric triplet, ϵ must be invariant. The same cannot be said about δ_{ab} . Remember this term corresponds to the triplet where the three components can mix into each other.

Let's take a moment to dig into some math again, we'll need the idea of a "dual space," but as usual we'll avoid too much math language in favor of a more physical understanding.

2. Lorentz Invariant Lagrangians

A **dual space** is the space of linear maps V^* , which are vectors in their own right, mapping the vector space V onto a Field (in the math sense! which for us generally will mean the real or complex numbers). So it is both the vectors being contracted with those in V and the operation defining that contraction. For example V could be column vectors while V^* is the row vectors along with the rule of how to contract row and column vectors.

Consider a rotation in two dimensions, we have the transformations for some $v \in V$ and some $w \in V^*$:

$$v \rightarrow Rv, \quad w \rightarrow wR^T = wR^{-1}. \quad (2.50)$$

Or similarly x^μ is in the space V (we call X^μ covariant) and x_μ is in the dual space V^* (we call x_μ contravariant):

$$x^\mu \rightarrow \Lambda^\mu{}_\nu x^\nu, \quad x_\mu \rightarrow (\Lambda^{-1})_\mu{}^\nu x_\nu. \quad (2.51)$$

For our spinors this means:

$$\psi_a \rightarrow L_a{}^b \psi_b, \quad \psi^a \rightarrow \psi^b (L^{-1})_b{}^a. \quad (2.52)$$

Similarly, considering Eq. 2.49,

$$\epsilon^{cd} (L^{-1})_c{}^a (L^{-1})_d{}^b = \epsilon^{ab}, \quad (2.53)$$

which establishes that ϵ^{ab} is an invariant symbol in the dual space. Consider the contraction of two Levi-Civita tensors (using the convention $\epsilon_{12} = \epsilon^{12} = 1$),

$$\epsilon^{ab} \epsilon_{bc} = \delta^a{}_c, \quad (2.54)$$

under a Lorentz transformation we have:

$$\epsilon^{ab} \epsilon_{bc} \rightarrow [\epsilon^{de} (L^{-1})_d{}^a (L^{-1})_e{}^b] [L_b{}^f L_c{}^g \epsilon_{fg}] \quad (2.55)$$

$$= \delta_e{}^f \epsilon^{de} (L^{-1})_d{}^a L_c{}^g \epsilon_{fg} \quad (2.56)$$

$$= \delta^d{}_g (L^{-1})_d{}^a L_c{}^g \quad (2.57)$$

$$= \delta^a{}_c \quad (2.58)$$

The invariance of Eq. 2.54 as demonstrated above, tells us that $\delta^a{}_c$ is also an invariant symbol.

Taking this a step further, we saw that the $(2, 2)$ irreducible representation corresponded to a Lorentz vector. This means we can relate an object with both a left-handed and right-handed spinor symbol to a Lorentz vector via the equation:

$$A_{a\dot{a}} = \sigma_{a\dot{a}}^\mu A_\mu. \quad (2.59)$$

This relation should hold in any frame, it just says the $(2, 2)$ irreducible representation is equal to the vector irreducible representation⁴. It interpolates between $(2, 2) \leftrightarrow (a, \dot{a})$ and μ . This leads to the conclusion,

$$\sigma_{a\dot{a}}^\mu = L_a{}^b \Lambda^\mu{}_\nu \sigma_{b\dot{b}}^\nu R_{\dot{a}}{}^{\dot{b}}, \quad (2.60)$$

i.e. σ is invariant under the simultaneous transformation of all its indices.

We have invariant symbols that relate spinor indices to spinor indices, as well as invariant symbols that relate one left-handed spinor index and one right-handed spinor index to a vector index. With this we are finally equipped to begin writing Lorentz invariant Lagrangians for Weyl spinors. In the next section we will write down invariant Lagrangians involving the $(2, 1)$ and $(1, 2)$ irreducible representations, but also define a Dirac spinor which combines the two irreps into a four-component spinor. The Dirac spinor is actually what we'll mostly use, we'll see that we can define projection operators that put us into the space of left- or right-handed spinors. By doing this we won't have to keep track of all of these spinor indices (a and $\dot{a}$) and instead can simplify our life to only one index, the Dirac index.

Lagrangians for Weyl spinors I: simple polynomials

Now it's nearly as simple to write Lorentz invariant Lagrangians for our spinor fields as it was for scalars. Remember, when we say Lorentz invariant we really mean up to a relabelling of x , which is integrated over in order to obtain the action and so plays the role of a dummy index. For simplicity of presentation we'll suppress the spacetime argument of our fields. All that's left to do is contract indices.

⁴From a formal math perspective, σ is an intertwiner between the $(2, 2)$ irrep and the vector representation.

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Let's start with simple polynomials in ψ_a and $\psi_a^\dagger$:

$$\mathcal{L} = a_1 \psi_a \epsilon^{ab} \psi_b + b_1 \psi_a^\dagger \epsilon^{\dot{a}\dot{b}} \psi_b^\dagger \quad (2.61)$$

$$\rightarrow a_1 L_a^c \psi_c \epsilon^{ab} L_b^d \psi_d + b_1 \psi_c^\dagger R_{\dot{c}}^{\dot{a}} \epsilon^{\dot{a}\dot{b}} \psi_d^\dagger R_{\dot{d}}^{\dot{b}} \quad (2.62)$$

$$= a_1 \psi_a \epsilon^{ab} \psi_b + b_1 \psi_a^\dagger \epsilon^{\dot{a}\dot{b}} \psi_b^\dagger \quad (2.63)$$

in the second line we implicitly used the invariance of the ϵ s under Lorentz transformations (Eqs. 2.49 and 2.53), and in the last line we again used the invariance to absorb the L and R dependence. The only real difference here from the scalar case is our fields now have indices, so we need things like the ϵ s and later σ to aid in forming the Lorentz invariants⁵.

The simplicity here should be no surprise, all indices are contracted and *everything with an index is a properly defined Lorentz tensor*. Letting $\psi^a = \epsilon^{ab} \psi_b$ and $\psi^{\dagger\dot{a}} = \epsilon^{\dot{a}\dot{b}} \psi_b^\dagger$, we can write as many polynomial terms as we'd like,

$$\begin{aligned} \mathcal{L} = & a_1 (\psi_a \psi^a) + b_1 (\psi_a^\dagger \psi^{\dagger\dot{a}}) + a_2 (\psi_a \psi^a)^2 + b_2 (\psi_a^\dagger \psi^{\dagger\dot{a}})^2 \\ & + c_2 (\psi_a \psi^a) (\psi_a^\dagger \psi^{\dagger\dot{a}}) + d_1 (\psi^{\dagger\dot{a}} \sigma_{\dot{a}a}^\mu \psi^a)^2 + \dots, \end{aligned} \quad (2.64)$$

all of which are Lorentz invariant. Notice the terms quadratic in ψ don't mix between ψ and $\psi^\dagger$. We'll come back to this for Lagrangians with derivatives, but save a more meaningful conversation for when we've defined Dirac spinors which will be the building blocks of the models we form later.

Lagrangians for Weyl spinors II: dynamical Lagrangians

As we mentioned for scalars, derivatives give dynamics. So we want to incorporate derivatives into our theory. Again, since all of our “ingredients” (fields, derivatives, invariant symbols) transform covariantly, it's just a matter of appropriately contracted indices. For example the simplest dynamical terms we can write are,

$$\mathcal{L} = a_1 (\partial_\mu \psi^a) (\partial^\mu \psi_a) + b_2 (\partial_\mu \psi^{\dagger\dot{a}}) (\partial^\mu \psi_a^\dagger). \quad (2.65)$$

⁵If you're paying very close attention, you may have noticed $\psi_a \epsilon^{ab} \psi_b$ looks like it's symmetric in $a \leftrightarrow b$ from the ψ terms and antisymmetric in $a \leftrightarrow b$ from the ϵ ! This term *isn't zero*, later we'll take our fermionic fields to be anticommuting variables, i.e. the $\psi_a \psi_b$ part is also antisymmetric in $a \leftrightarrow b$. We'll need to do this to satisfy Fermi statistics.

These terms will prove problematic for QFTs, as they produce states with “negative energy” which clearly isn’t physics. With our σ matrix we can also write some single-derivative Lagrangian terms:

$$\mathcal{L} = a_1 \psi^{\dagger\dot{a}} \sigma_{a\dot{a}}^\mu \partial_\mu \psi^a + b_1 (\partial^\mu \psi^{\dagger\dot{a}}) \sigma_{a\dot{a}}^\mu \psi^a . \quad (2.66)$$

Interestingly this kinetic term mixes the left-handed spinor and the right-handed spinor. Proving it is Lorentz invariant follows just as above, apply all of the corresponding Lorentz transformations for each index, then use the invariance of the symbol σ to “absorb” the transformations and we’re left with a Lorentz invariant quantity.

At the end of Chapter 2.1, we stressed we need a Hermitian Lagrangian, and Eq. 2.66 above is Hermitian. Notice, however, that integration by parts (IBP) allows us to rewrite the second term in terms of the first,

$$\mathcal{L} = (a_1 - b_1) \psi^{\dagger\dot{a}} \sigma_{a\dot{a}}^\mu \partial_\mu \psi^a , \quad (2.67)$$

in this specific case, as the Lagrangian is Hermitian up to IBP identities, we’re safe to use just the Lagrangian of Eq. 2.67 instead of the first Lagrangian.

Lagrangians for Weyl spinors III: vectors & scalars

We haven’t covered Lagrangians for vectors yet, but since we have σ which has a vector index, we can already incorporate vectors into our Lagrangians. The only way we know how, so far, is to replace ∂^μ with A^μ . The resulting Lagrangian is:

$$\mathcal{L} = a_1 \psi^{\dagger\dot{a}} \sigma_{a\dot{a}}^\mu A_\mu \psi^a . \quad (2.68)$$

Again, we can raise this to an arbitrary power and get a more complicated contribution to the Lagrangian, but it will still be Lorentz invariant. Later we’ll learn that this comes with a lot of subtleties due to Gauge invariance, but for now it’s a perfectly valid Lorentz invariant Lagrangian.

We could also include scalar interactions with a Weyl fermion, for example as:

$$\mathcal{L} = a_1 (\psi_a \psi^a) \phi + b_1 (\psi_a^\dagger \psi^{\dagger\dot{a}}) \phi + c_1 (\psi^{\dagger\dot{a}} \sigma_{a\dot{a}}^\mu \psi^a) \partial_\mu \phi \quad (2.69)$$

We can take the products of these different terms and find even more Lorentz invariant terms as well as many more interesting combinations as we include more fields.

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As was the case with scalars, we can always add more fields ψ , χ , etc, or combine them into a spinor with more components $\Psi_{a,i} = (\psi_a, \chi_a)$. Unlike scalars which we could take to be real, our spinors so far are implicitly complex, they live in complex representations of the Lorentz group.

2.3 Dirac Spinors

Now that we've established the transformation properties of Weyl spinors and have created Lorentz invariant Lagrangians from these spinors, we want to look at the Dirac Spinor. The **Dirac spinor** is going to be a four-component spinor that consists of two Weyl spinors. Like we mentioned earlier, we'll use Dirac spinors to write down all of our future models that contain fermions. We'll simply project out a specific Weyl component using projection operators when we need to irreps of the Proper Orthochronous Lorentz group (See the end of Chapter 1.3).

We'll define the Dirac spinor as,

$$\psi = \begin{pmatrix} \chi_a \\ \xi^{\dagger \dot{a}} \end{pmatrix}, \quad \psi^\dagger = (\chi_a^\dagger, \xi^a), \quad (2.70)$$

i.e. it is defined in terms of one left-handed Weyl spinor (χ) and one right-handed spinor ($\xi^\dagger$). If we took just the usual dot-product between these vectors, we would mix indices and would fail to produce a Lorentz invariant. We will need a special matrix to connect the components, which will match like indices in order to form a Lorentz invariant. Such a matrix can be taken to be:

$$\beta \equiv \begin{pmatrix} 0 & \delta_{\dot{a}}^a \\ \delta_a^c & 0 \end{pmatrix}. \quad (2.71)$$

Recalling the invariance of the Kronecker delta (with one index up and one down) from Eq. 2.58, we see that this can be used to form a Lorentz invariant quantity using the Dirac spinor,

$$\mathcal{L} = \bar{\psi}\psi \equiv \psi^\dagger \beta \psi = \chi_a^\dagger \xi^{\dagger \dot{a}} + \xi^a \chi_a, \quad (2.72)$$

where we have defined $\bar{\psi} = \psi^\dagger \beta$ as we will need $\bar{\psi}$ so frequently. In fact, for Dirac spinors it will play the role that $\psi^\dagger$ played for Weyl spinors.

Naturally, we'll also want a way to incorporate our σ matrix. To do this we'll need a second sigma matrix,

$$\bar{\sigma}^{\mu a \dot{a}} = \epsilon^{\dot{a} b} \epsilon^{ab} \sigma_{b \dot{b}}^\mu, \quad (2.73)$$

whose invariance follows from the separate invariance of ϵ and σ . We will combine σ and $\bar{\sigma}$ into a single matrix in the Dirac spinor space:

$$\gamma^\mu \equiv \begin{pmatrix} 0 & \sigma_{a \dot{c}}^\mu \\ \bar{\sigma}^{\mu c \dot{a}} & 0 \end{pmatrix} \quad (2.74)$$

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Allowing us to write another Lorentz invariant,

$$\mathcal{L} = \bar{\psi}\gamma^\mu\partial_\mu\psi = \xi^a\sigma_{ac}^\mu\partial_\mu\xi^{\dagger\dot{c}} + \chi_a^\dagger\bar{\sigma}^{\mu c\dot{a}}\partial_\mu\chi_c. \quad (2.75)$$

As we pointed out in earlier sections for other fields, there are an infinite number of possible polynomial terms or derivative terms involving the Dirac field that are invariant under the Lorentz group. We can also freely include scalars or vectors in these terms as long as all indices are contracted appropriately.

Gamma matrices and the Clifford algebra

As we'll be focusing on Dirac spinors for the majority of our discussions we need to take a moment to dig in deeper and understand more about the nature of these gamma matrices. To do so we'll make a specific choice for the σ^μ we've been using:

$$\sigma_{a\dot{a}}^\mu = (\delta_{a\dot{a}}, \sigma_{a\dot{a}}^i) \quad \Rightarrow \quad \bar{\sigma}^{\mu a\dot{a}} = (\delta^{a\dot{a}}, -\sigma^{i,a\dot{a}}). \quad (2.76)$$

If you want to convince yourself of this choice of $\sigma_{a\dot{a}}^\mu$ it is sufficient to show that Eq. 2.60 holds after applying the relevant transformations from Eqs. 1.100, 2.32, and 2.44. Having chosen this representation, our goal is to derive the algebra of the gamma matrices, also known as the “**Clifford algebra**.” We'll start by just multiplying two gamma matrices.

$$\gamma^\mu\gamma^\nu = \begin{pmatrix} 0 & \sigma_{a\dot{c}}^\mu \\ \bar{\sigma}^{\mu\dot{a}c} & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma_{cd}^\nu \\ \bar{\sigma}^{\nu\dot{c}d} & 0 \end{pmatrix} = \begin{pmatrix} \sigma_{a\dot{c}}^\mu\bar{\sigma}^{\nu\dot{c}d} & 0 \\ 0 & \bar{\sigma}^{\mu\dot{a}c}\sigma_{cd}^\nu \end{pmatrix}. \quad (2.77)$$

Next can determine the product of the σ s component by component using the explicit representation in Eq. 2.76,

$$\sigma^\mu\bar{\sigma}^\nu = \begin{cases} \mu\nu = 00, & \delta_a^{\dot{d}} \\ \mu\nu = 0i, & -\sigma_a^{i\dot{d}} \\ \mu\nu = i0, & +\sigma_a^{i\dot{d}} \\ \mu\nu = ij, & -(\sigma^i\sigma^j)_a^{\dot{d}} = -(\delta^{ij}\delta_a^{\dot{d}} + i\epsilon^{ij}_k\sigma_a^{k\dot{d}}) \end{cases}, \quad (2.78)$$

doing the same for $\bar{\sigma}\sigma$ gives the same result, just the undotted indices are swapped with dotted and the $0i$ and $i0$ components swap signs. Using this

gives for *the anticommutator* of the gamma matrices:

$$\{\gamma^\mu, \gamma^\nu\} = \begin{cases} \mu\nu = 00, & 2 \begin{pmatrix} \delta_a^d & 0 \\ 0 & \delta_{\dot{a} \dot{d}} \end{pmatrix} \\ \mu\nu = 0i, & \begin{pmatrix} -\sigma_a^{i d} + \sigma_a^{i \dot{d}} & 0 \\ 0 & \sigma^{i \dot{a}}_{\dot{d}} - \sigma^{i a}_d \end{pmatrix} = 0 \\ \mu\nu = i0, & 0 \text{ similar to } 0i \\ \mu\nu = ij, & \begin{pmatrix} -2\delta^{ij}\delta_a^d & 0 \\ 0 & -2\delta^{ij}\delta_{\dot{a} \dot{d}} \end{pmatrix} \end{cases} \quad (2.79)$$

$$= 2\eta^{\mu\nu} \quad (2.80)$$

This defines the *associative algebra* referred to as the **Clifford algebra**, i.e. the algebra of gamma matrices under anticommutation. That it is associative will prove very important when we finally calculate scattering amplitudes and their squares in QFT as it will ensure that the product of three or more gamma matrices is unambiguous and we can still use a lot of familiar math. These gamma matrices are still connected to the Lorentz group, in fact we can write the generators of the Lorentz group in the Dirac space as,

$$S^{\mu\nu} \equiv \frac{i}{4} [\gamma^\mu, \gamma^\nu] , \quad (2.81)$$

you can check that these generators satisfy the Lorentz algebra of Eq. 1.105. Indeed to do so you'll need the result of the Clifford algebra derived in Eq. 2.80.

Another useful matrix we can define is γ_5 ⁶. It is defined as,

$$\gamma_5 = \begin{pmatrix} -\delta_a^c & 0 \\ 0 & \delta_{\dot{a} \dot{c}} \end{pmatrix} , \quad (2.82)$$

whose square gives the identity matrix,

$$\gamma_5^2 = \begin{pmatrix} -\delta_a^b & 0 \\ 0 & \delta_{\dot{a} \dot{b}} \end{pmatrix} \begin{pmatrix} -\delta_b^c & 0 \\ 0 & \delta_{\dot{b} \dot{c}} \end{pmatrix} = \begin{pmatrix} \delta_a^c & 0 \\ 0 & \delta_{\dot{a} \dot{c}} \end{pmatrix} . \quad (2.83)$$

With the γ_5 matrix we can further define:

$$P_L = P_- = \frac{1}{2} (\mathbb{1} - \gamma_5) = \begin{pmatrix} \delta_a^c & 0 \\ 0 & 0 \end{pmatrix} , \quad (2.84)$$

$$P_R = P_+ = \frac{1}{2} (\mathbb{1} + \gamma_5) = \begin{pmatrix} 0 & 0 \\ 0 & \delta_{\dot{a} \dot{c}} \end{pmatrix} . \quad (2.85)$$

⁶The perplexing name dating back to a time when $\mu = \{1, 2, 3, 4\}$ instead of our use $\mu = \{0, 1, 2, 3\}$, such that γ_5 was the fifth gamma matrix.

2. Lorentz Invariant Lagrangians

The double-naming of these matrices is not a mistake. We will use them so much that being able to write $P_{\pm}$ will be critical for general proofs, but as we'll see the labels L (Left) and R (Right) are physically intuitive. A little algebra gives,

$$P_L + P_R = \mathbb{1} , \quad (2.86)$$

$$P_{\pm}^2 = P_{\pm} , \quad (2.87)$$

$$P_{\pm} P_{\mp} = 0 . \quad (2.88)$$

Any matrices satisfying these algebraic conditions are, by definition, **projection operators**. Applying them to our Dirac spinor gives:

$$\psi_L \equiv P_L \psi = \begin{pmatrix} \chi_a \\ 0 \end{pmatrix} , \quad (2.89)$$

$$\psi_R \equiv P_R \psi = \begin{pmatrix} 0 \\ \xi^{\dagger a} \end{pmatrix} , \quad (2.90)$$

demonstrating they project out the left-handed spinor ($P_L = P_-$) or right-handed spinor ($P_R = P_+$) components. This is precisely why we'll largely only consider Dirac fermions moving forward, we can always project out the relevant component as needed. Some physicists prefer to work with Weyl spinors, but that won't be our approach in these notes.

When speaking of the gamma matrices, we can speak of bases that span the space of 4×4 complex matrices that act on Dirac spinors. There are two bases used often in the literature. The first is the **V-A basis**, or vector-axial vector basis:

$$\Gamma_{VA} = \left\{ \mathbb{1}, \gamma^{\mu}, \sigma^{\mu\nu} = \frac{i}{2} [\gamma^{\mu}, \gamma^{\nu}], \gamma_5, \gamma_5 \gamma^{\mu} \right\} . \quad (2.91)$$

This basis is preferred by nuclear physicists as it has some very nice properties when dealing with the strong interactions. For example, when speaking of nucleon decay (Fermi theory) the vector current (V) is nearly conserved, but the axial current (A) receives large corrections from the strong interactions. Here V refers to currents $\bar{\psi} \gamma^{\mu} \psi$ while A to $\bar{\psi} \gamma_5 \gamma^{\mu} \psi$, we'll cover this in more detail later.

Notice from Eqs. 2.82 and 2.89–2.90 that the matrix γ_5 is the “chirality operator”. For left-handed Dirac spinors it has eigenvalue -1 , while for

right-handed Dirac spinors it has eigenvalue $+1$:

$$\gamma_5 P_L \psi = -\psi_L = \begin{pmatrix} -\delta_a^c & 0 \\ 0 & \delta_a^c \end{pmatrix} \begin{pmatrix} \chi_c \\ 0 \end{pmatrix} = - \begin{pmatrix} \chi_a \\ 0 \end{pmatrix} \quad (2.92)$$

$$\gamma_5 P_R \psi = +\psi_R \quad (2.93)$$

Alternatively, physicists dealing with the Electroweak Standard Model will often prefer the **Chiral basis** as the Standard Model treats left- and right-handed fermions differently:

$$\Gamma_{\text{chiral}} = \{P_L, P_R, \gamma^\mu P_L, \gamma^\mu P_R, \sigma^{\mu\nu}\} . \quad (2.94)$$

An added benefit is that as $P_\pm$ are projection operators, orthogonality quickly simplifies a lot of problems in the chiral basis.

With the projection operators identity $\mathbb{1} = P_L + P_R$ we can take a moment to clarify how each of these basis project onto left- and right-handed spinors. This will be very useful later when we write down the Standard Model, and also relates our Dirac fermions to their Weyl spinor components. Let's decompose the Dirac fermion as follows,

$$\psi = (P_L + P_R)\psi \equiv \psi_L + \psi_R , \quad (2.95)$$

i.e. into its left and right handed components. Notice, with these projection operators we don't really need to think about Weyl spinors anymore, we can just let the L and R subscripts do the work. This way we don't have to track dotted and undotted indices.

Consider the following identities, which follow from the definitions of $P_\pm$ and β :

$$\psi_L = P_L \psi = (\chi_a, \ 0) , \quad (2.96)$$

$$\psi_L^\dagger = [P_L \psi]^\dagger = (\chi_a^\dagger, \ 0) , \quad (2.97)$$

$$\psi_L^\dagger \beta = (0, \ \chi_c^\dagger) , \quad (2.98)$$

$$\bar{\psi} P_L = (\xi^a, \ 0) , \quad (2.99)$$

$$\bar{\psi} P_R = (0, \ \chi_c^\dagger) . \quad (2.100)$$

We see that $\bar{\psi} P_R \equiv \bar{\psi}_L$ corresponds to the conjugate of the *left-handed spinor* in ψ , this is because of the matrix β wedged between $\psi^\dagger$ and P_R . Similarly

2. Lorentz Invariant Lagrangians

$\bar{\psi}P_L = \bar{\psi}_R$. Let's consider the bilinear combining two different Dirac spinors ψ and χ ,

$$\bar{\psi}\chi = \bar{\psi} \underbrace{(P_L + P_R)}_{=1} \chi, \quad (2.101)$$

$$= \bar{\psi} \left(\underbrace{P_L^2}_{=P_L} + \underbrace{P_R^2}_{=P_R} \right) \chi, \quad (2.102)$$

$$= \bar{\psi}P_L\chi_L + \bar{\psi}P_R\chi_R, \quad (2.103)$$

$$= \bar{\psi}_R\chi_L + \bar{\psi}_L\chi_R, \quad (2.104)$$

We have found that the term $\bar{\psi}\chi$ mixes between left- and right-handed spinors⁷. This will be very important in the future, we will see this corresponds to mass terms. In the Standard Model the left-handed and right-handed fermions are charged differently under the SM gauge group so these terms are forbidden by symmetry. This is why the Standard Model needs the Higgs mechanism.

Next we need to know how to commute γ_5 and γ^μ ,

$$\gamma_5\gamma^\mu = \begin{pmatrix} -\delta_a^b & 0 \\ 0 & \delta_{\dot{a}}^{\dot{b}} \end{pmatrix} \begin{pmatrix} 0 & \sigma_{b\dot{c}}^\mu \\ \bar{\sigma}^{\mu c\dot{b}} & 0 \end{pmatrix} \quad (2.105)$$

$$= \begin{pmatrix} 0 & -\sigma_{a\dot{c}}^\mu \\ \bar{\sigma}^{\mu c\dot{a}} & 0 \end{pmatrix}, \quad (2.106)$$

$$= - \begin{pmatrix} 0 & \sigma_{a\dot{b}}^\mu \\ \bar{\sigma}^{\mu\dot{a}b} & 0 \end{pmatrix} \begin{pmatrix} -\delta_b^c & 0 \\ 0 & \delta_{\dot{c}}^{\dot{b}} \end{pmatrix}, \quad (2.107)$$

$$= -\gamma^\mu\gamma_5, \quad (2.108)$$

so γ_5 and γ^μ anticommute. From this result it follows,

$$\gamma^\mu P_\pm = \gamma^\mu \frac{1}{2} (\mathbb{1} \pm \gamma_5) = \frac{1}{2} (\mathbb{1} \mp \gamma_5) \gamma^\mu = P_\mp \gamma^\mu. \quad (2.109)$$

⁷Okay, more honestly it mixes between the left- and right-handed components of the Dirac spinor ψ . One of the components is conjugated, meaning in terms of Weyl spinors it's still the product of two (different!) LH or RH Weyl spinors consistent with Eq. 2.64. Saying it mixes between left- and right-handed spinors is common nomenclature we'll continue to use, and won't comment on again.

With this in mind we can also demonstrate:

$$\bar{\psi}\gamma^\mu\chi = \bar{\psi}\gamma^\mu(P_L^2 + P_R^2)\chi, \quad (2.110)$$

$$= \bar{\psi}\gamma^\mu P_L \chi_L + \bar{\psi}\gamma^\mu P_R \chi_R, \quad (2.111)$$

$$= \bar{\psi}P_R\gamma^\mu\chi_L + \bar{\psi}P_L\gamma^\mu\chi_R, \quad (2.112)$$

$$= \bar{\psi}_L\gamma^\mu\chi_L + \bar{\psi}_R\gamma^\mu\chi_R. \quad (2.113)$$

So the vector (the V in the V - A basis) doesn't mix between left- and right-handed spinors. Table 2.1 contains a detailed list of projections of spinor bilinears onto **chiralities** or what we've been referring to as left- and right-handedness. In the Chiral basis the only bilinear that decomposes into a sum of bilinears is $\sigma^{\mu\nu}$, in practice we frequently use $\sigma^{\mu\nu}P_\pm$ which avoids this issue. This is not done in the Table to emphasize the structure of the *chiral basis* of gamma matrices.

The concept of chirality is a *very common source of confusion* among new students of QFT and particle physics. A similar concept that is more physically intuitive is **helicity**: for a particle with spin if the spin is aligned with its momentum it has “right-handed” helicity, if it is anti-aligned it is referred to as “left-handed.” For a massless particle this is the same as chirality. If a particle is massive, however, we can perform a Lorentz boost to a frame where the particle's momentum switches direction and therefore swaps helicities. Chirality is the frame independent version of helicity. With all the work we did in Chapter 1 it is hopefully clear: our spinors are in irreducible representations, so associating chirality with their left- and right-handedness is frame independent. This common source of confusion can be reduced to the simple statement that chirality corresponds to irreps of the Lorentz group.

2.4 Vectors

At this point we can immediately write down a multitude of Lorentz invariant Lagrangians for vector fields. What is missing from this conversation, and we won't have a chance to talk about until later, is which of these is physical⁸. Let's give some examples of terms purely polynomial in some vector A_μ to begin:

$$\mathcal{L} = a_1 A_\mu A^\mu + a_2 (A_\mu A^\mu)(A_\nu A^\nu) + \dots \quad (2.114)$$

⁸Indeed we glossed over this for the spinor case too, but fermions have fewer complications in this regard.

Element	Bilinear	Bilinear in terms of LR	mixes chirality?
$\mathbb{1}$	$\bar{\psi}\chi$	$\bar{\psi}_L\chi_R + \bar{\psi}_R\chi_L$	yes
γ^μ	$\bar{\psi}\gamma^\mu\chi$	$\bar{\psi}_L\gamma^\mu\chi_L + \bar{\psi}_R\gamma^\mu\chi_R$	no
$\sigma^{\mu\nu}$	$\bar{\psi}\sigma^{\mu\nu}\chi$	$\bar{\psi}_L\sigma^{\mu\nu}\chi_R + \bar{\psi}_R\sigma^{\mu\nu}\chi_L$	yes
γ_5	$\bar{\psi}\gamma_5\chi$	$\bar{\psi}_L\gamma_5\chi_R + \bar{\psi}_R\gamma_5\chi_L$ $= +\bar{\psi}_L\chi_R - \bar{\psi}_R\chi_L$	yes
$\gamma_5\gamma^\mu$	$\bar{\psi}\gamma_5\gamma^\mu\chi$	$\bar{\psi}_R\gamma_5\gamma^\mu\chi_R + \bar{\psi}_L\gamma_5\gamma^\mu\chi_L$ $= -\bar{\psi}_R\gamma^\mu\chi_R + \bar{\psi}_L\gamma^\mu\chi_L$	no
P_L	$\bar{\psi}P_L\chi$	$\bar{\psi}_R\chi_L$	yes
P_R	$\bar{\psi}P_R\chi$	$\bar{\psi}_L\chi_R$	yes
$\gamma^\mu P_L$	$\bar{\psi}\gamma^\mu P_L\chi$	$\bar{\psi}_L\gamma^\mu\psi_L$	no
$\gamma^\mu P_R$	$\bar{\psi}\gamma^\mu P_R\chi$	$\bar{\psi}_R\gamma^\mu\psi_R$	no
$\sigma^{\mu\nu}$	$\bar{\psi}\sigma^{\mu\nu}\chi$	$\bar{\psi}_L\sigma^{\mu\nu}\chi_R + \bar{\psi}_R\sigma^{\mu\nu}\chi_L$	yes

Table 2.1: Projections onto right- and left-handed Dirac spinors of the V - A and chiral bases of gamma matrices. The first column is the element of either basis under consideration. The V - A basis elements are above the double line while the chiral basis elements are below. In the case of the terms involving γ_5 , we can use γ_5 to project out the chirality eigenvalue as per Eqs. 2.92 and 2.93. The last column indicates whether the term mixes left- and right-handed spinors.

All of these will prove to violate gauge invariance and so we won't be able to use them, at least not in this form. We could also write some dynamical terms like,

$$\mathcal{L} = a_1(\partial^\mu A_\mu)(\partial^\nu A_\nu) + b_1(\partial_\mu A^\nu)(\partial^\mu A_\nu) + \dots, \quad (2.115)$$

terms like $A_\mu \partial^\mu \partial^\nu A_\nu$ are related by **IBP** relations. Again, these terms, unless squeezed into a particular form, will lead to unphysical properties of our theory⁹. Consider the dynamical term for vectors we know from electrodynamics coming from the field-strength tensor:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad (2.116)$$

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}, \quad (2.117)$$

$$= -\frac{1}{2}[\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A^\mu]. \quad (2.118)$$

This corresponds to $a_1 = -1/2$ (after integration by parts) and $b_1 = -1/2$. The field strength will help us organize derivatives of the vector field in such a manner as to maintain gauge symmetry and keep our theory physical. In fact, the antisymmetry of $F_{\mu\nu}$ projects out the unwanted components of A_μ leaving us with physical polarizations.

If we have four different vectors we could also use the four-dimensional Levi-Civita tensor, which is another invariant symbol, to write:

$$\mathcal{L} = a_1 \epsilon^{\mu\nu\rho\sigma} A_\mu B_\nu C_\rho D_\sigma. \quad (2.119)$$

As a small spoiler, we can already write the Lagrangian for **QED** as:

$$\mathcal{L}_{\text{QED}} = \underbrace{-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\text{gauge kinetic}} + \underbrace{i\bar{\psi}\gamma^\mu\partial_\mu\psi}_{\text{fermion kinetic}} - \underbrace{m\bar{\psi}\psi}_{\text{fermion mass}} - \underbrace{e\bar{\psi}\gamma^\mu\psi A_\mu}_{\text{interaction}} \quad (2.120)$$

We've marked each term in the Lagrangian with what it corresponds to physically. A reasonable question is, "why don't we include any of these other terms mentioned above" such as $(\bar{\psi}\gamma^\mu\psi)^2$. This is because QED is what we refer to as a "renormalizable theory" meaning only a finite number of Lagrangian terms are required to describe the theory. We could add these extra terms $(\bar{\psi}\psi)^2$, but they will have a specific physical meaning beyond what is usually considered QED. In fact, these terms will correspond to

⁹Such as having unphysical degrees of freedom that propagate.

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some heavy physics beyond QED that imprint on the “low energy theory” that QED will represent. Our focus in these notes is on general QFTs, so “renormalizable” QFTs will be treated as a corollary.

If we wanted to describe charged pions interacting with the photon field, we could consider e.g.,

$$\begin{aligned}\mathcal{L}_{\pi\text{-QED}} = & -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \partial_\mu\pi^\dagger\partial^\mu\pi - m^2|\pi|^2 \\ & + ie(\partial_\mu\pi^\dagger)\pi A^\mu - ie\pi^\dagger(\partial_\mu\pi)A^\mu + e^2|\pi|^2 A_\mu A^\mu, \quad (2.121)\end{aligned}$$

where π is a complex scalar that corresponds to the charged pions. This is another renormalizable theory. Notice that in the fermion case there is only a term $\bar{\psi}\psi A$ while for pions we had $\pi^\dagger\pi A$ and $\pi^\dagger\pi A^2$ interactions, this is a consequence of gauge symmetry. Later when we explore the *nonrenormalizable* theory Chiral perturbation theory, we’ll see that the theory that describes pions isn’t renormalizable, and that what we’ve written above isn’t the full story.

2.5 Big Picture

In this Chapter we explored how to write “Lorentz invariant Lagrangians”. This phrasing comes with a caveat, we really mean Lagrangians that only differ in the spacetime-point at which they are evaluated, which is a dummy variable that is integrated over to find the action.

For scalars we were extra careful, we transformed the field (trivially) and its spacetime argument $x \rightarrow \bar{x}$ as well as showing that the Jacobian for the change of variables was 1 and therefore could show that *action was invariant*. We also revisited the concepts of covariant (x^μ) and contravariant (∂_μ) Lorentz transformations so we would get the transformations of derivatives correct. We distinguished polynomials in ϕ from terms we deemed dynamical involving derivatives, and gave a few examples among the infinitely many terms that can be written down.

For Weyl spinors we carefully derived their infinitesimal transformations under the Lorentz group and then developed the idea of Invariant Symbols. When applying a Lorentz transformation to these symbols, they returned the original symbol. This was our first step beyond simple contraction of indices and required us to improve our understanding of Group theory one step further. These tools allowed for the construction of Lorentz invariant Lagrangians in terms of Weyl fields.

We introduced the Dirac field as a field composed of both a left- and right-handed Weyl spinor. By defining projection operators we could easily reduce the Dirac field to either a left- or right-handed spinor, but maintain the Dirac notation. This will allow us to consider only one Dirac index in the future instead of keeping track of two Weyl spinor indices. We wrote some Lorentz invariant Lagrangians in terms of Dirac fields, then focused on how we can project bilinears in Dirac fields onto two different bases: the Chiral and $V-A$ bases. These two bases will have different applications later in our discussions, with $V-A$ favored in nuclear physics discussions and Chiral being crucial to our formulation of the Standard Model.

Finally we had a first look at Lorentz invariant Lagrangians for vectors. This came with some big caveats related to producing physical Lagrangians versus simply writing down Lorentz invariant Lagrangians. Ideas we’ll develop in more detail later. Finally we finished with two example Lagrangians (without explaining where they come from), one corresponding to QED and the other to the interaction of charged pions with the photon field. This showed that the Lagrangians between the photon and electrons and the charged scalars

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and photons differ in more than just the electron and pion's representations of the Lorentz group.

Everything so far has been classical! Indeed, we can study classical solutions of these field theories and find some first order approximations with physical meaning, although we won't do so in these notes. Our next step is to employ path integrals to transition from classical to quantum theories. The next chapter will introduce path integrals and generating functionals to create a foundation from which we can start making physical predictions for a given Lagrangian.

Part II

Path Integrals for QFT

In this chapter we will start by looking at a simple example of a path integral in the context of the more familiar quantum mechanics. Once we have developed these tools in QM, we'll look at how to construct a path integral for a scalar field. We'll see how certain operations on the path integral then yields time-ordered products of fields which we will in a later chapter directly relate to the familiar scattering amplitudes from QM.

Next we will visit the idea of generating functionals. In the course of doing this we will develop diagrammatic techniques for calculations (Feynman diagrams) and we will see that the different generating functionals we consider will correspond to different types of “graphs” or Feynman diagrams such as disconnected and connected diagrams (See e.g. Tab. 4.1). Another generating functional will allow us to generate one-particle irreducible diagrams, we will use this to obtain (most of) the Feynman rules necessary for the calculations we perform throughout these notes.

With path integrals for scalars in hand, we will then consider Fermionic path integrals. As fermions must obey Fermi statistics we will find we need to impose anticommutation relations on our fermions. This will require us to develop the idea of “Grassman variables.” Anticommutation will give us a structure for the path integral which is unique from that of bosonic fields and will have major implications later.

Finally we will look at vector fields. Vectors will parallel scalars in many ways, but will require a great deal of care when dealing with gauge symmetry. To handle this we will learn to remove some gauge redundancy with “gauge fixing” which will require us to introduce a new type of field, the ghost. Ghosts are *scalar fields obeying Fermi statistics*. They are not physical, but are used to exactly cancel the unphysical degrees of freedom of our vector fields (in gauge theories).

With the tools we develop in this Chapter we will finally be in a place where we can consider QFTs and their predictions.

Chapter 3

Path integrals in QM

Now we will learn how to transition from a Hamiltonian-centric view of quantum mechanics to one focused on the Lagrangian. This will naturally bring about the concept of path integrals and will set us up for their usage in quantum field theory. In the QM picture we'll relate transition amplitudes to integrals over all paths of the exponential of the action ($\sim$ Lagrangian). We'll then relate this to *time-ordered products of operators* ($\hat{x}(t)$ or $\hat{p}(t)$).

In contrast, the *time-ordered product of fields* will be directly related to scattering amplitudes in QFT. In the next section we will directly parallel this section's discussion to find out exactly how to get what we need from path integrals over the Lorentz invariant Lagrangians of the last section.

3.1 Defining the path integral

We'll start with a Hamiltonian with the standard kinetic term and a potential dependent only on position and time,

$$\hat{H}(t) = \frac{\hat{p}^2}{2m} + V(\hat{x}, t), \quad (3.1)$$

here $\hat{H}$, $\hat{p}$, and $\hat{x}$ are operators acting on the Hilbert space, and t is a number labelling time. We'll label the initial and final position eigenstates, corresponding to initial and final times (t_i and t_f) as:

$$|i\rangle = |x_i\rangle, \quad (3.2)$$

$$|f\rangle = |x_f\rangle. \quad (3.3)$$

If our Hamiltonian were *time independent* we could write the probability amplitude for a state $|i\rangle$ to transition to some state $|f\rangle$ after some time $(t_f - t_i)$ as:

$$\langle f|i\rangle = \langle x_f | \exp \left[-i(t_f - t_i)\hat{H} \right] |x_i\rangle . \quad (3.4)$$

In dealing with a time-dependent Hamiltonian, we'll break up $(t_f - t_i)$ into n small intervals which we treat as constant in t :

$$t_j = t_i + j\delta t \quad \text{with } t_n = t_f . \quad (3.5)$$

We proceed by evolving $|i\rangle$ step-by-step in δt until we reach the state $|f\rangle$ at t_f :

$$\begin{aligned} \langle f|i\rangle &= \int dx_n \cdots dx_1 \langle x_f | e^{-i\hat{H}(t_f)\delta t} |x_n\rangle \langle x_n | \cdots |x_2\rangle \langle x_2 | e^{-i\hat{H}(t_2)\delta t} |x_1\rangle \\ &\quad \times \langle x_1 | e^{-i\hat{H}(t_1)\delta t} |x_i\rangle . \end{aligned} \quad (3.6)$$

Now let's zoom in on a single one of these steps, we'll insert a complete set of momenta eigenstates,

$$1 = \int \frac{dp}{2\pi} |p\rangle \langle p| , \quad (3.7)$$

and do some simplification:

$$\langle x_{j+1} | e^{-iH\delta t} |x_j\rangle = \int \frac{dp}{2\pi} \underbrace{\langle x_{j+1} | p\rangle}_{\exp(ipx_{j+1})} \langle p | e^{-i\left[\frac{\hat{p}^2}{2m} + V(\hat{x}_j, t_j)\right]\delta t} |x_j\rangle , \quad (3.8)$$

$$= \int \frac{dp}{2\pi} e^{ipx_{j+1}} e^{-i\frac{p^2}{2m}\delta t} e^{-iV(x_j, t_j)\delta t} \langle p | x_j\rangle , \quad (3.9)$$

$$= e^{-iV(x_j, t_j)\delta t} \int \frac{dp}{2\pi} e^{-i\frac{p^2}{2m}\delta t} e^{ip(x_{j+1} - x_j)} . \quad (3.10)$$

Be careful following that second line! The exponential in $\hat{p}^2$ has acted *to the left* on the bra $\langle p|$ to replace $\hat{p}$ with its eigenvalue p ($\hat{p}|p\rangle = p|p\rangle$). Similarly $V(\hat{x})$ has acted *to the right* to replace $\hat{x}$ with x .

Integrals are hard, among the very few we know how to solve are Gaussian integrals. And so we begin the standard QFT game of writing every integral we possibly can as a Gaussian:

$$\int dp \exp \left(-\frac{1}{2}ap^2 + Jp \right) = \sqrt{\frac{2\pi}{a}} \exp \left(\frac{J^2}{2a} \right) . \quad (3.11)$$

3. Path integrals in QM

In the future, if we can't solve an integral we'll write it as a Gaussian plus a perturbation, then we can approximate the integral. So Eq. 3.11 is going to be very important for the rest of our development of QFT. Associating $a = i\delta t/m$ and $J = i(x_{j+1} - x_j)$ gives for Eq. 3.10:

$$\langle x_{j+1} | e^{-iH\delta t} | x_j \rangle = e^{-iV(x_j, t_j)\delta t} \sqrt{\frac{m}{2\pi i\delta t}} e^{-\frac{m(x_{j+1}-x_j)^2}{2i\delta t}}, \quad (3.12)$$

$$= N e^{-iV(x_j, t_j)\delta t} e^{\frac{im\delta t}{2} \frac{(x_{j+1}-x_j)^2}{\delta t^2}}, \quad (3.13)$$

$$= N e^{i\left(\frac{m\dot{x}^2}{2} - V(x_j, t_j)\right)\delta t}, \quad (3.14)$$

$$= N e^{iL(x, \dot{x})\delta t}. \quad (3.15)$$

In the second line, we've written the x - and t -independent radical as a constant N . We now have our first glimpse of how the Lagrangian formulation enters Quantum Mechanics, but we need to return to our calculation in Eq. 3.6 and take this further.

Plugging in our result from Eq. 3.15 into Eq. 3.6 gives:

$$\langle f|i \rangle = N^n \int dx_f \dots dx_i e^{iL(x_n, \dot{x}_n)\delta t} \dots e^{iL(x_1, \dot{x}_1)\delta t}. \quad (3.16)$$

In the limit that $\delta t \rightarrow 0$ we obtain an integral over $\delta t \rightarrow dt$,

$$\langle f|i \rangle = \bar{N} \int_{x(t_i)=x_i}^{x(t_f)=x_f} \mathcal{D}x(t) e^{i \int dt L(x(t), \dot{x}(t))}, \quad (3.17)$$

where our constant N^n is replaced with some formally infinite constant $\bar{N}$. We will see that this infinite constant drops out of observables and therefore doesn't present a problem¹. We have replaced our integral over the steps x_j with an integral over some path $x(t)$. Our differentials, dx_j , are then transformed from a product of discrete individual integrals to a *functional integral* over all of these possible continuous paths connecting x_i to x_f .

Notice, our path integral is written in terms of x and $\dot{x} = (x_{j+1} - x_j)/\delta t$. These are just variables in the traditional sense. So our action behaves classically! It's the path integral and how we will learn to manipulate it that takes us from the classical action to quantum mechanics.

¹For physicists, mathematicians really don't like this. Their real gripe is actually with the lack of a rigorous definition of the measure.

3.2 Aside: Functional calculus

To proceed we'll need to develop, or review, some mathematical background. While we may not be super familiar with the technical details of functional calculus/analysis, we are actually familiar with the concepts from the Hamiltonian and Lagrangian formulation of mechanics.

Consider the action defined in terms of the Lagrangian:

$$S[x] = \int_{t_i}^{t_f} dt L(x(t), \dot{x}(t)) . \quad (3.18)$$

$S[x]$ is a **functional**, a map of the space of functions $x(t)$ to the real numbers. This is analogous to $f(x)$ being a map of the variable x to the real numbers.

Hamilton's principle, or the principle of least action, dictates that *extrema of the action* lead to the classical trajectories. In terms of the functional, this means:

$$\frac{\delta S[x(t)]}{\delta x(t)} = 0 . \quad (3.19)$$

Notice the explicit usage of the argument t . We're stressing that we're extremizing the action (functional) with respect to the trajectory (function). This is our analogue to $df/dx = 0$, giving the extrema. If we take the variation a step further we arrive at the Euler Lagrange equations,

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = 0 , \quad (3.20)$$

L is a function of the trajectory x and its time derivative $\dot{x}$. This is the same analogue of $df/dx = 0$, re-expressed in terms of the function $L(x(t), \dot{x}(t), t)^2$.

Our path integral was expressed as,

$$\langle f|i \rangle \sim \int \mathcal{D}x(t) e^{iS[x(t)]} , \quad (3.21)$$

or a new type of integral over the functional $S[x(t)]$ with respect to a measure $\mathcal{D}x(t)$. In only the simplest case (Gaussian!) will we be able to solve this integral. Otherwise we will solve it perturbatively, which will require us to know how to perform calculus on this object.

²More or less, one can make a complaint about this claim based on boundary conditions which we generally will not consider in these notes.

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Recall from basic calculus that the definition of the derivative was,

$$\frac{df(x)}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \quad (3.22)$$

for functional calculus we'll instead be looking at how a functional, e.g. $F[x(t)]$, varies with small deformations of $x(t)$. For example, let $x(t) \rightarrow x(t) + \epsilon\eta(t)$ where ϵ is a quantity we will take to zero in analogy to h in Eq. 3.22 and $\eta(t)$ is an arbitrary deformation. Then we can define $F[x(t)]$ and its **variation** $\delta F[x(t)]$ as:

$$F[x(t)] = \int_{t_i}^{t_f} dt f(x(t), \dot{x}(t), t), \quad (3.23)$$

$$\delta F[x(t)] = \lim_{\epsilon \rightarrow 0} \frac{F[x(t) + \epsilon\eta(t)] - F[x(t)]}{\epsilon} = \int_{t_i}^{t_f} dt \frac{\delta F[x(t)]}{\delta x(t)} \eta(t). \quad (3.24)$$

As η is arbitrary, it isn't included in the definition of the derivative. As an example consider F quadratic in $x(t)$,

$$F[x(t)] = \int_{t_i}^{t_f} dt x(t)^2, \quad (3.25)$$

$$\delta F[x(t)] = \lim_{\epsilon \rightarrow 0} \frac{F[x(t) + \epsilon\eta(t)] - F[x(t)]}{\epsilon}, \quad (3.26)$$

$$= \lim_{\epsilon \rightarrow 0} \int_{t_i}^{t_f} dt \frac{[(x(t) + \epsilon\eta(t))^2 - x(t)^2]}{\epsilon}, \quad (3.27)$$

$$= \lim_{\epsilon \rightarrow 0} \left[2 \int_{t_i}^{t_f} dt x(t) \eta(t) + \mathcal{O}(\epsilon) \right], \quad (3.28)$$

$$= \int_{t_i}^{t_f} dt 2x(t) \eta(t), \quad (3.29)$$

$$= \int_{t_i}^{t_f} dt \frac{\delta F[x(t)]}{\delta x(t)} \eta(t). \quad (3.30)$$

In direct analogy with the calculus of functions we see the “power rule” at

play:

$$\frac{\delta F[x(t)]}{\delta x(t)} = \frac{\delta}{\delta x(t)} \int dt x(t)^2 = 2x(t) \quad (3.31)$$

$$\Rightarrow \frac{\delta F[x(t)]}{\delta x(t)} = \frac{\delta}{\delta x(t)} \int dt x(t)^n = nx(t)^{n-1}. \quad (3.32)$$

In fact all of the usual results of calculus will translate over to the calculus of functionals fairly seamlessly (think linearity of derivatives, product rule, and chain rule). This will be crucial very soon as we will need to take a lot of variations to generate amplitudes associated with scattering processes in QFT.

Let's abstract this a step without being as rigorous. What if our variation were with respect to a function of different argument? In this case, we'll define our functional derivative as:

$$\frac{\delta}{\delta f(y)} f(x) = \delta(x - y), \quad (3.33)$$

where $\delta(x - y)$ is the Dirac delta function. This delta function is critical, it maintains that the functional derivative is local in its argument: that is, varying $x(t)$ only affects the integrand at the specific point $t = t'$. Now let's take the functional derivative of the same F from Eq. 3.25:

$$\frac{\delta F[x(t)]}{\delta x(t')} = \int_{t_i}^{t_f} dt \frac{\delta x(t)^2}{\delta x(t')}, \quad (3.34)$$

$$= \int_{t_i}^{t_f} dt 2x(t) \delta(t - t'), \quad (3.35)$$

$$= 2x(t'), \quad (3.36)$$

where for the sake of the consistency of the δ we've assumed $t_i < t' < t_f$. In general for us we'll be integrating over all spacetime so this distinction won't even come up.

This concept along with linearity, the power, product, and chain rules, will be all we need to know about functional calculus to proceed.

As a final example of the calculus of functionals let's rederive the Euler-Lagrange equation, in a slightly roundabout way, to demonstrate what we've developed here and stress locality. We start with the action $S[x(t)]$, a functional of the Lagrangian $L(x(t), \dot{x}(t), t)$, which is a function of the trajectory

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$x(t)$, its time derivative $\dot{x}(t)$, and time. It sounds so complicated put this way, but the familiarity of the concept also stresses how simple this very complicated math becomes simply because of the parallels to familiar derivatives. We write the action as:

$$S[x(t)] = \int_{t_i}^{t_f} dt L(x(t), \dot{x}(t), t). \quad (3.37)$$

Hamilton's principle states $\delta S = 0$, let's do this slightly differently and use a functional derivative:

$$\frac{\delta S[x(t)]}{\delta x(t')} = 0. \quad (3.38)$$

We can evaluate this derivative with what we've developed in this section:

$$\frac{\delta S[x(t)]}{\delta x(t')} = \int_{t_i}^{t_f} dt \frac{\delta L(x(t), \dot{x}(t), t)}{\delta x(t')}, \quad (3.39)$$

$$= \int_{t_i}^{t_f} dt \left[\frac{\partial L(x(t), \dot{x}(t), t)}{\partial x(t)} \frac{\delta x(t)}{\delta x(t')} + \frac{\partial L(x(t), \dot{x}(t), t)}{\partial \dot{x}(t)} \frac{\delta \dot{x}(t)}{\delta x(t')} \right], \quad (3.40)$$

$$= \int_{t_i}^{t_f} dt \left[\frac{\partial L(x(t), \dot{x}(t), t)}{\partial x(t)} \delta(t - t') + \frac{\partial L(x(t), \dot{x}(t), t)}{\partial \dot{x}(t)} \frac{\delta \dot{x}(t)}{\delta x(t')} \right], \quad (3.41)$$

in the first line we're *abusing the idea of the variational derivative a bit by treating the Lagrangian as a functional*. It's not and we clear it up in the second equivalence where we applied the chain rule (as abstracted from the chain rule we know and love from the calculus of functions). Next we'll simply rewrite,

$$\frac{\delta \dot{x}(t)}{\delta x(t')} = \frac{d}{dt} \frac{\delta x(t)}{\delta x(t')} = \frac{d}{dt} \delta(t - t'), \quad (3.42)$$

this is a big deal. Remember below Eq. 3.33 we stressed how important it was that our functional derivatives maintain locality, now we're just extending that to variations of time derivatives. The functional derivative *only affects the integrand at the specific point $t = t'$: except* when derivatives are involved the functional derivative *only affects the integrand at the point $t = t'$ up to sensitivity to the slope at this point*, which is encoded by the derivative of the delta function. Remember, the delta function is 0 everywhere except at the point where its argument is 0. When the derivative, $\delta'(x - y)$ is

integrated against a test function $f(x)$ it returns minus the derivative of the test function evaluated at the point y , $-f'(y)$.

This short “proof” may seem like we’re saying the time derivative and the variation commute without explaining why. Ultimately it’s the linearity of both the derivative and variation that allows us to do this. If you want to understand how in more detail, consider $\delta\dot{x}(t)$ and work from the definition in Eq. 3.24 following our proof ending with Eq. 3.30.

Returning to our variation of the action we have:

$$\frac{\delta S[x(t)]}{\delta x(t')} = 0 = \int_{t_i}^{t_f} dt \left[\frac{\partial L(x(t), \dot{x}(t), t)}{\partial x(t)} + \frac{\partial L(x(t), \dot{x}(t), t)}{\partial \dot{x}(t)} \frac{d}{dt} \right] \delta(t - t'), \quad (3.43)$$

$$= \int_{t_i}^{t_f} dt \left[\frac{\partial L(x(t), \dot{x}(t), t)}{\partial x(t)} - \frac{d}{dt} \frac{\partial L(x(t), \dot{x}(t), t)}{\partial \dot{x}(t)} \right] \delta(t - t'). \quad (3.44)$$

In the second part, we used integration by parts to move the time derivative to acting on the derivative of L , hence the minus sign *which exactly matches the minus given by the derivative of the delta function we discussed above*. Integrating over the δ function gives the usual result for the Euler-Lagrange equation:

$$\frac{\partial L(x(t), \dot{x}(t), t)}{\partial x(t)} - \frac{d}{dt} \frac{\partial L(x(t), \dot{x}(t), t)}{\partial \dot{x}(t)} = 0 \quad (3.45)$$

You most likely did this proof in a classical mechanics course without all the extra language about functionals and locality. So the concepts and arguments should feel familiar, now we’ve just added some rigor. We also got a good look at how the delta functions appear in context, we’re going to see a lot more of this soon.

If not every step here clicked, that’s okay. It takes some getting used to, *the most important ally in tackling functional calculus is remembering it’s just a slight abstraction of the calculus of functions* and everything works analogous to the usual rules of calculus. As we’ll be using these new concepts extensively through the rest of the chapter, it is a good idea to make sure you’re comfortable with what’s happened so far before proceeding.

3.3 Time-ordered products in QM

Generally in QM we're not only interested in $\langle f|i \rangle$, but also in quantities like $\langle f|\hat{O}|i \rangle$. So we'll need to develop some further technology to understand this. Consider the operators $\hat{x}(t_j)$ and $\hat{p}(t_j)$,

$$\hat{x}(t_j) |x(t_j)\rangle = x(t_j) |x(t_j)\rangle, \quad \hat{p}(t_j) |p(t_j)\rangle = p(t_j) |p(t_j)\rangle, \quad (3.46)$$

that is, the kets at a time t_j are eigenkets of the operators at t_j . In our path integral formulation, we took small time slices where the Hamiltonian could be treated as time independent, so we also have:

$$\hat{x} |x_j\rangle = x_j |x_j\rangle, \quad \hat{p} |p_j\rangle = p_j |p_j\rangle, \quad (3.47)$$

keep in mind the subscript j means these time-independent kets are inserted at time slice j .

Our path integral was defined to return $\langle f|i \rangle$, but what if we instead want $\langle f|\hat{x}(t_j)|i \rangle$? Going from the Heisenberg picture to Schrödinger gives:

$$\langle f|\hat{x}(t_j)|i \rangle \equiv \langle x(t_f)|\hat{x}(t_j)|x(t_i)\rangle = \underbrace{\langle x_f|e^{-i\hat{H}(t_f)}}_{\langle x(t_f)|} \underbrace{e^{i\hat{H}(t_j)}\hat{x}e^{-i\hat{H}(t_j)}}_{\hat{x}(t_j)} \underbrace{e^{i\hat{H}(t_i)}}_{|x(t_i)\rangle} |x_i\rangle. \quad (3.48)$$

Following the same procedure as before, we split this into small time slices (δt) where the Hamiltonian can be treated as time independent,

$$\begin{aligned} \langle f|\hat{x}(t_j)|i \rangle &= \int dx_n \cdots dx_1 \langle x_f|e^{-i\hat{H}(t_f)\delta t}|x_n\rangle \langle x_n|\cdots\hat{x}|x_j\rangle \langle x_j|\cdots|x_1\rangle \\ &\quad \times \langle x_1|e^{-i\hat{H}(t_1)\delta t}|x_i\rangle, \end{aligned} \quad (3.49)$$

Notice time slicing the exponentials happens to either side of $\hat{x}$ and $\hat{x}$ is left in precisely the right position to return the eigenvalue x_j . Completing all the same steps from before we arrive at the path integral:

$$\langle f|\hat{x}(t_j)|i \rangle = \bar{N} \int_{x(t_i)}^{x(t_f)} \mathcal{D}x(t) x_j e^{i \int dt L(x(t), \dot{x}(t))}. \quad (3.50)$$

If we instead wanted $\langle f|\hat{p}(t_j)|i \rangle$ it works almost the same, but instead of acting on the $|x_j\rangle$ it will act to the left on the complete set of states we

inserted in Eqs. 3.7 and 3.8. So if we went through the calculation we would find,

$$\langle f | \hat{p}(t_j) | i \rangle = \bar{N} \int_{x(t_i)}^{x(t_f)} \mathcal{D}x(t) p_j e^{i \int dt L(x(t), \dot{x}(t))}, \quad (3.51)$$

in complete analogy with the $\hat{x}$ case.

The story changes a bit when we have two or more operators. For example,

$$\langle f | \hat{x}(t_j) \hat{x}(t_k) | i \rangle, \quad (3.52)$$

without specifying the ordering of t_j and t_k , they may not be in the correct order to act on their respective eigenkets at the appropriate time slices. In general in time-dependent QM the two operators also don't commute. For example for free particles in the Heisenberg picture we have,

$$[\hat{x}(t_j), \hat{x}(t_k)] = \frac{i}{m}(t_j - t_k) \quad [\hat{p}(t_j), \hat{p}(t_k)] = 0, \quad (3.53)$$

while for the harmonic oscillator we have:

$$[\hat{x}(t_j), \hat{x}(t_k)] = \frac{i}{m\omega} \sin(\omega[t_j - t_k]) \quad [\hat{p}(t_j), \hat{p}(t_k)] = -im\omega \sin(\omega[t_j - t_k]). \quad (3.54)$$

So we'll introduce **time ordering**, for two operators the operation of time ordering gives,

$$T[\hat{x}(t_j), \hat{x}(t_k)] = \begin{cases} \hat{x}(t_j) \hat{x}(t_k) & t_k < t_j \\ \hat{x}(t_k) \hat{x}(t_j) & t_j < t_k \end{cases}, \quad (3.55)$$

for more than two operators we simply extend this into placing them in order with the operator evaluated at the earliest time to the right and latest time to the left. We won't have to consider the case of equal time, but in this case a prescription of the original order or symmetrizing is generally used.

Following the same derivation as above, it then follows that:

$$\langle f | T[\hat{x}(t_j), \hat{x}(t_k)] | i \rangle = \bar{N} \int_{x(t_i)}^{x(t_f)} \mathcal{D}x(t) x_j x_k e^{i \int dt L(x(t), \dot{x}(t))}. \quad (3.56)$$

Remember, the time ordering puts the two operators in temporal order, then when we slice up our time evolution operators (exponentials of $\hat{H}$) the operators will be in the correct places to act on the correct *time-independent* eigenkets and return x_j and x_k . In the same way, this carries over in a straightforward manner to time-ordered products of momenta, or time-ordered products

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involving both $\hat{x}$ and $\hat{p}$. Indeed, time ordering is even more important in this case where $[\hat{x}(t_j), \hat{p}(t_k)]$ in general will not vanish.

Our path integral involves the functional $S[x(t)]$. It turns out we can use our concepts of variational calculus to obtain time-ordered products directly. To see this we will extend the exponential with some new terms. These terms will be set to zero at the end, so our path integral doesn't change:

$$\langle f|i \rangle = \bar{N} \int_{x_i}^{x_f} \mathcal{D}x(t) e^{i \int dt [L(x(t), \dot{x}(t)) + f(t)x(t) + h(t)p(t)]} \Big|_{f, g \rightarrow 0}. \quad (3.57)$$

Now consider taking a variation (recall Eq. 3.33) with respect to, for example $f(t_j)$:

$$\frac{1}{i} \frac{\delta}{\delta f(t_j)} \langle f|i \rangle = \bar{N} \int_{x_i}^{x_f} \mathcal{D}x(t) \frac{1}{i} \frac{\delta}{\delta f(t_j)} e^{i \int dt [L(x(t), \dot{x}(t)) + f(t)x(t) + h(t)p(t)]} \Big|_{f, g \rightarrow 0}, \quad (3.58)$$

$$= \bar{N} \int_{x_i}^{x_f} \mathcal{D}x(t) e^{i \int dt [L(x(t), \dot{x}(t)) + f(t)x(t) + h(t)p(t)]} \int dt x(t) \delta(t - t_j) \Big|_{f, g \rightarrow 0}, \quad (3.59)$$

$$= \bar{N} \int_{x_i}^{x_f} \mathcal{D}x(t) x_j e^{i \int dt [L(x(t), \dot{x}(t)) + f(t)x(t) + h(t)p(t)]} \Big|_{f, g \rightarrow 0}, \quad (3.60)$$

$$= \langle f | \hat{x}(t_j) | i \rangle. \quad (3.61)$$

where the extra factor of $1/i$ is used to remove the factor of i coming from the variation of the exponential. We've relabelled $x(t_j) = x_j$ as here when applying the variation we're referring to the specific slice of time t_j . In our more general discussion above we distinguished the two because we needed to look for a specific time in our sum over slices. When we take the variation, it acts on the exponential of the integral with respect to dt . Here dt is analogous to a dummy index, we're integrating over all values of t . So when the integral "pulls down" a factor of $x(t)$ it is accompanied by an integral that checks every t and a delta function (from Eq. 3.33) that makes sure we pick out the correct t (t_j). This is familiar from sums, the derivative dx_i/dx_j picks out j from x_i to gives δ_{ij} .

We can repeat the same calculation as many times as we want and find for example:

$$\begin{aligned} & \frac{1}{i} \frac{\delta}{\delta x(t_j)} \frac{1}{i} \frac{\delta}{\delta p(t_k)} \frac{1}{i} \frac{\delta}{\delta x(t_l)} \langle f|i \rangle \\ &= \bar{N} \int_{x_i}^{x_f} \mathcal{D}x(t) x_j p_k x_l e^{i \int dt [L(x(t), \dot{x}(t)) + f(t)x(t) + h(t)p(t)]} \Bigg|_{f, g \rightarrow 0} \end{aligned} \quad (3.62)$$

$$= \langle f| T[\hat{x}(t_j), \hat{p}(t_k), \hat{x}(t_l)] |i \rangle . \quad (3.63)$$

We've effectively reduced the problem of evaluating quantum expectation values to a problem in functional analysis.

Earlier we mentioned that our action can now be treated classically. From what we covered in this section, we can see how the classical action encodes quantum correlations through the path integral. Our time-ordered products of operators, which are quantum mechanical concepts, are derived from the classical action through these variations acting on the path integral. When we look at path integrals for scalars, this will allow us to obtain time ordered products of fields in a Lorentz covariant way. We can simply making use of our classical Lagrangian densities from Chapter 2! Instead of manipulating operators acting on Hilbert space, we'll use classical Lagrangians and functional derivatives to compute expectation values.

3.4 Aside: The action as a weight

When we wrote the path integral, each x_j represented a position at some specified slice in time. As we take the continuum limit of those time slices, every intermediate point is integrated over independently, so the amplitude receives contributions from every possible trajectory connecting x_i and x_f . If every possible path feels problematic physically, look at the integrand and rest assured: the action, $S = \int dt L$, acts as a weight on each path, so Hamilton's principle of minimizing the action corresponds to the idea that in the classical limit classical trajectories dominate.

Let's see this more exactly. We expand the action about the classical path,

$$x(t) = x_{cl}(t) + \eta(t), \quad \eta(t_f) = \eta(t_i) = 0, \quad (3.64)$$

where η is some variation away from the classical path, when it's small we're near the classical limit. For the cases where it's sufficiently small we can

3. Path integrals in QM

Taylor expand the action as:

$$S[x] = S[x_{\text{cl}}] + \underbrace{\int dt \frac{\delta S}{\delta x(t)} \Big|_{x_{\text{cl}}}}_{=0} \eta(t) + \frac{1}{2} \int dt dt' \eta(t) \frac{\delta^2 S}{\delta x(t) \delta x(t')} \Big|_{x_{\text{cl}}} \eta(t') + \dots \quad (3.65)$$

Remember, in Eq. 3.17 we've switched to a functional integral which is why we write $S[x]$ with square brackets, S is a functional over the space of all trajectories $x(t)$. We've set the term linear in the variations to zero, by the Euler-Lagrange equations this is the classical solution, so in evaluating it on the classical trajectory it is identically zero.

Using a change of variables, the path integral of Eq. 3.17 can be rewritten in terms of this perturbation around x_{cl} as:

$$\int \mathcal{D}\eta e^{iS[x_{\text{cl}}+\eta]} \sim e^{iS[x_{\text{cl}}]} \int \mathcal{D}\eta \exp \left[i \frac{1}{2} \int dt dt' \eta(t) \frac{\delta^2 S}{\delta x \delta x'} \eta(t') + \dots \right]. \quad (3.66)$$

Near the classical path, $\eta \sim 0$, the phase from the path integral varies slowly and contributions from nearby paths will add constructively. As we allow η to become larger, i.e. consider paths deviating from what is expected from classical physics, the phase will vary more rapidly leading to nearby paths interfering destructively, as a result the contributions will be small.

Chapter 4

Path integrals for scalars

Now that we have a prescription for the path integral, we can jump immediately to the path integral for scalar fields:

$$\langle \Omega | \Omega \rangle = \int \mathcal{D}\phi \exp \left[i \int d^4x \mathcal{L}(\phi(x)) \right]. \quad (4.1)$$

Some things have changed: Now instead of a probability of a transition from $|i\rangle$ to $|f\rangle$ we have the expectation of the vacuum. We'll understand this by analogy a little later in the context of the "free theory" (i.e. scalars with no interactions). For now, we can think of it as an integral over all configurations of the field ϕ , absent sources (f and g from the QM example) the action acts as a weight giving the vacuum state. We've also replaced $\int dt L$ with $\int d^4x \mathcal{L}$.

By introducing sources, $J(x)$, we can then use our methods from the calculus of functionals to obtain time-ordered products of fields:

$$\langle \Omega | \Omega \rangle = \int \mathcal{D}\phi \exp \left[i \int d^4x (\mathcal{L}(\phi(x)) + J(x)\phi(x)) \right] \Big|_{J \rightarrow 0}, \quad (4.2)$$

$$\langle \Omega | T[\phi(y), \phi(z)] | \Omega \rangle = \frac{1}{i^2} \frac{\delta}{\delta J_y} \frac{\delta}{\delta J_z} \int \mathcal{D}\phi \exp \left[i \int d^4x (\mathcal{L}(\phi(x)) + J(x)\phi(x)) \right] \Big|_{J \rightarrow 0}. \quad (4.3)$$

We've introduced the shorthand $J_x = J(x)$ because, as we'll see, we're going to need a lot of these variations and they often aren't going to fit nicely on a page. Each variation is accompanied by a factor of $1/i$, just as in the QM case, hence the $1/i^2$ term.

4.1 Free scalar field theory

Let's consider the simplest example we can actually solve, where the exponential forms a (you guessed it) Gaussian. This corresponds to a Lagrangian quadratic in the fields:

$$\mathcal{L} = a_1 \phi \square \phi + a_2 \phi^2 . \quad (4.4)$$

Yes, we could include a linear term in ϕ as well as a constant term. Constant terms translate to an overall constant of the path integral, which we've seen doesn't change the physics. A linear term (a *tadpole*) can simply be replaced by making a redefinition $\phi \rightarrow c + \phi$ and choosing c such that it cancels the tadpole term (up to an overall constant). So despite appearances, we are considering the most general Lagrangian (up to more derivatives, which we'll discuss when we discuss effective field theories).

Our path integral takes the form:

$$\langle \Omega | \Omega \rangle = \int \mathcal{D}\phi \exp \left[i \int d^4x (a_1 \phi(x) \square \phi(x) + a_2 \phi(x)^2 + J(x) \phi(x)) \right] \Big|_{J \rightarrow 0} , \quad (4.5)$$

$$= \int \mathcal{D}\phi \exp \left[i \int d^4x \left(\frac{1}{2} \phi(x) (2a_1 \square + 2a_2) \phi(x) + J(x) \phi(x) \right) \right] \Big|_{J \rightarrow 0} , \quad (4.6)$$

$$= \int \mathcal{D}\phi \exp \left[i \int d^4x d^4y \frac{1}{2} \phi(x) A(x, y) \phi(y) \delta^4(x - y) + i \int d^4x J(x) \phi(x) \right] \Big|_{J \rightarrow 0} . \quad (4.7)$$

In the second line we defined the operator $A(x, y) = 2a_1 \square_x + 2a_2$. If we compare with the Gaussian integral in n dimensions,

$$\int_{-\infty}^{\infty} d\vec{p} \exp \left[-\frac{1}{2} p_i A_{ij} p_j + J_i p_i \right] = \sqrt{\frac{(2\pi)^n}{\det A}} \exp \left[\frac{1}{2} J_i A_{ij}^{-1} J_j \right] , \quad (4.8)$$

we see that the integrals over x and y are direct analogues over the sums over dummy indices i and j . We'll apply Eq. 4.8 to our path integral, which requires us to understand that we're going from something like a lattices of points in spacetime to the continuum limit. In this case $d\vec{p}$ goes from a fixed number of ps (n) to the product of infinitely many in the continuum limit, for our scalar field we've been calling this $\mathcal{D}\phi$. The determinant on the right becomes a "functional determinant," which is maybe not familiar. And A^{-1}

4. Path integrals for scalars

is the inverse of the operator A which involves $\square$. This transition may seem tricky, but as physicists we're usually comfortable abstracting in this way: numerous proofs in undergraduate physics depend on going from a sum to an integral by taking the continuum limit, we've just taken this a step further. We leave the details to the Mathematicians.

Applying our Gaussian integral in Eq. 4.8 we arrive at:

$$\langle \Omega | \Omega \rangle = N \exp \left[-\frac{i}{2} \int d^4x \int d^4y J(x) A^{-1}(x, y) J(y) \right]_{J=0}. \quad (4.9)$$

N is that nasty infinite constant we also encountered in QM, but now it also includes the functional determinant. We'll come back to these functional determinants a bit later. The extra minus sign, relative to the discrete Gaussian integral, in Eq. 4.9 comes from carefully tracking the factor i that multiplies $J\phi$ in Eq. 4.7. If we want to get a closed form for A^{-1} we need to understand what it means a bit better, when we say it's the inverse of A we really mean:

$$\underbrace{2(a_1 \square_x + a_2)}_{A(x, y)} A^{-1}(x, y) = \delta^4(x - y). \quad (4.10)$$

The solution, A^{-1} to Eq. 4.10 is referred to as a Green's function solution. It's the solution to some linear differential equation with a delta function source (i.e. the right hand side of Eq. 4.10). This doesn't actually provide a ton of useful insight, but the phrase "Green's function" is used so much in QFT it's good to have this in the back of your mind. We don't need to do any work to solve this one, we can guess by inspection that the solution is:

$$A^{-1}(x, y) = A^{-1}(x - y) = \int \frac{d^4p}{(2\pi)^4} \frac{1}{2} \frac{-1}{a_1 p^2 - a_2} e^{ip \cdot (x - y)}. \quad (4.11)$$

Without digging deeply into the specifics, we can now fix a_1 and a_2 so that our Green's function solution corresponds to physical scattering. We'll start by understanding the time-ordered product of two fields. In order to work this out, and fit the calculations on the page, we need to invent some shorthand notation, let's agree:

$$J(x_i) \rightarrow J_{x_i} \quad (4.12)$$

$$\frac{\delta}{\delta J_{x_i}} = \delta_{x_i} \quad (4.13)$$

$$A^{-1}(x - y) \rightarrow A_{x, y}^{-1} \quad (4.14)$$

$$\delta^4(x - y) \rightarrow \delta_{(x - y)}^4 \quad (4.15)$$

With the shorthand for the functional derivative we need to be careful, it really resembles the Dirac delta functions. Remember the Dirac delta function has a subscript like $(x - y)$ to help distinguish. With this shorthand we can start to work out the time-ordered product of two fields:

$$\begin{aligned}
 & \langle \Omega | T[\phi(y), \phi(z)] | \Omega \rangle \\
 &= \frac{1}{i^2} \delta_z \delta_y \left[N \exp \left[-\frac{i}{2} \int d^4 x_1 \int d^4 x_2 J_{x_1} A_{x_1, x_2}^{-1} J_{x_2} \right] \right] \Big|_{J=0} \quad (4.16) \\
 &= \frac{1}{i^2} \delta_z \left[\langle \Omega | \Omega \rangle \left(\frac{-i}{2} \right) \int d^4 x_3 d^4 x_4 A_{x_3, x_4}^{-1} \left[\delta_{(x_3-y)}^4 J_{x_4} + (x_3 \leftrightarrow x_4) \right] \right] \Big|_{J=0}. \quad (4.17)
 \end{aligned}$$

The second line follows from acting with the variation δ_y , it hits the exponential, and brings down two factors (product rule) from each of J_{x_1} and J_{x_2} , the new integrals of x_3 and x_4 are the same kind of dummy variable integrals we had in Eq. 3.59 (they are really just bookkeeping!)¹. To fit the line on the page we had to write $(x_3 \leftrightarrow x_4)$, being able to do this requires us to remember that $\delta_{z-x_4} = \delta_{x_4-z}$.

Next note that $A^{-1}(x-y)$ is symmetric in the exchange of x and y . This isn't obvious immediately from looking at Eq. 4.11. To see the symmetry, notice swapping x and y in the exponential flips the sign of the exponential, but we can make a change of variables $p^\mu \rightarrow -p^\mu$ which has Jacobian 1 effectively reversing that sign flip. Using this symmetry we can simplify Eq. 4.17 as:

$$\begin{aligned}
 & \langle \Omega | T[\phi(y), \phi(z)] | \Omega \rangle \\
 &= i \delta_z \left[\langle \Omega | \Omega \rangle \int d^4 x_3 d^4 x_4 A_{x_3, x_4}^{-1} \delta_{(x_3-y)}^4 J_{x_4} \right] \Big|_{J=0}. \quad (4.18)
 \end{aligned}$$

$$\begin{aligned}
 &= i \left[\langle \Omega | \Omega \rangle \left(\int d^4 x_3 d^4 x_4 A_{x_3, x_4}^{-1} \delta_{(x_3-y)}^4 \delta_{(x_4-z)}^4 \right. \right. \\
 &\quad \left. \left. - i \int d^4 x_3 d^4 x_4 A_{x_3, x_4}^{-1} \delta_{(x_3-y)}^4 J_{x_4} \int d^4 x_5 d^4 x_6 A_{x_5, x_6}^{-1} \delta_{(x_5-z)}^4 J_{x_6} \right) \right] \Big|_{J=0} \quad (4.19)
 \end{aligned}$$

When we set $J = 0$ the second line of the second equivalence vanishes. In the future we'll neglect to write these terms (again, for space) and just

¹This is an annoyance that won't go away, even later when we switch to Feynman rules. We'll never have enough labels for all the indices/variables. Since acting with variations will require new dummy indices to the right of the path integral, we're going to label those x_i since we can just increase the subscript.

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write “ $0_{J=0}$ ” to indicate the remaining terms vanish when we set $J = 0$. Integrating x_3 and x_4 against the delta functions gives,

$$\langle \Omega | T[\phi(y), \phi(z)] | \Omega \rangle = iA^{-1}(y - z)N, \quad (4.20)$$

Even after setting the sources to zero we’re left with that pesky infinite constant. We’ll get rid of it soon.

The key point here is that the time-ordered product of two fields is exactly iA^{-1} (up to the infinite constant), our Green’s function solution. We interpret the time-ordered product of two fields as the response at y given a source at z (assuming $z^0 < y^0$), if y^0 is before z^0 then the time ordered product tells us that instead its the response at z given a source at y . It describes how a particle or disturbance *propagates from one point to another*. We’ll refer to iA^{-1} from now on as the **propagator**, frequently labelled $\Delta_F(x - y)$ with the subscript crediting Feynman. We’ll just call it Δ for the sake of saving space for our longer equations.

With this idea of the time-ordered product of two fields giving the propagation of a particle at one spacetime point to another, we can recognize that the quadratic terms in the Lagrangian correspond to the “free field theory.” That is, the theory without interactions. Again, this corresponded to a Gaussian which is basically the only integral we know how to solve. Everything else, in these notes, will require perturbation theory (an expansion in a small parameter).

Now we invoke some knowledge from scattering theory without proof:

1. the pole of the propagator should correspond to the physical mass, this fixes $a_1 m^2 - a_2 = 0$.
2. the residue at the pole should be i , which has to do with unitarity, or that probabilities add to 1. The condition on the residue fixes $-i/(2a_1) = i$, together the conditions give $a_1 = -1/2$ and $a_2 = -m^2/2$.

For the time being these notes are not intended to cover these topics. These restrictions will be really important later when we calculate quantum corrections to the propagator which will skew these relations and require us to *renormalize* in order to fix these conditions again². With these conditions in place we have for the Feynman propagator:

$$\Delta_{(x-y)} \equiv \Delta_F(x - y) = iA^{-1}(x - y) = \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p^2 - m^2} e^{ip \cdot (x-y)}. \quad (4.21)$$

²Technically this is not necessary to perform calculations, we just need physical observables to work like this. We will discuss these nuances a bit when we get to renormalization in QED.

The conditions on a_1 and a_2 dictate that our free Lagrangian is given by,

$$\mathcal{L} = -\frac{1}{2}\phi\Box\phi - \frac{m^2}{2}\phi^2, \quad (4.22)$$

and now we can understand that by “free” we mean absent interactions. We’ll refer to the way this Lagrangian is normalized as “canonically” normalized. That is, the choice $a_1 = -1/2$ and $a_2 = -m^2/2$ corresponds to the canonically normalized propagator which corresponds to physical scattering.

It is not mandatory we work with canonically normalized fields. When we discuss observables later in Chapter 8.5, we’ll see that with some modifications to the “LSZ reduction formula” we can work with arbitrary a_1 and a_2 . The LSZ formula will relate these more general normalizations back to observables. We’ll re-encounter this when we discuss renormalization; loop diagrams will lead to corrections to a_1 and a_2 and we won’t always be able to make choices that return us to canonical normalization.

4.2 Euler Lagrange equation for Lagrangian densities

If you don’t feel fully convinced by the argument about the time-ordered product of two particles giving the free propagation of the field, and therefore that the terms quadratic in fields from the Lagrangian give propagators, let’s look back to classical field theory.

First we’ll need the Euler Lagrange equations for Lagrangian densities, writing the action as,

$$S[\phi(x)] = \int d^4x \mathcal{L}(\phi(x), \partial_\mu \phi(x)), \quad (4.23)$$

we can just follow the steps outlined in Section 3.2 where we developed functional analysis. So we’ll take the variation of S with respect to x^μ as:

$$\frac{\delta S}{\delta \phi(x')} = 0 = \int d^4x \frac{\delta \mathcal{L}}{\delta \phi(x')}, \quad (4.24)$$

$$= \int d^4x \left(\frac{\partial \mathcal{L}}{\partial \phi(x)} \frac{\delta \phi(x)}{\delta \phi(x')} + \frac{\partial \mathcal{L}}{\partial \partial_\nu \phi(x)} \frac{\delta \partial_\nu \phi(x)}{\delta \phi(x')} \right). \quad (4.25)$$

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Integration by parts removes the four-derivative from $\phi(x)$, giving:

$$\frac{\delta S}{\delta \phi(x')} = 0 = \int d^4x \left(\frac{\partial \mathcal{L}}{\partial \phi(x)} - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \phi(x)} \right) \underbrace{\frac{\delta \phi(x)}{\delta \phi(x')}}_{\delta^4(x-x')}. \quad (4.26)$$

Integrating over the delta function, we conclude:

$$\frac{\partial \mathcal{L}}{\partial \phi(x)} - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \phi(x)} = 0. \quad (4.27)$$

While this is a classical result, it will prove useful for our QFT discussions. Particularly when we get to effective field theories where the **(Euler Lagrange) Equations of Motion (EOM)** will allow us to make complicated field redefinitions very simple.

If we start with Eq. 4.22 and apply the Euler Lagrange equations we find:

$$\frac{\partial \mathcal{L}}{\partial \phi(x)} - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \phi(x)} = (\square + m^2) \phi(x) = 0 \quad (4.28)$$

This is the **Klein-Gordon equation** which has plane wave solutions for the dispersion of a relativistic wave of mass m . In the presence of a source term, $\mathcal{L}_{\text{source}} = J(x)\phi(x)$, we find:

$$(\square + m^2) \phi(x) = J(x) \quad (4.29)$$

So in the presence of a source, the Klein-Gordon equation is modified by the source term. Absent a source we have the free propagation of a wave. That is, the quadratic terms in $\mathcal{L}$ results in solutions to the free propagation of a wave, while the presence of a source modifies this (formerly) free propagation.

4.3 Time-ordered product of 3 or 4 fields: free theory

We can write our time-ordered product of N fields in the free theory as:

$$\begin{aligned} & \langle \Omega | T[\phi(y_1) \cdots \phi(y_N)] | \Omega \rangle \\ &= \left(\prod_{i=1}^N \frac{1}{i} \frac{\delta}{\delta J_{y_i}} \right) N \int \mathcal{D}\phi \exp \left[-\frac{1}{2} \int d^4x_1 \int d^4x_2 J(x_1) \Delta(x_1 - x_2) J(x_2) \right] \Big|_{J=0} \end{aligned} \quad (4.30)$$

below I had a mistake in the PI where a factor of i appeared in the exponential. This means in what follows I need to divide every instance of Δ by i to get the correct phase. This does not change the physics as an overall phase has no effect in QM. This will be fixed in a future version of the notes.

Consider the time-ordered product of three scalar fields:

$$i^3 \langle \Omega | T[\phi(y_3), \phi(y_2), \phi(y_1)] | \Omega \rangle , \quad (4.31)$$

$$= \delta_{y_3} \delta_{y_2} \delta_{y_1} \langle \Omega | \Omega \rangle \Big|_{J=0} , \quad (4.32)$$

$$= \delta_{y_3} \delta_{y_2} \left[\langle \Omega | \Omega \rangle \left(-i \int d^4 x_3 d^4 x_4 \Delta_{x_3, x_4} J_{x_4} \delta_{(x_3 - y_1)}^4 \right) \right] \Big|_{J=0} , \quad (4.33)$$

$$= \delta_{y_3} \left[\langle \Omega | \Omega \rangle \left(-i \Delta_{y_1, y_2} - \int d^4 x_3 d^4 x_4 d^4 x_5 d^4 x_6 \Delta_{x_3, x_4} J_{x_4} \delta_{(x_3 - y_1)}^4 \Delta_{x_5, x_6} J_{x_6} \delta_{(x_5 - y_2)}^4 \right) \right] \Big|_{J=0} , \quad (4.34)$$

$$\begin{aligned} &= \langle \Omega | \Omega \rangle \left(-\Delta_{(y_1 - y_2)} \int d^4 x_3 d^4 x_4 \Delta_{x_3, x_4} J_{x_4} \delta_{(y_3 - x_4)}^4 \right. \\ &\quad \left. - \int d^4 x_3 d^4 x_4 d^4 x_5 d^4 x_6 \Delta_{x_3, x_4} \delta_{(x_4 - y_3)}^4 \delta_{(x_3 - y_1)}^4 \Delta_{x_5, x_6} J_{x_6} \delta_{x_5, y_2}^4 \right. \\ &\quad \left. - \int d^4 x_3 d^4 x_4 d^4 x_5 d^4 x_6 \Delta_{x_3, x_4} J_{x_4} \delta_{(x_3 - y_1)}^4 \Delta_{x_5, x_6} \delta_{(x_6 - y_3)}^4 \delta_{(x_5 - y_2)}^4 + 0_{J=0} \right) \Big|_{J=0} , \end{aligned} \quad (4.35)$$

$$= \langle \Omega | \Omega \rangle \left(-\Delta_{y_1, y_2} \int d^4 x_3 \Delta_{y_3, x_4} J_{x_4} - \int d^4 x_6 \Delta_{y_1, y_3} \Delta_{y_2, x_6} J_{x_6} - \int d^4 x_4 \Delta_{y_1, x_4} J_{x_4} \Delta_{y_2, y_3} \right) \Big|_{J=0} , \quad (4.36)$$

$$= 0 . \quad (4.37)$$

4. Path integrals for scalars

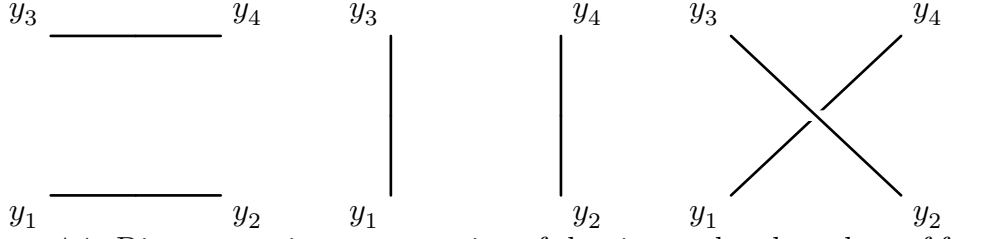


Figure 4.1: Diagrammatic representation of the time-ordered product of four scalars in the free theory. In the last diagram a white space is used to indicate that the lines do not interact at the crossing.

Remember: the term “ $0_{J=0}$ ” indicates terms we don’t include because they’ll be 0 after setting the sources to zero. In fact, as we state in the last line, after setting $J = 0$ this whole thing vanishes. We’ve used the symmetry of the delta functions and propagator to simplify our lines. Also notice that we’ve moved those factors of $1/i$ to the left-hand side of the equation to save space. There are a couple observations to make here:

1. The Lagrangian of Eq. 4.22 has a $\mathbb{Z}_2$ symmetry, i.e. it is invariant under $\phi \rightarrow -\phi$. This symmetry is preserved, and therefore the time-ordered product of three fields vanishes. Indeed, the time-ordered product of any odd number of fields vanishes for this simple theory.
2. In more physical terms, the two-point function (propagator) of Eq. 4.21 connects two fields, in the absence of some *interaction* that can connect three fields, this time ordered product must vanish.

In order to get the time-ordered product of four fields we need only take one more variation on Eq. 4.36 (and properly include the factor of i we’ve moved to the left-hand side). Performing one more variation and integrating over the dummy variables leaves:

$$\langle \Omega | T[\phi(y_4), \phi(y_3), \phi(y_2), \phi(y_1)] | \Omega \rangle , \quad (4.38)$$

$$= -N \left(\Delta_{y_1, y_2} \Delta_{y_3, y_4} + \Delta_{y_1, y_3} \Delta_{y_2, y_4} + \Delta_{y_1, y_4} \Delta_{y_2, y_3} \right) \quad (4.39)$$

Look at the first term, we have $\Delta(y_1 - y_2)\Delta(y_3 - y_4)$. This is just the propagation of a particle from y_1 to y_2 and separately another particle from y_3 to y_4 (or vice versa, since Δ is symmetric in the arguments, this is the built in time-ordering). Diagrammatically we can write this as in Fig. 4.1. Eventually, these **Feynman diagrams** will come to represent equations we can write down directly just by applying a set of Feynman rules.

4.4 Path integral for interacting scalar fields, ϕ^3 theory

Let's add the simplest possible interaction to this theory. We'll add an interaction Lagrangian,

$$\mathcal{L}_I = \frac{g}{3!} \phi^3, \quad (4.40)$$

where the normalization factor $3!$ will balance a $3!$ we get later. Our path integral becomes:

$$\begin{aligned} \langle \Omega | \Omega \rangle &= \int \mathcal{D}\phi \exp \left[i \int d^4x \left(-\frac{1}{2} \phi(\square + m^2) \phi + J\phi + \frac{g}{3!} \phi^3 \right) \right] \Big|_{J=0}, \quad (4.41) \\ &= \int \mathcal{D}\phi \exp \left[i \int d^4x \frac{g}{3!} \phi^3 \right] \exp \left[i \int d^4x \left(-\frac{1}{2} \phi(\square + m^2) \phi + J\phi \right) \right] \Big|_{J=0}. \end{aligned} \quad (4.42)$$

The second line is just rearranging the exponentials. Now we're going to assume g is a small number. This puts us into the realm of **perturbation theory**: solving hard integrals by Taylor expanding in small parameters. In these notes we'll only consider perturbation theory. So assuming g is sufficiently small (eventually we'll talk about what that actually means), we expand our first exponential to obtain:

$$\begin{aligned} \langle \Omega | \Omega \rangle &= \int \mathcal{D}\phi \left[1 + i \int d^4x_1 \frac{g}{3!} \phi^3(x_1) + \frac{i^2}{2} \int d^4x_1 d^4x_2 \frac{g^2}{(3!)^2} \phi^3(x_1) \phi^3(x_2) \right] \\ &\quad \times \exp \left[i \int d^4x \left(-\frac{1}{2} \phi(\square + m^2) \phi + J\phi \right) \right] \Big|_{J=0}. \end{aligned} \quad (4.43)$$

Wait, isn't this expansion nonlocal? While it may appear nonlocal because we have, for example $\phi^3(x_1) \phi^3(x_2)$, we'll soon see these will correspond to separate interaction vertices connected by propagators which maintain locality. Essentially, in this form locality is not manifest in our path integral, but its equivalence to the form with manifest locality shows we've just obscured the locality not removed it. A nonlocal interaction would couple two distinct spacetime points.

Notice that the parts coming from expanding the potential are integrated over the relevant dx_i . The x_i are also dummy indices (variables), later we'll make use of this as we can sum terms symmetric in e.g. $x_1 \leftrightarrow x_2$.

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Now, we rewrite each term separately as a path integral. These polynomial terms we've expanded out front can then be rewritten as variations acting on the *path integral for the free theory* giving us:

$$\begin{aligned} \langle \Omega | \Omega \rangle &= \langle 0 | 0 \rangle + \frac{ig}{3!} \int d^4x_1 \left(\frac{1}{i} \frac{\delta}{\delta J(x_1)} \right)^3 \langle 0 | 0 \rangle \Big|_{J=0} \\ &+ \left(\frac{ig}{3!} \right)^2 \int d^4x_1 d^4x_2 \left(\frac{1}{i} \frac{\delta}{\delta J(x_1)} \right)^3 \left(\frac{1}{i} \frac{\delta}{\delta J(x_2)} \right)^3 \langle 0 | 0 \rangle \Big|_{J=0} + \dots \end{aligned} \quad (4.44)$$

Where we've conveniently renamed the vacuum of the free theory $|0\rangle$. What's happened here is we've taken a theory we know (and can solve), the free theory, and we've done a perturbative expansion around this solvable case. A classic for physicists. Now if we want to know how a particle goes from y to z in the presence of an interacting field theory we can just use the same tools we've used all along: more variations!

Indeed we've related the vacuum of the full/interacting theory $\langle \Omega | \Omega \rangle$ to the vacuum of the free theory. At 0th order they're the same vacuum, at leading order in g the correction vanishes (remember the $\mathbb{Z}_2$ symmetry of the free theory³), and at next-to-leading order we expect the first non-vanishing correction.

Let's agree on a change of notation and approach, now that we'll be doing a lot more variations. In terms of the free theory:

$$\delta_y \left[N \exp \left[-\frac{i}{2} \int d^4x_1 \int d^4x_2 J_{x_1} A_{x_1, x_2}^{-1} J_{x_2} \right] \right] \Big|_{J=0} \quad (4.45)$$

$$= -i \langle 0 | 0 \rangle \int d^4x_3 d^4x_4 \Delta_{x_3, x_4} \delta_{(x_3 - y)}^4 J_{x_4} \Big|_{J=0}, \quad (4.46)$$

$$\rightarrow -i \langle 0 | 0 \rangle \Delta_{y, x_4} J_{x_4} \Big|_{J=0}. \quad (4.47)$$

So what we're doing now is:

1. Automatically incorporating that $\Delta(x - y)$ is symmetric in $x \leftrightarrow y$, so we just add terms like $\Delta(x - y) + \Delta(y - x) = 2\Delta_{x, y}$.
2. Immediately integrating over delta functions.

³i.e. the $\phi \rightarrow -\phi$ symmetry.

3. Integrals over a spacetime point are *implied by repeated indices*, they're just dummy variables anyway so this is basically Einstein summation notation in the continuous case. The terms from Taylor expanding the potential may have indices which are repeated more times⁴, in our example of ϕ^3 theory we'll have 3 repeated indices for the interaction point.

4.5 Interaction between 3 scalars, order g

We'll start with the simplest interaction we can have in our theory, the interaction between three particles. In our notation this corresponds to the time-ordered product,

$$\langle \Omega | T[\phi(y_3), \phi(y_2), \phi(y_1)] | \Omega \rangle = \left(\prod_{i=1}^3 \frac{1}{i} \frac{\delta}{\delta J(y_i)} \right) \langle \Omega | \Omega \rangle \Big|_{J=0}, \quad (4.48)$$

remember the time-ordered product of three fields is just three variations with respect to those three fields. We haven't invoked perturbation theory yet, hence the appears on $|\Omega\rangle$ on both sides of the equation. We showed in Eq. 4.37 that the leading contribution vanishes due to the $\mathbb{Z}_2$ symmetry of the free theory. The next order correction comes from the order g correction as follows:

$$\langle \Omega | T[\phi(y_3), \phi(y_2), \phi(y_1)] | \Omega \rangle \sim \left(\prod_{i=1}^3 \frac{1}{i} \frac{\delta}{\delta J(y_i)} \right) \frac{ig}{3!} \left(\frac{1}{i} \frac{\delta}{\delta J(z)} \right)^3 \langle 0 | 0 \rangle \Big|_{J=0}. \quad (4.49)$$

Now we're in perturbation theory: we're treating the interaction as part of the Taylor expansion of the interaction Lagrangian (the interaction occurs at the spacetime point z) and we're perturbing the free theory's vacuum. Notice we used $\sim$ instead of an equality, since we're just approximating the interacting theory. Remember, there's also an $\int d^4z$ we've suppressed for conciseness as well.

What we'll find is that we have disconnected and connected diagrams, this is going to lead to a more physical intuition of what is meant by the interacting vacuum. When we discuss generating functionals we'll learn how to remove these.

⁴or fewer if we allow a tadpole term in the Lagrangian, i.e. a term that only goes as $\phi^1(x)$.

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Switching to our shorthand notation and moving all of our factors of i and $3!$ to the left-hand side (for convenience):

$$(3!)i^5 \langle \Omega | T[\phi(y_3), \phi(y_2), \phi(y_1)] | \Omega \rangle \quad (4.50)$$

$$\sim \delta_{y_1} \delta_{y_2} \delta_{y_3} \delta_z^3 \langle 0|0 \rangle \Big|_{J=0} \quad (4.51)$$

$$= -i \delta_{y_1} \delta_{y_2} \delta_{y_3} \delta_z^2 \langle 0|0 \rangle \Delta_{z,x_1} J_{x_1} \Big|_{J=0} \quad (4.52)$$

$$= -i \delta_{y_1} \delta_{y_2} \delta_{y_3} \delta_z \langle 0|0 \rangle \left[\Delta_{z,z} - i \Delta_{z,x_1} J_{x_1} \Delta_{z,x_2} J_{x_2} \right] \Big|_{J=0} \quad (4.53)$$

$$= (-i)^2 \delta_{y_1} \delta_{y_2} \delta_{y_3} \langle 0|0 \rangle \left[3 \Delta_{z,z} \Delta_{z,x_1} J_{x_1} - i \Delta_{z,x_1} J_{x_1} \Delta_{z,x_2} J_{x_2} \Delta_{z,x_3} J_{x_3} \right] \Big|_{J=0} \quad (4.54)$$

In the fourth equivalence (including $\sim$), the factor of 3 comes from the $\delta_{z,z}$ term multiplied by another term brought down by the path integral in the prior line and setting $x_1 = z$ or $x_2 = z$ in the second term of the previous line as well as relabelling the dummy variables x_i appropriately.

We've performed all the variations with respect to the interaction vertex, what's left is to take the variations with respect to the "external spacetime points." That is, those corresponding to the fields from the time-ordered product. Doing so and dropping terms that will contain too many sources at the end of the calculation (when J is set to zero):

$$(3!)i^5 \langle \Omega | T[\phi(y_3), \phi(y_2), \phi(y_1)] | \Omega \rangle \quad (4.55)$$

$$\sim (-i)^2 \delta_{y_1} \delta_{y_2} \langle 0|0 \rangle \left[3 \Delta_{z,z} \Delta_{z,y_3} - 3i \Delta_{z,z} \Delta_{z,x_1} J_{x_1} \Delta_{y_3,x_2} J_{x_2} - 3i \Delta_{z,y_3} \Delta_{z,x_2} J_{x_2} \Delta_{z,x_3} J_{x_3} + 0_{J=0} \right] \Big|_{J=0} \quad (4.56)$$

$$= (-i)^3 \delta_{y_1} \langle 0|0 \rangle \left[3 \Delta_{z,z} \Delta_{z,y_3} \Delta_{y_2,x_1} J_{x_1} + 3 \Delta_{z,z} \Delta_{z,y_2} \Delta_{y_3,x_2} J_{x_2} + 3 \Delta_{z,z} \Delta_{z,x_1} J_{x_1} \Delta_{y_3,y_2} + 6 \Delta_{z,y_3} \Delta_{z,y_2} \Delta_{z,x_3} J_{x_3} + 0_{J=0} \right] \Big|_{J=0}$$

$$= (-i)^3 \langle 0|0 \rangle \left[3 \Delta_{z,z} \Delta_{z,y_3} \Delta_{y_2,y_1} + 3 \Delta_{z,z} \Delta_{z,y_2} \Delta_{y_3,y_1} \right. \quad (4.57)$$

$$\left. + 3 \Delta_{z,z} \Delta_{z,y_1} \Delta_{y_3,y_2} + 6 \Delta_{z,y_3} \Delta_{z,y_2} \Delta_{z,y_1} + 0_{J=0} \right] \Big|_{J=0} \quad (4.58)$$

Again, in the lines above we have combined terms by renaming dummy variables. This gave us the 3 in the second line of the first equivalence, and the 6 in the second equivalence. It's very easy to make mistakes here, the only viable approach is to keep trying until it starts to feel natural.

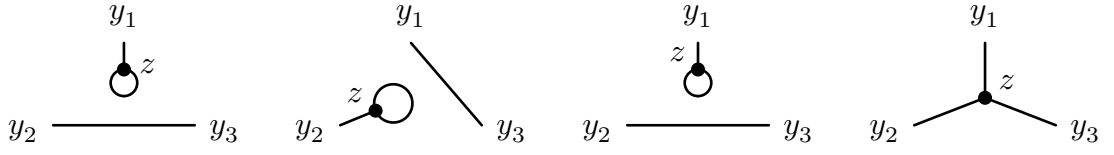


Figure 4.2: Diagrams resulting from calculating the time-ordered product of three scalars in ϕ^3 theory. The left three diagrams are referred to as disconnected and the right-most is referred to as connected.

Setting the sources to zero, moving those extra factors on the left back, and restoring the integrals over the dummy variable z gives:

$$\begin{aligned}
 & \langle \Omega | T[\phi(y_3), \phi(y_2), \phi(y_1)] | \Omega \rangle \\
 &= N \int d^4 z \left[\frac{1}{2} \Delta_{z,z} \Delta_{z,y_1} \Delta_{y_3,y_2} + \frac{1}{2} \Delta_{z,z} \Delta_{z,y_2} \Delta_{y_3,y_1} + \frac{1}{2} \Delta_{z,z} \Delta_{z,y_3} \Delta_{y_2,y_1} \right. \\
 & \quad \left. + \Delta_{z,y_3} \Delta_{z,y_2} \Delta_{z,y_1} \right] + \mathcal{O}(g^2)
 \end{aligned} \tag{4.59}$$

Notice in the last line we wrote $\mathcal{O}(g^2)$: our result is only a partial result. This is also why we reinstated “=” instead of employing “ $\sim$ ”, this distinction is common in perturbation theory. We could go and perform even more variations and find a (small) correction to this leading contribution. Diagrammatically we can express this result as in Fig. 4.2.

Let’s take a moment to understand what we’ve found:

1. The three diagrams on the left are called **disconnected**, while the last diagram on the right is referred to as **connected**.
2. The three diagrams on the left include a **loop**, we will see this corresponds to an unspecified momentum that must be integrated over which leads to problems called ultraviolet (UV) divergences and infrared (IR) divergences. The UV divergences will result in our need to renormalize in QFTs, the IR divergences will require us to carefully define observables.
3. The three diagrams on the right describe a field propagating from one spacetime point to another, and a tadpole **tadpole diagram**. The tadpole is representative of the fact we’re expanding around the *free vacuum* and is representative of the fact the vacuum is interacting. Contributions from tadpoles like these require us to shift scalar fields when calculating one-loop corrections to properly incorporate the fact the true vacuum is not the free vacuum.
4. Separately each of the disconnected diagrams is accompanied by a factor of $1/2$. When we develop Feynman rules, we’ll refer to this as a **symmetry factor**.

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5. The factor of $1/3!$ in the definition of the interaction Lagrangian exactly cancelled the factor $6 = 3!$ coming from applying our variations for the right-most diagram. This is a choice we often make so that the Feynman rules come out nicer⁵.
6. As promised, our interactions occur at one spacetime point (z) and are connected by propagators $\Delta(z - y_i)$. In this manner locality is preserved.
7. The free vacuum has a $\mathbb{Z}_2$ symmetry, our interaction breaks this symmetry. Still, if we had calculated the vacuum expectation value (in the free theory) of an odd number of variations we would find zero. The interaction gave us three variations while the time-ordered product gave us three more for a total of six, hence the nonzero result despite us calculating the time-ordered product of three fields.

We'll discuss the Feynman rules in more detail after a couple more examples.

4.6 The tadpole diagram, order g

We mentioned that the ϕ^3 term (explicitly) breaks the $\mathbb{Z}_2$ symmetry of the free Lagrangian. While we don't include an explicit term in our interaction Lagrangian of the form,

$$\mathcal{L}_I = \delta v \phi(x), \quad (4.60)$$

that doesn't mean that this term isn't generated by perturbative corrections. Let's have a look.

The tadpole diagram can first be generated at order g : the interaction connects three scalars, one pair will form a loop while the other will connect to the external field $\phi(y_1)$ giving a tadpole term. This corresponds to the vacuum expectation value of the field $\phi(y_1)$:

$$\langle \Omega | \phi(y_1) | \Omega \rangle = \frac{1}{i} \delta_{y_1} \left(\frac{ig}{3!} \right) \int d^4 z \left(\frac{1}{i} \delta_z \right)^3 \langle 0 | 0 \rangle \Big|_{J=0}, \quad (4.61)$$

$$= (-i)^2 \left(\frac{1}{i} \right)^4 \frac{ig}{3!} \delta_{y_1} \langle 0 | 0 \rangle \int d^4 z \left[3 \Delta_{z,z} \Delta_{z,x_1} J_{x_1} + 0_{J=0} \right], \quad (4.62)$$

$$= -\frac{ig}{2} \langle 0 | 0 \rangle \int d^4 z \Delta_{z,z} \Delta_{z,y_1}. \quad (4.63)$$

⁵Later we'll see that Lagrangian parameters are not physical, and that we must re-express them in terms of observables. So any choice of normalization makes no physical difference.

$$\langle \Omega | \phi | \Omega \rangle = \frac{1}{2} \text{---} \bigcirc$$

Figure 4.3: The order g tadpole induced by perturbative corrections in $g\phi^3$ theory.

In the second line we’ve made use of our partial result in Eq. 4.54, the $(-i)^2$ comes from the $-i$ in the exponential being “brought down” by variations twice. This result is expressed diagrammatically in Fig. 4.3.

This **vacuum expectation value** has a loose physical interpretation of the amount of the field permeating space. If the propagator tells us how a disturbance at y_1 moves through the field to y_2 and the background expectation value tells us how much of the field is already there. These sorts of heuristic interpretations are like reading a pamphlet versus the 100 page legal text it cites: the math itself, built into the QFT, is the final word on the interpretation; explanations like these come with many caveats and tend to upset purists in the field. Real understanding comes from diving deeper into the intricacies of QFT.

This simple result is nonetheless important. *In our Lagrangian we must include all operators consistent with our symmetries.* Since our $\mathbb{Z}_2$ symmetry is broken we need to include the linear term $\phi(x)$. When we discuss renormalization later we’ll learn more about how this works and when we can restrict ourselves to a subset of operators.

4.7 The interacting vacuum, order g^2

Our next calculation will be focused on only how the vacuum behaves, therefore we won’t calculate a time-ordered product. This will provide a setup for calculating the one-loop correction to the propagator later, so we’ll keep some extra terms we could normally set to “0 _{$J=0$} .”

We’ll calculate the vacuum corrections at order g^2 . Why not at order g ? Like we discussed, the vacuum is still $\mathbb{Z}_2$ symmetric, so the order g correction, which is odd in variations, vanishes exactly. This time we’re going to

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completely remove the following factors and reintroduce them at the end:

$$\begin{aligned} \text{interactions} &\rightarrow \left(\frac{ig}{3!}\right)^2, & \text{variations} &\rightarrow \left(\frac{1}{i}\right)^6, & \text{Taylor expansion} &\rightarrow \frac{1}{2}, \\ &\Rightarrow \text{overall} \rightarrow C = \frac{g^2}{72}. \end{aligned} \quad (4.64)$$

So we'll need to multiply the result we derive by C at the end. Our calculation is then (up to C):

$$\begin{aligned} \langle \Omega | \Omega \rangle &\sim \delta_{z_2}^3 \delta_{z_1}^3 \langle 0 | 0 \rangle \\ &= (-i)^2 \delta_{z_2}^3 \langle 0 | 0 \rangle \left[3\Delta_{z_1, z_1} \Delta_{z_1, x_1} J_{x_1} - i\Delta_{z_1, x_1} J_{x_1} \Delta_{z_1, x_2} J_{x_2} \Delta_{z_1, x_3} J_{x_3} \right] \Big|_{J=0}, \end{aligned} \quad (4.65)$$

$$(4.66)$$

where we've copied the second line from our previous result in Eq. 4.54. Proceeding with the other variations gives:

$$\begin{aligned} \langle \Omega | \Omega \rangle &\sim (-i)^2 \delta_{z_2}^2 \langle 0 | 0 \rangle \left[3\Delta_{z_1, z_1} \Delta_{z_1, z_2} - 3i\Delta_{z_1, z_1} \Delta_{z_1, x_1} J_{x_1} \Delta_{z_2, x_2} J_{x_2} \right. \\ &\quad \left. - 3i\Delta_{z_1, z_2} \Delta_{z_1, x_2} J_{x_2} \Delta_{z_1, x_3} J_{x_3} \right. \\ &\quad \left. - \Delta_{z_1, x_1} J_{x_1} \Delta_{z_1, x_2} J_{x_2} \Delta_{z_1, x_3} J_{x_3} \Delta_{z_2, x_4} J_{x_4} \right] \Big|_{J=0}, \end{aligned} \quad (4.67)$$

$$\begin{aligned} &= (-i)^3 \delta_{z_2} \langle 0 | 0 \rangle \left[6\Delta_{z_1, z_1} \Delta_{z_1, z_2} \Delta_{z_2, x_1} J_{x_1} + 3\Delta_{z_1, z_1} \Delta_{z_2, z_2} \Delta_{z_1, x_1} J_{x_1} \right. \\ &\quad + 6\Delta_{z_1, z_2} \Delta_{z_1, z_2} \Delta_{z_1, x_1} J_{x_1} - 6i\Delta_{z_1, z_2} \Delta_{z_1, x_1} J_{x_1} \Delta_{z_1, x_2} J_{x_2} \Delta_{z_2, x_3} J_{x_3} \\ &\quad - 3i\Delta_{z_1, z_1} \Delta_{z_1, x_1} J_{x_1} \Delta_{z_2, x_2} J_{x_2} \Delta_{z_2, x_3} J_{x_3} \\ &\quad \left. - i\Delta_{z_2, z_2} \Delta_{z_1, x_1} J_{x_1} \Delta_{z_1, x_2} J_{x_2} \Delta_{z_1, x_3} J_{x_3} + 0_{J=0} \right] \Big|_{J=0}, \end{aligned} \quad (4.68)$$

$$\begin{aligned} &= (-i)^3 N \left[9\Delta_{z_1, z_1} \Delta_{z_1, z_2} \Delta_{z_2, z_2} + 6\Delta_{z_1, z_2} \Delta_{z_1, z_2} \Delta_{z_1, z_2} \right. \\ &\quad - 18i\Delta_{z_1, z_1} \Delta_{z_1, z_2} \Delta_{z_2, x_1} J_{x_1} \Delta_{z_2, x_2} J_{x_2} \\ &\quad - 18i\Delta_{z_1, z_2}^2 \Delta_{z_1, x_1} J_{x_1} \Delta_{z_2, x_2} J_{x_2} \\ &\quad \left. - 9i\Delta_{z_1, z_1} \Delta_{z_2, z_2} \Delta_{z_1, x_1} J_{x_1} \Delta_{z_2, x_2} J_{x_2} + 0_{J=0} \right] \Big|_{J=0} \end{aligned} \quad (4.69)$$

In Eq. 4.69 we kept the extra terms dependent on J because we'll need it next for the correction to the propagator. Restoring the factor of C we hid,

$$\langle \Omega | \Omega \rangle = \langle 0 | 0 \rangle + \frac{1}{8} \text{---}\bigcirc\text{---}\bigcirc\text{---} + \frac{1}{12} \text{---}\bigcirc\text{---}\bigcirc\text{---} + \mathcal{O}(g^4)$$

Figure 4.4: Diagrams corresponding to the order g^2 corrections to the perturbative calculation of the vacuum. The vertices correspond to interaction points z_1 and z_2 . The diagrams (and vacuum corrections) are symmetric in the exchange of z_1 and z_2 . We've included $\langle 0|0 \rangle$ here as opposed to in Eq. 4.70 which refers strictly to the g^2 corrections. As the free vacuum is $\mathbb{Z}_2$ symmetric, the next order correction is $\mathcal{O}(g^4)$.

the integrals over the dummy variables z_1 and z_2 , and taking $J \rightarrow 0$ we find:

$$\langle \Omega | \Omega \rangle = ig^2 N \int d^4 z_1 d^4 z_2 \left[\frac{1}{8} \Delta_{z_1, z_1} \Delta_{z_1, z_2} \Delta_{z_2, z_2} + \frac{1}{12} \Delta_{z_1, z_2} \Delta_{z_1, z_2} \Delta_{z_1, z_2} \right] \quad (4.70)$$

These corrections to the vacuum can be reexpressed diagrammatically as in Fig. 4.4. The next order corrections to the vacuum occur at order g^4 , again due to the $\mathbb{Z}_2$ symmetry of the free Lagrangian. We'll learn what to do with these diagrams later in Sec. 5, where we'll discuss generating functionals and obtain the generating functional for *connected diagrams*. In general, physical observables will only depend on connected diagrams, so we won't have to do this sort of vacuum calculation again. Next we'll apply two more variations to obtain the time-ordered product of two fields at order g^2 , i.e. the leading correction to the propagator.

4.8 The one-loop correction to the propagator, order g^2

As promised, we'll now move on to calculating the leading correction to the propagator. As the title of this section suggests, this will be a one-loop result of order g^2 . As we discussed earlier, the time-ordered product of two fields corresponds to the inverse propagator. So we'll start by applying two more variations to Eq. 4.69, again reinstating the overall constants at the end for clarity:

$$\begin{aligned}
 & \delta_{y_2} \delta_{y_1} \langle \Omega | \Omega \rangle \\
 & \sim (-i)^4 \delta_{y_2} \langle 0 | 0 \rangle \left[9 \Delta_{z_1, z_1} \Delta_{z_1, z_2} \Delta_{z_2, z_2} \Delta_{y_1, x_1} J_{x_1} + 6 \Delta_{z_1, z_2}^3 \Delta_{y_1, x_1} J_{x_1} \right. \\
 & \quad + 36 \Delta_{z_1, z_1} \Delta_{z_1, z_2} \Delta_{z_2, y_1} \Delta_{z_2, x_1} J_{x_1} + 18 \Delta_{z_1, z_2}^2 \Delta_{z_1, y_1} \Delta_{z_2, x_1} J_{x_1} \\
 & \quad + 18 \Delta_{z_1, z_2}^2 \Delta_{z_1, x_1} J_{x_1} \Delta_{z_2, y_1} + 9 \Delta_{z_1, z_1} \Delta_{z_2, z_2} \Delta_{z_1, y_1} \Delta_{z_2, x_1} J_{x_1} \quad (4.71) \\
 & \quad \left. + 9 \Delta_{z_1, z_1} \Delta_{z_2, z_2} \Delta_{z_1, x_1} J_{x_1} \Delta_{z_2, y_1} + 0_{J=0} \right] \Big|_{J=0}
 \end{aligned}$$

Notice here, the points z_i correspond to the interaction points and so appear in triplicate, while the points y_i correspond to external points and occur only once each. Next we take the last remaining variation, set $J = 0$, and restore the overall normalization, which is given by:

$$C' = \left(\frac{1}{i} \right)^2 C = -\frac{g^2}{72}. \quad (4.72)$$

We also add in the integrals over the dummy variables z_1 and z_2 . Notice all we needed to do was add the factors of $1/i$ for the two extra variations. We find:

$$\begin{aligned}
 & \langle \Omega | T[\phi(y_2) \phi(y_1)] | \Omega \rangle \\
 & = -g^2 N \int d^4 z_1 d^4 z_2 \left[\frac{1}{8} \Delta_{z_1, z_1} \Delta_{z_1, z_2} \Delta_{z_2, z_2} \Delta_{y_1, y_2} + \frac{1}{12} \Delta_{z_1, z_2}^3 \Delta_{y_1, y_2} \right. \\
 & \quad + \frac{1}{2} \Delta_{z_1, z_1} \Delta_{z_1, z_2} \Delta_{z_2, y_1} \Delta_{z_2, y_2} + \frac{1}{2} \Delta_{z_1, z_2}^2 \Delta_{z_1, y_1} \Delta_{z_2, y_2} \\
 & \quad \left. + \frac{1}{4} \Delta_{z_1, z_1} \Delta_{z_2, z_2} \Delta_{z_1, y_1} \Delta_{z_2, y_2} \right] \quad (4.73)
 \end{aligned}$$

By now we're pretty used to this tedious algebra. Incase it isn't obvious how we arrived at the expression above, let's look at a specific simplification we made:

$$\Delta_{z_1, z_1} \Delta_{z_2, z_2} [\Delta_{z_1, y_1} \Delta_{z_2, y_2} + \Delta_{z_1, y_2} \Delta_{z_2, y_1}] = 2 \Delta_{z_1, z_1} \Delta_{z_2, z_2} \Delta_{z_1, y_1} \Delta_{z_2, y_2} \quad (4.74)$$

The equivalence follows from the fact the left-hand side of the equation is symmetric in $z_1 \leftrightarrow z_2$ and that these two variables are dummy variables (both integrated over all spacetime, see Eq. 4.44 for a reminder), so relabelling them doesn't change the result. This is no different from relabeling the variable

$$\begin{aligned}
 \langle \Omega | T[\phi(y_2), \phi(y_1)] | \Omega \rangle = & \frac{1}{8} \text{---}\bigcirc\text{---}\bigcirc\text{---} + \frac{1}{12} \text{---}\bigcirc\text{---}\bigcirc\text{---} \\
 & + \frac{1}{2} \text{---}\bigcirc\text{---} + \frac{1}{2} \text{---}\bigcirc\text{---} + \frac{1}{4} \text{---}\bigcirc\text{---}\bigcirc\text{---}
 \end{aligned}$$

Figure 4.5: Diagrams corresponding to the order g^2 corrections to the propagator in ϕ^3 theory. The two diagrams in the top line are disconnected and can be removed by considering the generating functional for connected diagrams. The last diagram can be removed by introducing a tadpole operator in the Lagrangian density.

x_2 with x_1 along the way in the calculations above, the only variables that aren't behaving as dummy variables in this whole calculation are y_1 and y_2 .

Figure 4.5 shows the diagrams corresponding to this time-ordered product. Notice that the disconnected graphs in the top line are *exactly those from the vacuum calculation* with the addition of a propagator. These contributions aren't physical. They can be cancelled by considering difference,

$$\underbrace{\langle \Omega | T[\phi_2, \phi_1] | \Omega \rangle}_\text{connected} = \underbrace{\langle \Omega | T[\phi_2, \phi_1] | \Omega \rangle}_\text{disconnected} - \underbrace{\langle \Omega | \Omega \rangle \langle 0 | T[\phi_2, \phi_1] | 0 \rangle}_\text{connected} \times \text{connected}, \quad (4.75)$$

where the subscript on ϕ_i indicates its argument $\phi(x_i)$. We've written underneath as a small spoiler of what's to come, that what we're subtracting off is a product of connected diagrams. Our next step will be to define a generating functional that produces exactly this sort of difference and removes the vacuum diagrams, as well as making this “connected×connect” bit make sense.

4.9 Diagram terminology

Let's quickly go over all the words we're using to describe diagrams. In Table 4.1 we write some of the terms we have used, or may use, to describe diagrams. These diagrams are classified into the following categories:

1. **connected:** all external lines are linked to each other via internal propagators and vertices.
2. **disconnected:** a single diagram composed of (technically products of) smaller connected diagrams.
3. **tadpole:** a diagram where the external line loops back on itself, the external line is only connected to itself.

4. Path integrals for scalars

4. **vacuum bubble:** closed diagrams with no external legs at all.
5. **one-particle irreducible (1PI):** connected diagrams which cannot be separated into two disconnected pieces by cutting a single internal line (propagator). By stating “internal line” we explicitly exclude external lines.

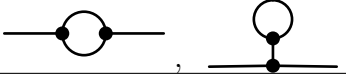
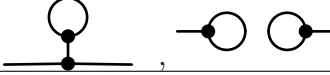
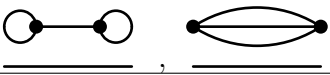
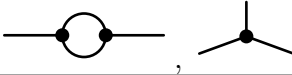
connected:	
disconnected:	
tadpoles:	
vacuum bubbles:	
one-particle irreducible:	

Table 4.1: Terminology for types of diagrams. Described in greater detail in the text.

4.10 On Complex and/or multiple scalar fields

When we discussed Lorentz invariant Lagrangians, we mentioned we weren't limited for real scalar fields. We could also consider complex scalar fields, or scalar fields with multiple components. How does this change the picture we've formed in the last few sections?

The answer is, really it doesn't. We were general enough with our discussions to pretty easily extend things to other cases of scalar fields. For example, let's look at the free Lagrangian for a complex scalar field:

$$\mathcal{L} = a_1 \phi^\dagger \square \phi + a_2 \phi^\dagger \phi + J^\dagger \phi + J \phi^\dagger \quad (4.76)$$

$$\rightarrow -\phi^\dagger \square \phi - m^2 \phi^\dagger \phi + J^\dagger \phi + J \phi^\dagger. \quad (4.77)$$

Where the $\rightarrow$ indicates that after going through the same procedure from before we find these values for a_1 and a_2 . More precisely, we'll find the same propagator as before except that requiring the pole be at $p^2 = m^2$ and the residue is i will give us $a_1 = -1$, $a_2 = -m^2$. Before we had an extra factor

of $\frac{1}{2}$. That factor of one half is simply a counting factor to balance the factor of 2 that comes from $\delta\phi^2 = 2\phi\delta\phi$ (where delta is shorthand for a variation).

We've also included the sources, which are conjugates of each other, $J^\dagger$ sources ϕ while J sources $\phi^\dagger$. This is the most common choice, but not necessary. What is necessary is that our Lagrangian be hermitian, and we need a separate source for ϕ and $\phi^\dagger$ since they are treated as independent fields.

It turns out that this complex scalar field is just a rewriting of two real scalar fields:

$$\mathcal{L} = -\frac{1}{2}\phi_1(\square + m^2)\phi_1 - \frac{1}{2}\phi_2(\square + m^2)\phi_2 + J_1\phi_1 + J_2\phi_2. \quad (4.78)$$

Which can be seen to be precisely the case of the complex scalar after making the field redefinition,

$$\phi_1 = \frac{1}{\sqrt{2}}(\phi + \phi^\dagger) \quad \phi_2 = \frac{1}{\sqrt{2}i}(\phi - \phi^\dagger), \quad (4.79)$$

it's straightforward to check that the determinant of this field redefinition is 1 meaning this change of variables doesn't change our path integral. From this we can see the justification of treating ϕ and $\phi^\dagger$ as independent fields: they really are.

We can go further, and write $\phi_i = (\phi_1, \dots, \phi_n)$, so n components which could be real or complex. Our free Lagrangian for the real case is:

$$\mathcal{L} = -\frac{1}{2}\phi_i(\square + m^2)\phi_i + J_i\phi_i \quad (4.80)$$

And the only thing that's changed is that we can now treat variations of the resulting action in terms of a general index:

$$\frac{\delta}{\delta J_i(x)} Z[J_i] = \dots, \quad (4.81)$$

This works as long as we treat all components of ϕ_i the same everywhere; we never treat a single component independently of the others. If we wanted something a little weird we could write a mass term like $m_{12}\phi_1\phi_2$ or even a kinetic term mixing two scalars. As these terms are strictly quadratic, we can always make a field redefinition that will diagonalize them: A hermitian matrix can always be diagonalized by a unitary transformation: $U^\dagger K U = \text{diag}(\lambda_i)$ where the lambda are the eigenvalues of the kinetic matrix K .

4. Path integrals for scalars

Notice that, if we just called $\phi \rightarrow A$ and the index $i \rightarrow \mu$ this would look an awful lot like a vector field! In fact what we've done with our path integral formulation combined with Eq. 4.81 is “enough” to treat path integrals for Vectors. When we do discuss path integrals for vectors, our focus will be more on Lagrangians with vectors that are physical and how to handle gauge symmetry in a QFT.

One last note: Treating ϕ_i as a set of n scalar fields looks an awful lot like a vector (in terms of general symmetries, not Lorentz vector) formed of scalars. This makes it perfect for us to impose new symmetries on our field and study the physics of QFTs with various symmetries imposed on them. The discussions from Chapter 1 will not be going away. Instead of focusing on the Lorentz group, we'll be focusing on Unitary groups. For example, the Higgs boson of the standard model is a complex scalar in the “fundamental” representation of $SU(2)_{\text{Left}}$, the singlet representation of $SU(3)_{\text{Color}}$, and has hypercharge $\frac{1}{2}$ under $U(1)_{\text{hypercharge}}$.

Chapter 5

Generating Functionals (scalar theory as an example)

So far we've been taking variations of the state $\langle\Omega|\Omega\rangle$ to obtain time ordered products. This is, in essence, the basis of the concept of **generating functionals**. A generating functional is a functional of the source J , and variations of this functional can generate all correlation functions¹ of the theory. We'll rewrite Eq. 4.48 as,

$$\langle\Omega| T[\phi(y_3), \phi(y_2), \phi(y_1)] |\Omega\rangle = \left(\prod_{i=1}^3 \frac{1}{i} \frac{\delta}{\delta J(y_i)} \right) Z[J] \Big|_{J=0}, \quad (5.1)$$

where we've rewritten $\langle\Omega|\Omega\rangle = Z[J]$. Really, the vacuum was always $Z[0]$, so we've been abusing notation by letting the vacuum implicitly depend on J . We upgrade this notion of the vacuum as a functional of J , and refer to it as $Z[J]$: the generating functional of disconnected diagrams. Why? Because it generates series like show in Fig. 4.5, where the diagrams are combination of connected and disconnected diagrams.

Physically, disconnected diagrams correspond to independent processes. For example, the first diagram of Fig. 4.5 could be described as a particle propagating from y_1 to y_2 and *separately* a particle popping out of the vacuum at z_1 , interacting with itself (the loop), propagating to z_2 , interacting with itself, and returning to the vacuum. These are different phenomena, and for

¹The use of the term “Green’s function” or “ n -point Green’s function” for a time-ordered product of n fields is very common. It comes from the fact that the propagator is the Green’s function solution of the free theory, and just extending the nomenclature. I will avoid this terminology here, as it is a slight, but commonplace, abuse of language.

what we discuss in these notes we'll focus on scattering experiments which corresponds to one process at a time. So we want to calculate correlation functions where all external spacetime points are connected by propagators and vertices. In other areas of QFT and particle physics disconnected diagrams can be relevant, such as when interested in thermodynamical properties (quark gluon plasma/neutron stars) or lattice gauge theory which deals with the large coupling limit (i.e. no perturbation theory).

5.1 Generating functional of connected diagrams $W[J]$

Recall again Eqs. 4.73 and 4.75 where we pointed out that the disconnected diagrams looked like a connected diagram plus the product of two connected diagrams. If we call the generating functional of connected diagrams $W[J]$, a functional that looks like this to leading order would be:

$$Z[J] \sim e^{W[J]} \sim \langle 0|0 \rangle + W[J] + \frac{1}{2}W[J]W[J] + \dots \quad (5.2)$$

Indeed, if we calculated the two point function to order g^4 we would find the product of three disconnected diagrams. We can define $W[J]$ in terms of $Z[J]$:

$$W[J] = -i \ln Z[J]. \quad (5.3)$$

Now you may be concerned about what a logarithm means in this context, but despair not! As we discussed, the linearity of derivative and integral operations in functional analysis means we can use all our usual tricks. The factor of $-i$ is simply convention.

Let's start with the statement that the tadpole diagram is connected, up to vacuum bubbles. That is: by definition a single line is connected, in that all external lines are connected to each other. Then we could also have bubbles from the vacuum. This gives us for a single variation on $Z[J]$:

$$\frac{1}{i} \delta_{y_1} Z[J] \Big|_{J=0} = \langle \Omega | \phi(y_1) | \Omega \rangle = \text{---} \bigcirc C \text{---} + (\text{bubbles}). \quad (5.4)$$

Here (bubbles) denotes purely vacuum bubbles, like those appearing in Fig. 4.4. We saw explicitly that the two-point function was disconnected:

$$\frac{1}{i^2} \delta_{y_2} \delta_{y_1} Z[J] \Big|_{J=0} = \langle \Omega | T[\phi(y_2), \phi(y_1)] | \Omega \rangle = \text{---} \bigcirc D \text{---}. \quad (5.5)$$

5. Generating Functionals (scalar theory as an example)

In this case we're including the bubbles in the diagram consistent with our notation in e.g. Fig. 4.5. Similarly the three-, four-, and higher-point functions are disconnected. For the three point function:

$$\frac{1}{i^3} \delta_{y_3} \delta_{y_2} \delta_{y_1} Z[J] \Big|_{J=0} = \langle \Omega | T[\phi(y_3), \phi(y_2), \phi(y_1)] | \Omega \rangle = \text{---} \bigcirc D \text{---} . \quad (5.6)$$

Now, let's see what happens when we apply these same variations to $W[J]$. The $-i$ in the definition of $W[J]$ allows us to relate $W[J]$ directly to the tadpole:

$$\delta_{y_1} W[J] \Big|_{J=0} = \frac{1}{i} \frac{\delta_{y_1} Z[J]}{Z[J]} \Big|_{J=0} = \frac{\langle \Omega | \phi(y_1) | \Omega \rangle}{\langle \Omega | \Omega \rangle} \quad (5.7)$$

What's happening here is we're dividing by the vacuum, we can think of this as removing all those bubbles. Diagrammatically, we're only left with the (connected) tadpole:

$$\delta_{y_1} W[J] \Big|_{J=0} = \text{---} \bigcirc C \quad (5.8)$$

We'll see how it works explicitly in perturbation theory in a moment when we look back at our calculation of the g^2 correction to the propagator.

Next let's look at the double variation, this time we include a $1/i$ so we get time-ordered products on the right without phases:

$$\frac{1}{i} \delta_{y_2} \delta_{y_1} W[J] \Big|_{J=0} = \frac{\delta_{y_2} \delta_{y_1} Z[J]}{i^2 Z[J]} - \frac{\delta_{y_2} Z[J] \delta_{y_1} Z[J]}{i^2 Z[J]^2} \Big|_{J=0} \quad (5.9)$$

$$= \frac{\langle \Omega | T[\phi(y_2), \phi(y_1)] | \Omega \rangle}{\langle \Omega | \Omega \rangle} - \frac{\langle \Omega | \phi(y_1) | \Omega \rangle \langle \Omega | \phi(y_2) | \Omega \rangle}{\langle \Omega | \Omega \rangle} \quad (5.10)$$

The second term, which comes with a minus sign thanks to the second derivative of the logarithm, is exactly the *connected tadpole squared*. Diagrammatically we have:

$$\text{---} \bigcirc D \text{---} - \text{---} \bigcirc C \bigcirc C \text{---} = \text{---} \bigcirc C \text{---} \quad (5.11)$$

Intuitively, the connected two-point function is the disconnected two-point function less the only other possible diagram with two external legs: that with two connected tadpoles. Again, the vacuum bubbles are technically there, but that's taken care of (cancelled) by normalizing to the vacuum.

Next we can consider the three-point function to really convince ourselves this is working. To do this, for the sake of brevity, we're going to shorthand

our expressions and treat the variations loosely (as if they were at the same spacetime point), the interpretation maps naturally back to having separate spacetime points (i.e. just symmetrize in the sum over the label permutations of y_i . Consider:

$$\left. \frac{1}{i^2} \delta^3 W[J] \right|_{J=0} = \left[\frac{\delta^3 Z[J]}{i^3 Z[J]} - 3 \frac{\delta^2 Z[J] \delta Z[J]}{i^3 Z[J]^2} + 2 \frac{\delta Z[J] \delta Z[J] \delta Z[J]}{i^3 Z[J]^3} \right]_{J=0} \quad (5.12)$$

In our last example, we could just jump right in and remove the contribution from two connected tadpoles (which is a disconnected diagram). In this case $\delta^2 Z/Z$ is not the connected two-point function. So we first have to sub in our relation from Eq. 5.10. Relating $\delta^2 Z/Z$ to the connected two point function we find:

$$\begin{aligned} \left. \frac{1}{i^2} \delta^3 W[J] \right|_{J=0} &= \left[\frac{\delta^3 Z[J]}{i^3 Z[J]} - 3 \left(\frac{1}{i} \delta^2 W[J] + \frac{\delta Z[J]}{i Z[J]} \frac{\delta Z[J]}{i Z[J]} \right) \frac{\delta Z[J]}{i Z[J]} \right. \\ &\quad \left. + 2 \frac{\delta Z[J] \delta Z[J] \delta Z[J]}{i^3 Z[J]^3} \right]_{J=0} \end{aligned} \quad (5.13)$$

$$= \left[\frac{\delta^3 Z[J]}{i^3 Z[J]} - 3 \frac{1}{i} \delta^2 W[J] \frac{\delta Z[J]}{i Z[J]} - \frac{\delta Z[J] \delta Z[J] \delta Z[J]}{i^3 Z[J]^3} \right]_{J=0} \quad (5.14)$$

That is, the connected three-point function is the disconnected three point function less:

- 3 different connected two point functions with a tadpole. Remember: we're being loose with our variations, they technically should be three different spacetime points, distributing them properly would have given the three different combinations.
- 1 instance of three tadpoles: there's only one way to write three tadpoles with three different spacetime points.

Diagrammatically this looks like:

$$\text{Diagram (D)} - 3 \text{Diagram (C-C)} - \text{Diagram (C-C-C)} = \text{Diagram (C)} \quad (5.15)$$

5. Generating Functionals (scalar theory as an example)

Now, if this still isn't fully clear, we have the perfect example to check against. And it includes those bubbles we've only tangentially addressed. Let's look at our one-loop correction to the two-point function from the perspective of $W[J]$:

$$\begin{aligned} & \frac{1}{i} \delta_{y_2} \delta_{y_1} W[J] \\ &= \frac{\langle \Omega | T[\phi(y_2), \phi(y_1)] | \Omega \rangle}{\langle \Omega | \Omega \rangle} - \frac{\langle \Omega | \phi(y_1) | \Omega \rangle}{\langle \Omega | \Omega \rangle} \frac{\langle \Omega | \phi(y_2) | \Omega \rangle}{\langle \Omega | \Omega \rangle} \end{aligned} \quad (5.16)$$

$$\begin{aligned} &= \frac{\langle 0 | T[\phi(y_2), \phi(y_1)] | 0 \rangle + \left(\frac{ig}{3!}\right)^2 \langle 0 | T[\phi(y_2), \phi(y_1), \phi(z_2)^3, \phi(z_1)^3] | 0 \rangle + \dots}{\langle 0 | 0 \rangle + \left(\frac{ig}{3!}\right)^2 \langle 0 | T[\phi(z_2)^3, \phi(z_1)^3] | 0 \rangle + \dots} \\ &\quad - \left(\frac{ig}{3!}\right)^2 \frac{\langle 0 | T[\phi(y_1), \phi(z_1)^3] | 0 \rangle \langle 0 | T[\phi(y_2), \phi(z_2)^3] | 0 \rangle}{\left(\langle 0 | 0 \rangle + \left(\frac{ig}{3!}\right)^2 \langle 0 | T[\phi(z_2)^3, \phi(z_1)^3] | 0 \rangle + \dots\right)^2} + \dots \end{aligned} \quad (5.17)$$

The first equivalence is just Eq. 5.10, the second comes from expanding the numerators and denominators in our small g expansion. Notice that means we're perturbing around the *free vacuum* $|0\rangle$, instead of dealing with the interacting vacuum $|\Omega\rangle$. Next we need to expand properly to order g^2 , which gives:

$$\begin{aligned} & \frac{1}{i} \delta_{y_2} \delta_{y_1} W[J] \\ &= \frac{\langle 0 | T[\phi(y_2), \phi(y_1)] | 0 \rangle}{\langle 0 | 0 \rangle} + \left(\frac{ig}{3!}\right)^2 \frac{\langle 0 | T[\phi(y_2), \phi(y_1), \phi(z_2)^3, \phi(z_1)^3] | 0 \rangle}{\langle 0 | 0 \rangle} \\ &\quad - \left(\frac{ig}{3!}\right)^2 \frac{\langle 0 | T[\phi(y_2), \phi(y_1)] | 0 \rangle \langle 0 | T[\phi(z_2)^3, \phi(z_1)^3] | 0 \rangle}{\langle 0 | 0 \rangle \langle 0 | 0 \rangle} \\ &\quad - \left(\frac{ig}{3!}\right)^2 \frac{\langle 0 | T[\phi(y_1), \phi(z_1)^3] | 0 \rangle \langle 0 | T[\phi(y_2), \phi(z_2)^3] | 0 \rangle}{\langle 0 | 0 \rangle \langle 0 | 0 \rangle} + \mathcal{O}(g^3) \end{aligned} \quad (5.18)$$

We've already evaluated these, we can write their sums diagrammatically as in Figure 5.1. The first line is the free field result, the next two lines come from our calculation of the order g^2 corrections to the two-point function, the next line cancels the two vacuum bubble diagrams and is the product of the vacuum calculation and the two-point function, while the final line cancels the two-tadpole diagram and is the product of two tadpole diagrams. This can be seen exactly, without diagrams, by plugging each of our respective calculations into Eq. 5.18.

$$\begin{aligned}
 & \text{---} \bigcirc \text{---} \\
 &= \text{---} \\
 &+ \frac{1}{8} \text{---} \bigcirc \text{---} \bigcirc \text{---} + \frac{1}{12} \text{---} \bigcirc \text{---} \bigcirc \text{---} \\
 &+ \frac{1}{2} \text{---} \bigcirc \text{---} + \frac{1}{2} \text{---} \bigcirc \text{---} + \frac{1}{4} \text{---} \bigcirc \text{---} \bigcirc \text{---} \\
 &- \left(\frac{1}{8} \text{---} \bigcirc \text{---} \bigcirc \text{---} + \frac{1}{12} \text{---} \bigcirc \text{---} \bigcirc \text{---} \right) \times \left(\text{---} \right) \\
 &- \left(\frac{1}{2} \text{---} \bigcirc \text{---} \right) \left(\frac{1}{2} \text{---} \bigcirc \text{---} \right).
 \end{aligned}$$

Figure 5.1: Demonstration of the cancellation of disconnected diagrams in $\delta^2 W[J]$ using the worked out example from ϕ^3 theory.

Now one last very important result: I promised that ugly infinite constant (N) would cancel out, and it does. For every time-ordered product in the numerators of Eq. 5.18 there is an accompanying factor of $N = \langle 0|0 \rangle^2$ in the denominator.

We’ve established that $W[J]$ generates the sum of all connected diagrams and in particular that n -variations applied to $W[J]$ (with appropriate factors of i) i.e.,

$$\frac{1}{i^{n-1}} \frac{\delta^n W[J]}{\delta J(x_1) \cdots \delta J(x_n)} \Big|_{J=0}, \quad (5.19)$$

gives the sum of all connected n -point diagrams.

Let’s take a moment to stress that this result is *exact*, we did not rely on perturbation theory to prove it. We just used it as a nice example afterward. This also doesn’t rely on “canonical quantization of fields³”, so no assumptions on the fields is made. Path integrals allow us to prove things that we otherwise may not be able to prove, and do so independent of perturbation theory (“nonperturbatively”).

²Before we were treating $\langle 0|0 \rangle$ as a functional of J so it may be more familiar to state $N = \langle 0|0 \rangle_{J=0}$, but now that we’ve defined $Z[J]$ we know the difference.

³i.e. the coefficients of the kinetic and mass terms in $\mathcal{L}_{\text{free}}$.

5.2 Generating functional for 1PI diagrams

Having seen that $W[J]$ generates all connected diagrams, we're interested in a generating functional that will isolate 1PI diagrams. We'll achieve this by making a Legendre transformation of $W[J]$. We start by defining:

$$\bar{\phi} \equiv \frac{\delta W[J]}{\delta J}. \quad (5.20)$$

That's effectively defining $\bar{\phi}$ to be the connected tadpole. The Legendre transformation we're going to make is the one that replaces J with $\bar{\phi}$. So instead of expanding around the vacuum with an external source, we're expanding around the actual quantum background field $\bar{\phi}$, i.e. we're sourcing our time-ordered products from something physical.

We refer to this generating functional as the “effective action,” and will turn out to generate 1PI diagrams. We define the effective action Γ to be:

$$\Gamma(\bar{\phi}) = W[J] - \int d^4x J \bar{\phi}. \quad (5.21)$$

Now we'll need some variations of Γ , for simplicity we're not going to label the spacetime point of each variation just like in our result for $\delta^3 W[J]$. Reinstating the spacetime argument is straightforward, but tedious, and our proof here is solid enough for our standards. As with $W[J]$, our results here are exact, they are independent of perturbation theory.

If we take a variation of Γ with respect to $\bar{\phi}$ we find,

$$\frac{\delta \Gamma}{\delta \bar{\phi}} = -J. \quad (5.22)$$

We can take the variation of $-J$ with respect to $\bar{\phi}$, and then relate that to variations of the generating functional of connected diagrams via Eq. 5.20:

$$-\frac{\delta J}{\delta \bar{\phi}} = \left[-\frac{\delta \bar{\phi}}{\delta J} \right]^{-1}, \quad (5.23)$$

$$= \left[-\frac{\delta^2 W}{\delta J^2} \right]^{-1}, \quad (5.24)$$

$$= [-iD_C]^{-1} = iD_C^{-1}. \quad (5.25)$$

In going from the second line to third we just look back to the double variation of $W[J]$ in Eqs. 5.10 and 5.11, and can see that D_C is just the connected two point function from our discussion of $W[J]$. From Eq. 5.22 we can immediately relate this to the second variation of Γ with respect to our new source $\bar{\phi}$:

$$\frac{1}{i} \frac{\delta^2 \Gamma}{\delta \bar{\phi}^2} = D_C^{-1}. \quad (5.26)$$

It's important to stress at this stage, that our definitions of $W[J]$ and $\Gamma[\bar{\phi}]$ and our manipulations of them are exact. We make no reliance of perturbation theory in manipulating our generating functionals. Multiplying Eq. 5.26 from the left and right by D_C gives the result,

$$\frac{1}{i} D_C \frac{\delta^2 \Gamma}{\delta \bar{\phi}^2} D_C = D_C, \quad (5.27)$$

which tells us that the exact connected two-point function is something dressed on either side by the exact connected two-point functions. The only thing that can be “left in the middle” is the 1PI two-point function. We can express this result diagrammatically as:

$$\text{---} \bigcirc_C \text{---} \text{1PI} \text{---} \bigcirc_C \text{---} = \text{---} \bigcirc_C \text{---} \quad (5.28)$$

Let's return to Eq. 5.25 for a moment. We can rewrite this as an operator identity by using the chain rule:

$$\frac{\delta}{\delta \bar{\phi}} = \underbrace{\frac{\delta J}{\delta \bar{\phi}} \frac{\delta}{\delta J}}_{\text{chain rule}} = \frac{1}{i} D_C^{-1} \frac{\delta}{\delta J}. \quad (5.29)$$

The benefit of this operator identity is that it elucidates how Γ generates 1PI diagrams. Remember variations with respect to J acting on $W[J]$ result in an external line on a connected diagram, this was the subject of the last section. Now we see variations of $\Gamma(\bar{\phi})$ with respect to $\bar{\phi}$ replaces an external exact connected two-point function with the usual external line.

We're going to need to go back and forth between two variations of $W[J]$ and two variations of $\Gamma[\bar{\phi}]$ quite a bit in the next two proofs, so let's take a moment to stress,

$$\frac{1}{i} \frac{\delta^2 W[J]}{\delta J^2} = D_C, \quad (5.30)$$

$$\frac{\delta^2 \Gamma[\bar{\phi}]}{\delta \bar{\phi}^2} = i D_C^{-1} = - \left(\frac{\delta^2 W[J]}{\delta J^2} \right)^{-1}, \quad (5.31)$$

5. Generating Functionals (scalar theory as an example)

funny terms like that last part with the inverse of the variation of $W[J]$ will make it hard to keep track of factors of i , so we need to be extra careful.

Now we're ready to push forward. Consider the 1PI three point function:

$$\frac{\delta^3 \Gamma}{\delta \bar{\phi}^3} = -\frac{\delta}{\delta \bar{\phi}} \left(\frac{\delta^2 W}{\delta J^2} \right)^{-1} \quad (5.32)$$

$$= -D_C^{-1} \frac{1}{i} \frac{\delta}{\delta J} \left(\frac{\delta^2 W}{\delta J^2} \right)^{-1}, \quad (5.33)$$

$$= \frac{1}{i} D_C^{-1} \left(\frac{\delta^3 W}{\delta J^3} \right) \Big/ \left(\frac{\delta^2 W}{\delta J^2} \right)^2. \quad (5.34)$$

In the first line we rewrite $\delta^2 \Gamma$ in terms of W using Eq. 5.31, in the next line we apply our new operator identity in Eq. 5.29. In the third line we take the variation with respect to J . Ugly as this expression may be, we already know what each of these variations of W are so we can rewrite this line as:

$$\frac{\delta^3 \Gamma}{\delta \bar{\phi}^3} = \frac{1}{i} D_C^{-1} (-i D_C^{-1})^2 \frac{\delta^3 W}{\delta J^3}, \quad (5.35)$$

$$= i D_C^{-3} \frac{\delta^3 W}{\delta J^3}. \quad (5.36)$$

It is certainly uncomfortable to divide by a functional, and perhaps more uncomfortable to deal with something that looks like a non-linear differential equation defined over a functional! Remember: physics makes it OK, if it's still uncomfortable, remember how it worked out in our perturbative expansion for connected diagrams. If that's still not enough you're ready to be a mathematical physicist!

Minor rearrangement of the last equation gives us:

$$\left(\frac{1}{i} \right)^2 \frac{\delta^3 W}{\delta J^3} = D_C^3 \left(i \frac{\delta^3 \Gamma}{\delta \bar{\phi}^3} \right) \quad (5.37)$$

The argument works the same as before: on the left we have the connected three-point function, stripping three connected two-point functions off that leaves only something 1PI in the middle ($i \delta^3 \Gamma$). That factor i is consistent: it's analogous to the i we get from expanding the interaction Lagrangian to

first order. Diagrammatically we can write this expression as:

$$\text{Diagram: A circle with three external lines} = \text{Diagram: A circle with one external line connected to a '1PI' circle, which is then connected to two circles labeled 'C'}$$
(5.38)

So far this was very clean. It does get a little more complicated from the four-point function onward. The good news is, the added complication clarifies the physics even more. We'll apply our operator identity one more time to obtain the fourth variation of Γ , starting from Eq. 5.36 and rewriting D_C in terms of variations of $W[J]$ gives:

$$\frac{\delta^4 \Gamma}{\delta \phi^4} = i \frac{\delta}{\delta \phi} \left[\frac{\delta^3 W}{\delta J^3} \left/ \left(\frac{1}{i^3} \frac{\delta^2 W}{\delta J^2} \right)^3 \right. \right], \quad (5.39)$$

$$= \frac{1}{i} D_C^{-1} \frac{\delta}{\delta J} \left[\frac{\delta^3 W}{\delta J^3} \left/ \left(\frac{\delta^2 W}{\delta J^2} \right)^3 \right. \right], \quad (5.40)$$

$$= \frac{1}{i} D_C^{-1} \left[\frac{\delta^4 W}{\delta J^4} \left/ \left(\frac{\delta^2 W}{\delta J^2} \right)^3 - 3 \left(\frac{\delta^3 W}{\delta J^3} \right)^2 \left/ \left(\frac{\delta^2 W}{\delta J^2} \right)^4 \right. \right]. \quad (5.41)$$

The first two lines directly parallel what we did in the case of $\delta^3 \Gamma$. In the last line, $\delta^3 W$ is squared because the chain rule in the second term requires we multiply by $\delta(\delta^2 W[J])$. Now we rewrite each case of $\delta^2 W$ in terms of D_C , taking great care with the factors of i :

$$\frac{\delta^4 \Gamma}{\delta \phi^4} = \frac{1}{i} D_C^{-1} \left[\frac{\delta^4 W}{\delta J^4} (i D_C)^{-3} - 3 \left(\frac{\delta^3 W}{\delta J^3} \right)^2 (i D_C)^{-4} \right], \quad (5.42)$$

$$= D_C^{-4} \frac{\delta^4 W}{\delta J^4} + 3 i D_C^{-5} \left(\frac{\delta^3 W}{\delta J^3} \right)^2 \quad (5.43)$$

We can already see where this is going, a four-point function can be constructed of interactions involving three fields or four. Solving for $\delta^4 W$, multiplying by $1/i^3$ to get the physical meaning correct, and rewriting $\delta^3 W$ in

5. Generating Functionals (scalar theory as an example)

terms of our previous result for $\delta^3\Gamma$ gives:

$$\frac{1}{i^3} \frac{\delta^4 W}{\delta J^4} = D_C^4 \left(i \frac{\delta^4 \Gamma}{\delta \bar{\phi}^4} \right) + 3 D_C^{-1} \left(i^3 D^3 \frac{\delta \Gamma^3}{\delta \bar{\phi}^3} \right)^2, \quad (5.44)$$

$$= D_C^4 \left(i \frac{\delta^4 \Gamma}{\delta \bar{\phi}^4} \right) + 3 D_C^4 \left(i \frac{\delta^3 \Gamma}{\delta \bar{\phi}^3} \right) D_C \left(i \frac{\delta^3 \Gamma}{\delta \bar{\phi}^3} \right). \quad (5.45)$$

So the connected four-point function is the sum of:

1. The 1PI four-point function with the four external lines dressed by connected two-point functions.
2. Three contributions from two insertions of the 1PI three-point function with the four external lines and one internal line dressed by connected-two point functions.

Diagrammatically this looks like:

The diagrammatic equation (5.46) illustrates the decomposition of a four-point function into 1PI diagrams with external lines dressed by two-point functions. On the left, a four-point function is represented by a central circle labeled 'C' with four external lines. This is equal to the sum of two terms. The first term is a 1PI four-point function (a central circle labeled '1PI') with four external lines, each of which is dressed by a connected two-point function (a circle labeled 'C'). The second term is a 1PI three-point function (a central circle labeled '1PI') with three external lines, each dressed by a connected two-point function (a circle labeled 'C'), and one internal line that is connected to another 1PI three-point function (a central circle labeled '1PI') with three external lines, each dressed by a connected two-point function (a circle labeled 'C').

$$(5.46)$$

As we said, these $i\delta^{(n)}\Gamma$ represent n -field one-particle irreducible diagrams. This will be particularly useful for us as, at “tree level” (i.e. when there are no loops in the diagram), $i\delta^{(n)}\Gamma$ just corresponds to the Feynman rule for a given interaction. We’ll get into Feynman rules later when we look in depth at calculations of physical processes. We’ll quickly take a look at what it means to have complex scalars and/or multiple scalars, then move on to applying our path integral expertise to fermion and vector fields.

Chapter 6

Path Integrals for Dirac fields

Fermions are going to require some extra work. From statistical mechanics or quantum mechanics we know that identical fermions obey Fermi statistics. That is, the wavefunction for two identical fermions swaps sign under exchange of the two fermions:

$$\psi(x_1, x_2) = -\psi(x_2, x_1). \quad (6.1)$$

We need to impose this condition on our fermionic fields as well. We'll achieve this by making our Dirac fields **Grassmann variables**. After developing the calculus of Grassman variables we'll be able to quantize in a manner that parallels our treatment of scalars. The Grassman nature of fermions leads to some interesting consequences we'll outline, but won't have time to discuss in detail. They will also, surprisingly, play a role in how we quantize vector fields.

6.1 Grassmann variables

We define Grassmann variables as variables that anticommute under multiplication, commute additively, include a 0 element, and can be multiplied by complex scalars $a \in \mathbb{C}$:

$$\theta_i \theta_j = -\theta_j \theta_i, \quad \theta_i + \theta_j = \theta_j + \theta_i \quad (6.2)$$

$$\theta_i + 0 = \theta_i, \quad a\theta = \theta a \quad (6.3)$$

Our Grassmann variables anticommute with all other Grassmann variables *from the same set*. We can introduce a separate set of Grassman variables

η_i that anticommute with each other, but commute with the θ s:

$$\eta_i \theta_j = \theta_j \eta_i, \quad \eta_i \eta_j = -\eta_j \eta_i, \quad \theta_i \theta_j = -\theta_j \theta_i. \quad (6.4)$$

This will be relevant soon – most of our models will have more than one type of fermion (flavors), and different fermions should commute. Each type of fermionic field will be represented by its own set of Grassmann variables.

The set of all such objects and their linear combinations forms the Grassmann algebra, an Algebra over $\mathbb{C}$ generated by anticommuting elements. For a single Grassmann variable, θ , the most general element of the Algebra can be written as,

$$g = a + b\theta, \quad a, b \in \mathbb{C}, \quad (6.5)$$

this is simply because $\theta^2 = -\theta^2 = 0$. For two species, θ_1 and θ_2 , we can write the most general element as,

$$g = a + b\theta_1 + c\theta_2 + d\theta_1\theta_2, \quad (6.6)$$

and so on.

Now consider the product of four Grassmann variables, we can see that any pair of Grassmann variables commutes with any other pair:

$$(\theta_1\theta_2)(\theta_3\theta_4) = +\theta_3(\theta_1\theta_2)\theta_4 = (\theta_3\theta_4)(\theta_1\theta_2). \quad (6.7)$$

The products of two Grassmann variables forms the “bosonic subalgebra” for a given set of Grassmann variables, that is they behave like bosons. The product of two Grassmann variables also commutes with any single Grassmann variable as well. Indeed any product of an even number of Grassmann variables is bosonic. In the same vein, an odd number forms another fermionic variable, these define the fermionic subalgebra.

Soon we’re going to upgrade these variables to a field over spacetime, $\theta \rightarrow \psi(x)$. We’re going to need to understand integration over Grassmann variables so we can treat them in our path integral. Consider the case of a single Grassmann variable again, the most general integral is given by:

$$\int d\theta (a + b\theta) = a \int d\theta + b \int d\theta \theta. \quad (6.8)$$

Just like an integral over functions maps a function to the real or complex numbers, we want our integral to map Grassmann variables to $\mathbb{C}$. We’ll require the following (i.e. these are axiomatic):

$$\int d\theta 1 = 0, \quad \int d\theta \theta = 1. \quad (6.9)$$

So for Eq. 6.8 we get the very simple result:

$$\int d\theta (a + b\theta) = b \quad (6.10)$$

Then if we use the usual definition for derivatives we find,

$$\frac{d}{d\theta}(a + b\theta) = b, \quad (6.11)$$

which is really interesting: differentiation and integration are effectively the same operation. We'll retain our understanding of what the two operations mean physically (which is definitely different), but in the case of Grassmann variables both operations just pick out the coefficient of θ .

If we extend this to many Grassmann variables we have,

$$\int d\theta_1 \cdots d\theta_n X = \frac{\partial}{\partial\theta_1} \cdots \frac{\partial}{\partial\theta_n} X, \quad (6.12)$$

$$\int d\theta_1 \cdots d\theta_n \theta_n \cdots \theta_1 = 1, \quad (6.13)$$

but if the integral doesn't carry a θ_i for every $d\theta_i$ it will vanish.

As order matters because of the anticommuting nature of the Grassmann variables, we're going to need to choose a specific ordering for integration:

$$\int d\theta_1 d\theta_2 \theta_2 \theta_1 = - \int d\theta_1 d\theta_2 \theta_1 \theta_2 = 1. \quad (6.14)$$

We'll integrate "from inside out." That is, the integrate from the right-most differential ($d\theta_n$ in Eq. 6.13) to the left-most.

For functions integrated over the full real line, we have what's called shift symmetry. That is, it doesn't matter what we call the origin if we integrate over all reals. For functions this looks like:

$$\int_{-\infty}^{\infty} dx f(x) = \int_{-\infty}^{\infty} dx f(x + a). \quad (6.15)$$

This very naturally carries over to the Grassmann variables. If we redefine $\theta \rightarrow \theta + X$ where X is a commuting constant, we have

$$\int d\theta(a + b\theta) = \int d\theta[a + b(\theta + X)], \quad (6.16)$$

which follows because the X integral vanishes as it doesn't contain a factor of θ . This shift invariance is going to be relevant as it's necessary for us to "complete the square" and write our free fermionic path integral (in the presence of sources) as a Gaussian.

One last thing to discuss is how the transpose behaves. Consider some matrices A , B , C , the transpose of their product is the reversal of rows and columns:

$$A_{ij}B_{jk}C_{kl} = (ABC)_{il} \quad (6.17)$$

$$[(ABC)_{il}]^T = (ABC)_{li} = A_{lk}B_{kj}C_{ji} \quad (6.18)$$

For normal matrices we usually rearrange the last term and write,

$$[(ABC)_{il}]^T = C_{ij}^T B_{jk}^T A_{kl}^T, \quad (6.19)$$

and say something like "the transpose reverses the order and separately transposes each matrix."

For anticommuting variables, we need to stick with the arguably more fundamental definition of swapping rows and columns given in Eq. 6.18. If A and B are instead Grassmann, we use Eq. 6.18 to write:

$$[(AB)_{ik}]^T = (AB)_{ki}, \quad (6.20)$$

$$= A_{kj}B_{ji}, \quad (6.21)$$

$$= -B_{ij}^T A_{jk}^T. \quad (6.22)$$

So we need to be careful of the number of times we commute Grassmann variables in rearranging the transpose so that we keep track of the proper sign. Unfortunately and confusingly, we'll need to distinguish between this transpose and that contained in most hermitian conjugates we'll apply. When we work with *operators in a Lagrangian* the Hermitian conjugate acts on the operator itself, ignoring the Grassmann nature of the Dirac fields. In this case the transpose is not a Grassmann transpose. When we perform path integral manipulations, or derive identities relating to manipulating the Grassmann bilinears we'll need the signs coming from transposes.

6.2 Gaussian integrals over Grassmann variables and the propagator

With our new understanding of integrals over Grassmann variables we're in a position to move toward the free Dirac field theory. First let's look at how Gaussian integrals work over Grassmann variables. Starting with the simplest case we have:

$$\int d\theta_1 d\theta_2 e^{-\theta_1 A \theta_2} = \int d\theta_1 d\theta_2 (1 - \theta_1 A \theta_2) = A. \quad (6.23)$$

In the first equivalence we Taylor expanded, notice that this result is *exact* because any higher terms the Taylor expansion would give vanish due to the Grassmann variables identity $\theta^2 = 0$. Also the minus sign goes away because of our convention: integrate from the inside out.

This is a good start, but for Dirac fields we'll need analogues of $\bar{\psi}$ and ψ , so let's rewrite this integral as,

$$\int d\bar{\theta} d\theta e^{-\bar{\theta} A \theta} = A. \quad (6.24)$$

For two θ s and $\bar{\theta}$ s we have:

$$\int d\bar{\theta}_1 d\bar{\theta}_2 d\theta_2 d\theta_1 e^{-\bar{\theta}_i A_{ij} \theta_j} \quad (6.25)$$

$$= \int d\bar{\theta}_1 d\bar{\theta}_2 d\theta_2 d\theta_1 \left(1 - \bar{\theta}_i A_{ij} \theta_j + \frac{1}{2} \bar{\theta}_i A_{ij} \theta_j \bar{\theta}_k A_{kl} \theta_l \right), \quad (6.26)$$

$$= 0 + \int d\bar{\theta}_1 d\bar{\theta}_2 d\theta_2 d\theta_1 \left(\frac{1}{2} \bar{\theta}_1 A_{11} \theta_1 \bar{\theta}_2 A_{22} \theta_2 + \frac{1}{2} \bar{\theta}_1 A_{12} \theta_2 \bar{\theta}_2 A_{21} \theta_1 + \dots \right), \quad (6.27)$$

$$= \frac{1}{2} [(-1)^{1+2+1} A_{11} A_{22} + (-1)^{3+1+1} A_{12} A_{21} + \dots] \quad (6.28)$$

$$= \frac{1}{2} [A_{11} A_{22} - A_{12} A_{21} + A_{11} A_{22} - A_{21} A_{12}], \quad (6.29)$$

$$= \det A_{ij}. \quad (6.30)$$

In the third equivalence, the $(-1)^{n+m+l}$ indicates how many factors of -1 we get from our convention on integrating from the innermost differential to

the outermost. For $(-1)^{n+m+l}$, n indicates how many factors for moving θ_1 all the way to the left, then m for θ_2 , then l for $\bar{\theta}_2$. Notice the choice of ordering for the $d\bar{\theta}$ s counts up while for $d\theta$ counts down, this is on purpose. It gives us a $+1$ for the determinant regardless of how many θ s we consider.

With a great talent for combinatorics we could show this determinant holds for n θ s as well:

$$\int d\bar{\theta}_1 \cdots d\bar{\theta}_n d\theta_n \cdots d\theta_1 e^{-\bar{\theta}_i A_{ij} \theta_j} \quad (6.31)$$

$$= \int d\bar{\theta}_1 \cdots d\bar{\theta}_n d\theta_n \cdots d\theta_1 \left(1 - \theta_i A_{ij} \theta_j + \cdots + \frac{(-1)^n}{n!} (\bar{\theta}_i A_{ij} \theta_j)^n \right), \quad (6.32)$$

$$= \sum_{\text{pairs}} (-1)^{\text{something}} (A_{ij})^n, \quad (6.33)$$

$$= \det A_{ij}. \quad (6.34)$$

The important result is: for scalars we had $(\det A)^{-1/2}$, for fermions we have $\det A$. This is a direct result of using Grassmann variables and has huge repercussions on the physical validity of gauge theories. The determinant structure plays a role in anomalies in gauge theories which affect their consistency. We won't discuss this for lack of time.

This is exactly what we need to define path integrals for Dirac fields. The only thing left is to introduce sources, separately for $\bar{\psi}$ and ψ , so that we can obtain the time ordered products of fermionic fields we'll need to calculate observables.

We introduce the external currents in such a manner that the action remains hermitian, the next step is to rewrite the argument of the exponential as a Gaussian (i.e. complete the square), then perform the integration:

$$\begin{aligned} & \int d\bar{\theta}_1 \cdots d\bar{\theta}_n d\theta_n \cdots d\theta_1 e^{-\bar{\theta}_i A_{ij} \theta_j + \bar{\eta}_i \theta_i + \bar{\theta}_i \eta_i} \\ &= e^{\bar{\eta}_i A_{ij}^{-1} \eta_j} \int d\bar{\theta}_1 \cdots d\bar{\theta}_n d\theta_n \cdots d\theta_1 e^{-(\bar{\theta}_i - \bar{\eta}_j A_{ji}^{-1}) A_{ik} (\theta_k - A_{kl}^{-1} \eta_l)} \end{aligned} \quad (6.35)$$

$$= (\det A) e^{\bar{\eta}_i A_{ij}^{-1} \eta_j} \quad (6.36)$$

This is exactly what we're looking for. It looks like a direct analogue of our solution for the path integral of a free scalar, where J sourced $A^{-1} = \Delta$ from the exponential – the only difference is in the determinant.

6. Path Integrals for Dirac fields

When we start taking variations of our Grassmann path integral, we'll also need to know how to keep track of signs. Consider:

$$\begin{aligned} & \frac{\delta}{\delta\theta_{i_1}} e^{\bar{\theta}_{j_1} A_{j_1,j_2} \theta_{j_2}} \\ &= \frac{\delta}{\delta\theta_{i_1}} \left[1 + \bar{\theta}_{j_1} A_{j_1,j_2} \theta_{j_2} + \frac{1}{2} \bar{\theta}_{j_1} A_{j_1,j_2} \theta_{j_2} \bar{\theta}_{j_3} A_{j_3,j_4} \theta_{j_4} \right] \end{aligned} \quad (6.37)$$

$$= (-1) \bar{\theta}_{j_1} A_{j_1,i_1} + (-1) \frac{1}{2} \bar{\theta}_{j_1} A_{j_1,i_1} \bar{\theta}_{j_3} A_{j_3,j_4} \theta_{j_4} + (-1)^3 \frac{1}{2} \bar{\theta}_{j_1} A_{j_1,j_2} \theta_{j_2} \bar{\theta}_{j_3} A_{j_3,i_1} \quad (6.38)$$

The $(-1)^n$ indicate how many times we commuted the variation through a θ or $\bar{\theta}$. Pulling out that factor of (-1) and separately pulling out $\bar{\theta}_j A_{ji}$ and relabeling dummy indices we find:

$$\begin{aligned} & \frac{\delta}{\delta\theta_{i_1}} e^{\bar{\theta}_{j_1} A_{j_1,j_2} \theta_{j_2}} \\ &= \frac{\delta}{\delta\theta_{i_1}} \left[1 + \bar{\theta}_{j_1} A_{j_1,j_2} \theta_{j_2} + \frac{1}{2} \bar{\theta}_{j_1} A_{j_1,j_2} \theta_{j_2} \bar{\theta}_{j_3} A_{j_3,j_4} \theta_{j_4} \right] \end{aligned} \quad (6.39)$$

$$= (-1) \left[1 + \bar{\theta}_{j_1} A_{j_1,j_2} \theta_{j_2} + \dots \right] \bar{\theta}_{j_3} A_{j_3,i_1} \quad (6.40)$$

$$= (-1) e^{\bar{\theta}_{j_1} A_{j_1,j_2} \theta_{j_2}} \bar{\theta}_{j_3} A_{j_3,i_1} \quad (6.41)$$

After pulling out the $\bar{\theta}A$ term, we were left with exactly the form of the exponential on the left hand side. This required some more commuting, but since the argument of the exponential is even in Grassmann variables we have $(-1)^{2n}$ and no extra signs. When we take another variation, we can just think of things as following the usual product rule, except we need to remember the anticommutivity and that the Gaussian is effectively bosonic:

$$\begin{aligned} & \frac{\delta}{\delta\theta_{i_2}} \frac{\delta}{\delta\theta_{i_1}} e^{\bar{\theta}_{j_1} A_{j_1,j_2} \theta_{j_2}} \\ &= (-1) e^{\bar{\theta}_{j_1} A_{j_1,j_2} \theta_{j_2}} \left[\bar{\theta}_{j_3} A_{j_3,i_1} + (-1) \bar{\theta}_{j_4} A_{j_4,i_2} \bar{\theta}_{j_3} A_{j_3,i_1} \right] \end{aligned} \quad (6.42)$$

The relative sign comes from commuting the variation through an uneven number of $\bar{\theta}$ s in the exponential. If we had taken a variation with respect to $\bar{\theta}$ instead, we wouldn't have that sign. We are going to need to be pretty careful with this anticommutivity to get all the signs right.

Now we just need to upgrade our θ s to fields over spacetime, $\psi(x)$, and the discrete index i to a continuous index, x . The latter takes us from $d\theta \rightarrow \mathcal{D}\psi(x)$ as well as changing our discrete sum in the exponential to an integral over the continuous spacetime argument:

$$Z[\bar{\eta}, \eta] = \int \mathcal{D}\bar{\psi}(x) \mathcal{D}\psi(x) e^{\int d^4x [\bar{\psi}(x) \underbrace{i(a_1 \gamma^\mu \partial_\mu + a_2)}_{\equiv -A} \psi(x) + i\bar{\eta}(x)\psi(x) + i\bar{\psi}(x)\eta(x)]}, \quad (6.43)$$

$$= N \exp \left[\int d^4x d^4y \bar{\eta}(y) A^{-1} \eta(x) \right]. \quad (6.44)$$

The choice of A is such that the exponential has a minus sign as required by Eq. 6.36. The double integral over dx and dy simply comes from rewriting,

$$\int d^4x \bar{\psi}(x) \psi(x) = \bar{\psi}(x) \psi(y) \delta^4(x - y), \quad (6.45)$$

which reflects the number of discrete indices in Eq. 6.35.

This is exactly what we need to handle free Dirac theory, and then extend to interactions with exactly the same tools we developed for scalars (plus anticommutivity). For this reason we immediately named the path integral $Z[J]$, it is the generator of disconnected diagrams. The infinite constant N is different from that coming from the scalar path integral, especially in terms of its functional determinant structure, but in the last section we saw precisely how it drops out of observables so we won't worry about it or giving it a unique name.

Let's obtain the propagator and fix a_1 and a_2 , to do so we need the time-ordered product of two fields:

$$\langle 0 | T[\psi(y_2), \bar{\psi}(y_1)] | 0 \rangle = i \frac{\delta}{\delta \bar{\eta}(y_2)} \frac{1}{i} \frac{\delta}{\delta \eta(y_1)} Z[\bar{\eta}, \eta] \Big|_{\bar{\eta}, \eta \rightarrow 0} \quad (6.46)$$

Notice, we take a variation with respect to η to source $\bar{\psi}$ and with respect to $\bar{\eta}$ to source ψ ! Also notice the convention that the barred η variation is accompanied by a factor of i instead of $1/i$, this convention reflects that $\bar{\eta} = \eta^\dagger \beta$, i.e. that $\bar{\eta}$ is up to a factor of β the hermitian conjugate of ψ . This is exactly what we would have done if we had gone through the quantization of the complex scalar field, see Section 4.10.

Taking the variations we find:

$$\begin{aligned}
 & \langle 0 | T[\psi(y_2), \bar{\psi}(y_1)] | 0 \rangle \\
 &= (-1) \frac{\delta}{\delta \bar{\eta}(y_2)} Z[\bar{\eta}, \eta] \left[\int d^4 x_1 d^4 x_2 \bar{\eta}(x_1) A^{-1}(x_1, x_2) \delta^4(x_2 - y_1) \right] \Big|_{\bar{\eta}, \eta \rightarrow 0}
 \end{aligned} \tag{6.47}$$

$$= -Z[\bar{\eta}, \eta] \left[A_{y_2, y_1}^{-1} + A_{y_2, x_2}^{-1} \eta_{x_2} \bar{\eta}_{x_1} A_{x_1, y_1}^{-1} \right] \Big|_{\bar{\eta}, \eta \rightarrow 0} \tag{6.48}$$

$$= -N A^{-1}(y_2, y_1) \tag{6.49}$$

The explicit (-1) in the first equivalence comes from commuting $\delta\eta$ through $\bar{\eta}$ in the exponential. In the second term of the second line, no extra sign is introduced. This term comes from pulling down a factor from the exponential, and $\bar{\eta}$ is the left-most term in the exponential so no commuting of η s is required. At zero source we obtain the time ordered product and that it is $-A^{-1}$, meaning $-A^{-1}$ is the propagator.

We can solve the Green's function equation, $A(x, y)A^{-1}(x, y) = \delta^4(x - y)$, by inspection (i.e. guess) and find our propagator:

$$\Delta(x - y) \equiv -A^{-1}(x - y) \tag{6.50}$$

$$= -[i(a_1 \gamma^\mu \partial_\mu + a_2)]^{-1} \tag{6.51}$$

$$= - \int \frac{d^4 p}{(2\pi)^4} \frac{1}{a_1 \gamma^\mu p_\mu + i a_2} e^{-ip \cdot (x-y)} \tag{6.52}$$

This is bizarre with a matrix in the denominator. To make this a bit nicer, we'll multiply by 1 in the form of the ratio of matrices. Before that, let's work out an important identity:

$$\gamma^\mu p_\mu \gamma^\nu p_\nu = p_\mu p_\nu \frac{1}{2} [\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu], \tag{6.53}$$

$$= p_\mu p_\nu \frac{1}{2} 2\eta^{\mu\nu}, \tag{6.54}$$

$$= p^2. \tag{6.55}$$

The first equivalence makes us of the fact $p^\mu p^\nu$ is symmetric under exchange of the Lorentz indices and the second employs the Clifford Algebra (Eq. 2.80).

So we'll multiply our propagator by 1 in the following way:

$$\Delta(x-y) = - \int \frac{d^4p}{(2\pi)^4} \frac{a_1 \gamma^\nu p_\nu - i a_2}{a_1 \gamma^\nu p_\nu - i a_2} \frac{1}{a_1 \gamma^\mu p_\mu + i a_2} e^{-ip \cdot (x-y)} \quad (6.56)$$

$$= - \int \frac{d^4p}{(2\pi)^4} \frac{(a_1 \gamma^\nu p_\nu - i a_2)}{a_1^2 p^2 + a_2^2} e^{-ip \cdot (x-y)} \quad (6.57)$$

The sign of the exponential differs from the scalar case by a sign, this is arbitrary and is a convention we follow due to its prevalence in other texts that use canonical quantization. It has no physical consequences.

Again, if we want this propagator to correspond to physical scattering, we require that the pole corresponds to $p^2 = m^2$ giving the condition:

$$a_2 = \pm i m a_1. \quad (6.58)$$

Imposing this condition on the propagator gives:

$$\Delta(x-y) = - \int \frac{d^4p}{(2\pi)^4} \frac{\gamma^\nu p_\nu \pm m}{a_1(p^2 - m^2)} e^{-ip \cdot (x-y)} \quad (6.59)$$

This is a matrix equation now, so we don't have the ease of just requiring the residue is i , but instead have requirements on the matrix residue at the pole. This is achieved by requiring that the residue projects onto the positive-energy subspace (i.e. the spectral weight for excitations is positive). We won't discuss this in these notes, it can be understood from studying classical solutions to Euler Lagrange equations resulting from the Lagrangian. This requires that the numerator be $i(\gamma^\nu p_\nu + m)$. This fixes the sign $a_2 = +i m a_1$ and requires $a_1 = i$ giving for our propagator and associated free Lagrangian:

$$\Delta(x-y) = \int \frac{d^4p}{(2\pi)^4} \frac{i(\gamma^\nu p_\nu + m)}{p^2 - m^2} e^{-ip \cdot (x-y)} \quad (6.60)$$

$$\mathcal{L} = i \bar{\psi} \gamma^\mu \partial_\mu \psi - m \bar{\psi} \psi \quad (6.61)$$

Remember $\Delta = -A^{-1}$ so our generating functional for free Dirac fields is given by,

$$Z[\bar{\eta}_a, \eta_b] = N \exp \left[- \int d^4x d^4y \bar{\eta}_a(y) \Delta_{ab} \eta_b(x) \right], \quad (6.62)$$

where we've taken care to restore the Dirac indices, which are the indices e.g. on the gamma matrix in Eq. 6.60.

6.3 Time ordered product of four fermions, free theory

When we get to the Feynman rules, we'll have two new rules for fermions which come from their Grassmann nature. To see this let's first consider the time-ordered product of four fermions in the free theory. We'll extend the shorthand defined earlier to include our new sources and now Δ is the Dirac fermion propagator. We'll also need to start keeping track of the Dirac indices, that is the indices that accompany things like the γ matrices. For this purpose we'll write $\Delta(x-y) \rightarrow \Delta^{ab}(x-y) = \Delta_{x,y}^{a,b}$, and our variations will come with Dirac indices a, b , etc. Unlike spacetime indices Δ *is not symmetric under exchange of Dirac indices*. Generally we'll use lowered indices for Dirac indices, right now we're just raising them artificially on the propagator to help keep our expressions page sized.

The time ordered product is obtained by taking four variations on our free Dirac path integral:

$$\begin{aligned} \langle 0 | T[\psi(y_4)_d, \bar{\psi}(y_3)_c, \psi(y_2)_b, \bar{\psi}(y_1)_a] | 0 \rangle \\ = i \frac{\delta}{\delta \bar{\eta}_d(y_4)} \frac{1}{i} \frac{\delta}{\delta \eta_c(y_3)} i \frac{\delta}{\delta \bar{\eta}_b(y_2)} \frac{1}{i} \frac{\delta}{\delta \eta_a(y_1)} Z[\bar{\eta}, \eta] \Big|_{\bar{\eta}, \eta \rightarrow 0} \end{aligned} \quad (6.63)$$

$$= \frac{\delta}{\delta \bar{\eta}(y_4)} \frac{\delta}{\delta \eta(y_3)} \frac{\delta}{\delta \bar{\eta}(y_2)} Z[\bar{\eta}, \eta] \left[(-1)^2 \bar{\eta}_{x_1}^{a'} \Delta_{x_1, y_1}^{a', a} \right] \Big|_{\bar{\eta}, \eta \rightarrow 0} \quad (6.64)$$

Notice that, when the variation brings down a factor of $\Delta_{x_1, y_1}^{a', a}$ we have a sum over the contracted dummy index a' that was also pulled down from the exponential. All of the shorthand notation we agreed on below Eq. 4.47 is still in effect, that is we use the symmetry of Δ in its spacetime arguments, immediately integrate over delta functions, and integrals over spacetime are implied by repeated continuous indices (like x_1 above). Notice we have a minus sign coming from the anticommutation of the Dirac fields as well as a minus from the argument of the exponential, we need to be extra careful with minus signs from now on! We'll write each minus sign explicitly as (-1) , whether it comes from the exponential or anticommutation, for this example.

Continuing to take variations gives:

$$\begin{aligned} \langle 0 | T[\psi(y_4)_d, \bar{\psi}(y_3)_c, \psi(y_2)_b, \bar{\psi}(y_1)_a] | 0 \rangle \\ = \frac{\delta}{\delta \bar{\eta}(y_4)} \frac{\delta}{\delta \eta(y_3)} Z[\bar{\eta}, \eta] \left[\Delta_{y_2, y_1}^{b, a} + (-1) \Delta_{y_2, x_2}^{b, b'} \eta_{x_2}^{b'} \bar{\eta}_{x_1}^{a'} \Delta_{x_1, y_1}^{a', a} \right] \Bigg|_{\bar{\eta}, \eta \rightarrow 0} \end{aligned} \quad (6.65)$$

$$= \frac{\delta}{\delta \bar{\eta}(y_4)} Z[\bar{\eta}, \eta] \left[(-1)^2 \bar{\eta}_{x_1}^{c'} \Delta_{x_1, y_3}^{c', c} \Delta_{y_2, y_1}^{b, a} - \Delta_{y_2, y_3}^{b, c} \bar{\eta}_{x_1}^{a'} \Delta_{x_1, y_1}^{a', a} + 0_{\bar{\eta}, \eta \rightarrow 0} \right] \Bigg|_{\bar{\eta}, \eta \rightarrow 0} \quad (6.66)$$

$$= N \left(\Delta_{y_1, y_2}^{b, a} \Delta_{y_3, y_4}^{d, c} - \Delta_{y_1, y_4}^{d, a} \Delta_{y_2, y_3}^{b, c} \right) \quad (6.67)$$

For the time ordered product of four free scalars we instead had:

$$\langle 0 | T[\phi(y_4), \phi(y_3), \phi(y_2), \phi(y_1)] | 0 \rangle, \quad (6.68)$$

$$= -N \left(\Delta_{y_1, y_2} \Delta_{y_3, y_4} + \Delta_{y_1, y_3} \Delta_{y_2, y_4} + \Delta_{y_1, y_4} \Delta_{y_2, y_3} \right) \quad (6.69)$$

The relative minus sign in Eq. 6.67 is new, it is a direct consequence of the nature of fermions following Fermi statistics. We also have one fewer term, this is because now we have $\bar{\psi}$ and ψ , which has effectively removed the pairing (13)(24): there is no propagator that connects a $\bar{\psi}$ with a $\bar{\psi}$, this would violate Lorentz invariance. This also would have been the case had we studied the complex scalar field in more detail. This relative minus sign will need keeping track of, and will be incorporated into the Feynman rules for Fermions we'll discuss later.

We also now have a matrix equation, the Dirac propagator contains γ_{ab}^μ . That matrix isn't symmetric, so we can't simply reverse the order of (a, b) as we can with the spacetime arguments. *Now order matters.* Diagrammatically we can write Eq. 6.67 incorporating this order in the form of arrows as:

$$\langle 0 | T[\psi, \bar{\psi}, \psi, \bar{\psi}] | 0 \rangle = \begin{array}{c} y_3, c \longrightarrow y_4, d \\ y_1, a \longrightarrow y_2, b \end{array} - \begin{array}{c} y_3, c \longrightarrow y_4, d \\ y_1, a \longrightarrow y_2, b \end{array} \quad (6.70)$$

Notice the arrow points from barred to unbarred field index. This corresponds to our ordering e.g. $\bar{\eta}_b \Delta_{ba} \eta_a$, remember: $\bar{\eta}$ sources ψ , and η sources $\bar{\psi}$.

In QFT and particle physics there are always many different conventions. Our convention for Δ corresponds to the ordering apparent in the gamma matrix's indices when we write out the propagator as in Eq. 6.60. An alternate option is to write the propagator $S_{ab} = \Delta_{ba}$, in this case the ordering of indices corresponds to that in the time-ordered product in Eq. 6.46.

6.4 Vacuum bubbles from a four-fermion operator

We have one more minus sign to discover: we want to show that Dirac indices closed by a loop (a contraction over Dirac indices that forms a trace) result in a relative minus sign. We can do this in a straightforward manner without too much effort by introducing the interaction Lagrangian,

$$\mathcal{L}_I = c (\bar{\psi}_a(x) \psi_a(x)) (\bar{\psi}_b(x) \psi_b(x)) , \quad (6.71)$$

where a and b are Dirac indices we've written explicitly. With the interaction we can calculate the leading correction to the vacuum. We'll assume c is a small number and employ perturbation theory.

For fermions just as for scalars, we rewrite the interaction Lagrangian first as an exponential in front of the Free Lagrangian¹. We then expand in this exponential in the small coupling, and finally rewrite the explicit fields as variations acting on the free Lagrangian (see Eq. 4.44 and the subsequent discussion). The quantity we want to calculate is the order c term in:

$$\langle \Omega | \Omega \rangle = \langle 0 | 0 \rangle + i c \int d^4 z \left(i \frac{\delta}{\delta \bar{\eta}_a(z)} \frac{1}{i} \frac{\delta}{\delta \eta_a(z)} \right) \left(i \frac{\delta}{\delta \bar{\eta}_b(z)} \frac{1}{i} \frac{\delta}{\delta \eta_b(z)} \right) Z[\bar{\eta}, \eta] \Big|_{\bar{\eta}, \eta \rightarrow 0} + \dots \quad (6.72)$$

Fortunately, we've already essentially done this. We can take our result in Eq. 6.67, set $y_i \rightarrow z$, and take $b \rightarrow a$ then $c, d \rightarrow b$. This gives us:

$$\langle \Omega | \Omega \rangle = \langle 0 | 0 \rangle + i c N (\Delta_{z,z}^{a,a} \Delta_{z,z}^{b,b} - \Delta_{z,z}^{b,a} \Delta_{z,z}^{a,b}) + \dots \quad (6.73)$$

¹Because of Lorentz invariance our fermions always come in pairs, so the free and interaction Lagrangian are "bosonic" in the sense that they belong to the bosonic subalgebra of the Grassmann algebra.

Diagrammatically we can write this as:

$$\langle \Omega | \Omega \rangle = \langle 0 | 0 \rangle + \text{[Diagram 1]} - \text{[Diagram 2]} + \mathcal{O}(c^2) \quad (6.74)$$

In both diagrams the interaction takes place at the same spacetime point z indicated by the red vertex. In the left diagram there are two closed Dirac index loops ($\Delta_{z,z}^{aa} \Delta_{z,z}^{bb}$) this is indicated by the slight space between the two loops; they don't touch. In contrast, in the right diagram the two loops meet at the point which bisects them, the loop closes at this point, but the Dirac indices remain open carrying into the second loop before closing when they return to the second loop ($\Delta_{z,z}^{ab} \Delta_{z,z}^{ab}$).

The relative minus sign between the diagrams is indicative of the fact that each *closed loop of Dirac indices* comes with a factor of (-1) . Again, this is due to the Grassmann nature of fermions. Don't confuse this with a closed loop, it's the closed Dirac index loop that dictates the extra sign. This subtle point is generally neglected in most QFT texts, this is because we don't deal with four-fermion interactions in renormalizable theories like QED and the Standard Model.

Chapter 7

Path integrals for Vector fields

We already mostly have the necessary tools for quantizing Vector fields. In this section we'll focus on two important subtleties, we want to remove that pesky spin 0 component of the $(2, 2)$ irrep of the Lorentz group¹ and for massless vectors we'll find we can't define the propagator (i.e. invert the two-point function) without some extra machinery called "gauge fixing." The first issue will lead us directly to the familiar Field-strength tensor $F_{\mu\nu}$ from electrodynamics. The second will require some further manipulation of the path integral.

7.1 Massive vectors and the field-strength tensor

We want to introduce a vector A^μ which has four components. These four components are directly analogous to our four dimensional representation of the $\mathcal{H}_{1/2} \otimes \mathcal{H}_{1/2}$ which we wrote as $|\pm\pm\rangle$ and $|\pm\mp\rangle$. But remember we decomposed this four dimensional representation into spin 1 and spin 0 irreps that don't mix under rotations. Now consider the generator of rotations:

$$J_k = \frac{1}{2}\epsilon_{ijk}M^{ij}, \quad (7.1)$$

where $M^{\mu\nu}$ are the generators of the Lorentz group. Notice that spatial rotations belong strictly to the ij block of M . That is if we act with J on A^μ

¹Remember $\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$, analogously our $(2, 2)$ irrep which is composed of two spin $\frac{1}{2}$ irreps has a spin one and spin one-half component.

it will only mix between the three spatial components. Now remember the Pauli-Lubanski vector in the rest frame of a massive particle is given by,

$$W^i = mJ_i \quad (7.2)$$

and the eigenvalues of W^2 are the spin $s(s+1)$. Since J_i is zero when acting on the temporal component of A^μ we conclude A^0 corresponds to the spin zero part of the vector. That is it's the piece that only rotates into itself under spatial rotations, while the three spatial components correspond to the spin 1 part. As the photon is spin 1, it's this part we want to isolate for our QFTs. We'll soon see that this is precisely the part the field strength tensor, $F_{\mu\nu}$, projects out.

It's important to remember that we've only sketched this proof for the massive case. It does hold in the massless case as well. A rigorous treatment of this requires more development of the Lorentz group than we made in Chapter 1, which we won't cover in these notes.

Now we need a Lagrangian we can quantize that only depends on the spatial components of A^μ . Remember, for a propagator corresponding to physical scattering we want a pole at $p^2 = m^2$, meaning our Lagrangian should be at most quadratic in derivatives. The most general (up to IBP) Lagrangian quadratic in vector fields and at most quadratic in derivatives is given by:

$$\mathcal{L} = a_1 [(\partial_\mu A_\nu)^2 + a_2(\partial_\mu A^\mu)^2 + a_3 A_\mu A^\mu] \quad (7.3)$$

We've defined a_1 as an overall normalization. This will allow us to fix the residue of the propagator, while a_3 will fix the pole. a_2 is the only remaining part we can fix to require the temporal component doesn't propagate.

Neglecting the mass term for now, let's split up the temporal and spatial components of this Lagrangian:

$$\begin{aligned} & (\partial_\mu A_\nu)^2 + a_2(\partial_\mu A^\mu)^2 \\ &= (1 + a_2)(\partial_0 A_0)^2 - (\partial_i A_0)^2 - (\partial_0 A_i)^2 + (\partial_i A_j)^2 \\ & \quad - 2a_2(\partial_0 A_0)(\partial_i A_i) + a_2(\partial_i A_i)^2. \end{aligned} \quad (7.4)$$

Apply integration by parts, we'll remove the ∂_0 from a_0 in the second-to-last term,

$$\begin{aligned} & (\partial_\mu A_\nu)^2 + a_2(\partial_\mu A^\mu)^2 \\ &= (1 + a_2)(\partial_0 A_0)^2 - (\partial_i A_0)^2 - (\partial_0 A_i)^2 + (\partial_i A_j)^2 \end{aligned} \quad (7.5)$$

$$+ 2a_2 A_0(\partial_0 \partial_i A_i) + a_2(\partial_i A_i)^2. \quad (7.6)$$

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Now notice that if we fix $a_2 = -1$, no temporal derivatives act on A_0 ! That spin 0 component of A no longer propagates. We can then collect all the terms depending on A_0 as,

$$\begin{aligned} & (\partial_\mu A_\nu)^2 + a_2 (\partial_\mu A^\mu)^2 \\ &= -(\partial_0 A_i)^2 + (\partial_i A_j)^2 - (\partial_i A_i)^2 + A_0 \underbrace{[\partial_i^2 A_0 - 2\partial_0 \partial_i A_i]}_{\text{Lagrange multiplier}}, \quad (7.7) \end{aligned}$$

indeed the A_0 part of our Lagrangian, which is still there, is a Lagrange multiplier. This is possibly an unfamiliar form of a Lagrange multiplier which we usually write as $L = \lambda(\text{condition})$. In this case, however, since our field plays the role of the Lagrange multiplier, the constraint imposes conditions on the field A_0 .

From classical electrodynamics this Lagrange multiplier is precisely Gauss's law; i.e. it's a feature not an added annoyance. To see this, start by finding the Euler Lagrange equation for A_0 :

$$\frac{\partial \mathcal{L}}{\partial A_0} - \partial_\rho \frac{\partial \mathcal{L}}{\partial \partial_\rho A_0} + \partial_\sigma \partial_\rho \frac{\partial \mathcal{L}}{\partial \partial_\rho \partial_\sigma A_0} = 0 \quad (7.8)$$

$$= a_1 (\partial_i^2 A_0 - 0 - 2\partial_0 \partial_i A_i + \partial_i^2 A_0) \quad (7.9)$$

$$= 2a_1 (\partial_i^2 A_0 - \partial_0 \partial_i A_i) = 0 \quad (7.10)$$

Notice that to simplify our lives, we didn't "unwind" our integration by parts so that the Lagrangian depends only on a single derivative of fields. We just extended the Euler Lagrange equations to work with Lagrangians where up to two derivatives act on the fields. This follows directly from extending our proof in Eq. 4.27.

Now we rewrite $A_0 = \phi$ and associated A_i with the vector potential, then we have the usual relations:

$$E_i = F_{0i} = \partial_0 A_i - \partial_i A_0 \quad (7.11)$$

Gauss's law states $\vec{\nabla} \cdot \vec{E} = \rho$, so in the absence of a source (i.e. the free Lagrangian) we have,

$$\partial_i E_i = 0 = \partial_0 \partial_i A_i - \partial_i^2 \phi \quad (7.12)$$

This is precisely the constraint enforced by our Lagrange multiplier. Because, as we've pointed out already, ϕ has no time derivatives in our Lagrangian, we've fully determined $\phi = A_0$ in terms of A_i . Physically this enforces that

a massless vector (e.g. photon) has only two physical polarizations which are transverse.

Having set $a_2 = -1$ and recognized that the A_0 field is not dynamic, we've settled on a valid Lagrangian for our QFTs given by:

$$\mathcal{L} = a_1 [(\partial_\mu A_\nu)^2 - (\partial_\mu A^\mu)^2 + a_3 A_\mu A^\mu] , \quad (7.13)$$

$$= a_1 (F_{\mu\nu} F^{\mu\nu} + a_3 A_\mu A^\mu) . \quad (7.14)$$

The relation to the field strength, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, is imposed by requiring our Lagrangian describe a dynamical spin 1 field. We've chosen $a_2 = -1$ to force the temporal component, which is spin 0, not to participate.

7.2 Propagator for massive vector fields

We have everything we need to immediately obtain the propagator for massive vector fields. Let's quickly rewrite our Lagrangian using IBP to get all of the derivatives acting on one field. This lets us define an operator, $i\Delta^{-1}$, defined this way so that the propagator is Δ , just like in the scalar case. We've renamed the operator we called A for scalars as $i\Delta^{-1}$ for the vector case. This avoids using A for the operator which would be confusing as that's the name of our vector field. Our Lagrangian is:

$$\mathcal{L} = \frac{1}{2} A_\mu \underbrace{2a_1 [2\partial^\mu \partial^\nu - 2\partial^2 \eta^{\mu\nu} + a_3 \eta^{\mu\nu}]}_{\equiv i(\Delta^{-1})^{\mu\nu}} A_\nu . \quad (7.15)$$

Our generating functional for the free theory then looks like:

$$Z[J^\mu] = \int \mathcal{D}A_\mu \exp \left[i \int d^4x \left(\frac{1}{2} A_\mu(x) i(\Delta^{-1})^{\mu\nu} A_\nu(x) + J^\mu(x) A_\mu(x) \right) \right] \quad (7.16)$$

Performing the Gaussian integral, in direct analogy with the scalar case² (except now A has a Lorentz index), we obtain the result:

$$Z[J^\mu] = N \exp \left[-\frac{1}{2} \int d^4x d^4y J^\mu(x) \Delta(x, y)_{\mu\nu} J^\nu(y) \right] . \quad (7.17)$$

Taking two variations with respect to J gives the time-ordered product of two fields which we associate (as we did with scalars and fermions) with the

²See Eqs. 4.5 and 4.9 where this was worked out in detail in the scalar case.

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two point function. Let's take the time to do this carefully just to track the indices, it's the same concept as with the Dirac indices, except now everything is bosonic:

$$\langle \Omega | T[A_\sigma(y_2), A_\rho(y_1)] | \Omega \rangle = \frac{1}{i} \frac{\delta}{\delta J^\sigma(y_2)} \frac{1}{i} \frac{\delta}{\delta J^\rho(y_1)} Z[J^\mu] \quad (7.18)$$

Notice J^σ sources A_σ and similarly for A_ρ . We could play with raised or lowered indices, but that's effectively just adding some extra effort with $\eta^{\mu\nu}$ and δ_ν^μ ,

$$\frac{\delta}{\delta J_\mu} J^\nu = \eta^{\mu\nu}, \quad (7.19)$$

$$\frac{\delta}{\delta J^\mu} J^\nu = \delta_\mu^\nu, \quad (7.20)$$

which can be achieved by just applying $\eta^{\mu\nu}$ to the time ordered product after evaluating it.

Working through the variations gives us our time-ordered product in terms of Δ :

$$\langle \Omega | T[A_\sigma(y_2), A_\rho(y_1)] | \Omega \rangle \quad (7.21)$$

$$\begin{aligned} &= \frac{\delta}{\delta J^\sigma(y_2)} Z[J^\mu] \left(\int d^4x_1 d^4x_2 J_\mu(x_1) \Delta(x_1, x_2)_{\mu\nu} \delta^4(x_2 - y_1) \delta_\rho^\nu \right) \Big|_{J=0} \\ &= N \Delta(y_1, y_2)_{\sigma\rho} \end{aligned} \quad (7.22)$$

In the first equivalence we immediately applied that Δ is symmetric in both its spacetime and Lorentz index arguments to shorten the expression. We know this is the case because we're again dealing with bosonic theories, the variations commute. The symmetry in $x_1 \leftrightarrow x_2$ corresponds to translational invariance of the theory, symmetry in the Lorentz indices is simply a reflection of the fact the free Lagrangian is symmetric in the Lorentz indices as is manifest in Eq. 7.15

The first line shows how the variations work, so in the second line we perform the second variation and immediately perform the integrals as well. N is an infinite constant, again it is different from that for the scalars and fermions because of the functional determinant. Like in the scalar case it goes as $(\det \Delta)^{-1/2}$.

We've identified Δ with the propagator (i.e. time-ordered product of two

fields) and are just left with solving the corresponding Green's function equation:

$$(\Delta^{-1})^{\mu\nu} \Delta(x_1, x_2)_{\nu\rho} = \delta^\mu_\rho \delta^4(x_1 - x_2) \quad (7.23)$$

By inspection, the solution takes the form:

$$\Delta(x_1, x_2)_{\nu\rho} = \frac{i}{2a_1} \int \frac{d^4p}{(2\pi)^4} (-2p^\mu p^\nu + 2p^2 \eta^{\mu\nu} + a_3 \eta^{\mu\nu})^{-1} e^{-ip \cdot (x_1 - x_2)} \quad (7.24)$$

We need to figure out the inverse of the operator in parenthesis, which up to the factor of $2a_1/i$ is simply Δ^{-1} in momentum space. Consider:

$$\delta^\mu_\rho = (-2p^\mu p^\nu + 2p^2 \eta^{\mu\nu} + a_3 \eta^{\mu\nu}) (b_1 p_\nu p_\rho + b_2 \eta_{\nu\rho}) , \quad (7.25)$$

$$= (-2b_1 + 2b_1)p^2 p^\mu p_\rho + (-2b_2 + a_3 b_1)p^\mu p_\rho + (2p^2 b_2 + a_3 b_2)\delta^\mu_\rho . \quad (7.26)$$

Requiring this expression be δ^μ_ρ gives for b_1 and b_2 :

$$b_2 = \frac{1}{2(p^2 + a_3/2)} \quad b_1 = \frac{1}{a_3(p^2 + a_3/2)} . \quad (7.27)$$

We can see already that we need to require $a_3 = -2m^2$ in order for the pole of the propagator to correspond to the mass of our vector field. Our propagator then takes the form:

$$\Delta(x_1, x_2)_{\nu\rho} = \int \frac{d^4p}{(2\pi)^4} \frac{i}{4a_1} \frac{1}{p^2 - m^2} \left(\eta_{\nu\rho} - \frac{p_\nu p_\rho}{m^2} \right) e^{-ip \cdot (x_1 - x_2)} \quad (7.28)$$

Fixing $a_1 = -1/4$ properly positions the residue at the pole as i . No careful consideration of the matrix part is required, this matrix is real and symmetric. Naively, the residue appears to be $-i$ with this choice, that this choice corresponds to residue $+i$ requires an understanding of the physical (on-shell) polarization vectors, whose normalization is taken to be $\epsilon^{*\mu} \epsilon_\mu = -1$. Therefore fixing the residue to $+i$ corresponds to fixing the equation,

$$\epsilon^{*\mu} \left[\frac{i}{4a_1} \left(\eta_{\mu\nu} - \frac{p_\mu p_\nu}{m^2} \right) \right] \epsilon^\nu = i , \quad (7.29)$$

subject to $p \cdot \epsilon = 0$, which is the reason we have a $-i$ in our propagator.

Finally we have, for a massive vector field:

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{m^2}{2} A_\mu A^\mu , \quad (7.30)$$

$$\Delta(x_1, x_2)_{\mu\nu} = \int \frac{d^4p}{(2\pi)^4} \frac{-i}{p^2 - m^2} \left(\eta_{\mu\nu} - \frac{p_\mu p_\nu}{m^2} \right) e^{-ip \cdot (x_1 - x_2)} . \quad (7.31)$$

The first part of the Lagrangian is familiar from electrodynamics. If we want to describe massless spin-one fields like the photon, we want to work this out for $m^2 \rightarrow 0$. Notice that this is a problem in the propagator of Eq. 7.31, as the term dependent on $p_\mu p_\nu$ will blow up. This is rooted in the fact that the operator Δ^{-1} in the massless limit is not invertible,

$$\Delta^{-1}(m^2 \rightarrow 0)^{\mu\nu} = -[\partial^\mu \partial^\nu - \partial^2 \eta^{\mu\nu}] . \quad (7.32)$$

To see this simply act with Δ^{-1} on the gradient of a scalar:

$$-[\partial^\mu \partial^\nu - \partial^2 \eta^{\mu\nu}] \partial_\nu \theta = -[\partial^2 \partial^\mu - \partial^2 \partial^\mu] \theta , \quad (7.33)$$

$$= 0 . \quad (7.34)$$

Since some field (eigenvector) exists for which $(\Delta^{-1})^{\mu\nu} \partial_\nu \theta(x) = 0$, the operator Δ^{-1} has at least one zero eigenvalue, meaning its determinant is 0. Actually, it's infinitely many 0 eigenvalues as the scalar field $\theta(x)$ is representative of any smooth function θ . When we discuss QED as a gauge theory, we will have that the QED Lagrangian is invariant under the gauge transformation,

$$A_\mu \rightarrow A_\mu + \partial_\mu \theta , \quad (7.35)$$

with $\partial_\mu \theta$ corresponding to those zero eigenvalues. To work around this, we need to work out the idea of gauge fixing, which will introduce extra terms that allow us to invert this matrix, but won't affect physical observables.

7.3 Gauge Fixing I: R_ξ gauge for the photon

For the massless case,

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} , \quad (7.36)$$

we saw that the two-point function of vector fields was not invertible, and so we cannot define a propagator. Our solution lies in the Lagrange multipliers that we discussed in the context of the temporal component of A^μ .

If we apply the field redefinition, $A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu \theta(x)$, we can look at how the field strength tensor behaves under this transformation:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (7.37)$$

$$\rightarrow [\partial_\mu A_\nu + \partial_\mu \partial_\nu \theta(x)] - [\partial_\nu A_\mu + \partial_\nu \partial_\mu \theta(x)] = F_{\mu\nu} . \quad (7.38)$$

The field strength, and therefore the Lagrangian, are invariant under this symmetry. In terms of the $\vec{E}$ and $\vec{B}$ field, this same gauge transformation takes the form:

$$\vec{E} = -\nabla\phi - \partial_0\vec{A} \quad \rightarrow \quad -\vec{\nabla}(\phi - \partial_0\theta) - \partial_0(\vec{A} + \vec{\nabla}\theta) = \vec{E}, \quad (7.39)$$

$$\vec{B} = \nabla \times \vec{A} \quad \rightarrow \quad \nabla \times (\vec{A} + \nabla\theta) = \vec{B}, \quad (7.40)$$

Since $\theta(x)$ is arbitrary, there are an infinite number of transformations that leave the physics invariant. That α is arbitrary is referred to as *gauge redundancy*. Gauge fixing restricts only the redundant directions, i.e. we don't change $\vec{E}$ and $\vec{B}$. Remember, F can be fully determined in terms of electric and magnetic fields. Gauge fixing may not remove all gauge redundancy, but it can greatly simplify our life.

What we wish to do restrict some of that gauge redundancy, and in particular to do so in such a manner that allows us to invert the kinetic operator in the Lagrangian. To achieve this we'll employ a similar Lagrange multiplier approach to what we observed in the last section.

We start by introducing a new field $\lambda(x)$. We'll take it to be nondynamical in analogy with how A_0 played the role of a Lagrange multiplier. We can write our path integral as:

$$Z[J] = \int \mathcal{D}\lambda \mathcal{D}A_\mu \exp \left[i \int d^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + J^\mu A_\mu + \lambda \left[(\partial^\mu A_\mu) + \frac{\xi}{2} \lambda \right] \right) \right]. \quad (7.41)$$

Notice our new $\lambda(x)$ isn't dynamical, making it completely analogous to our discussion of A_0 . Its equation of motion is:

$$\xi\lambda + (\partial^\mu A_\mu) = 0. \quad (7.42)$$

If we take $\xi \rightarrow 0$ this enforces the constraint $(\partial^\mu A_\mu) = 0$. Applying our transformation for A to our constraint we find:

$$0 = (\partial^\mu A_\mu) + \xi\lambda, \quad (7.43)$$

$$\rightarrow [\partial^\mu A_\mu + \partial^2\theta(x)] + \xi\lambda. \quad (7.44)$$

We're still invariant under the transformation $A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu\theta(x)$, but subject to a further constraint that $\partial^2\alpha(x) = 0$. That is:

1. F^2 is invariant under the general transformation:
 $A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu\theta(x)$

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2. $(\partial^\mu A_\mu)^2$ is invariant under the more restrictive transformation:

$$A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu \theta(x) \text{ subject to } \partial^2 \theta(x) = 0$$

We've restricted the gauge symmetry in terms of which choices of $\theta(x)$ are valid, that is we've fixed part of the redundancy in which θ are viable. We *have not* destroyed the gauge symmetry altogether.

Now let's **integrate out** the λ field, which is to say solve the path integral for λ . Our integral in λ is just another Gaussian, so we can apply Eq. 4.8 to find:

$$Z[J] = N \int \mathcal{D}A_\mu \exp \left[i \int d^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + J^\mu A_\mu + \frac{1}{2\xi} (\partial_\mu A^\mu)^2 \right) \right]. \quad (7.45)$$

Here N is another of those infinite constants that don't affect physical processes. Notice that all we have left is our original free Lagrangian and additional term going as $(\partial_\mu A^\mu)^2$. This term, which we showed just restricts the gauge symmetry and not the field A_μ , allows us to invert the kinetic operator and obtain a propagator. The kinetic operator $i\Delta^{-1}$ is now given by:

$$\mathcal{L} = \frac{1}{2} A_\mu \left[\partial^2 \eta^{\mu\nu} - \partial^\mu \partial^\nu + \frac{1}{\xi} \partial^\mu \partial^\nu \right] A_\nu, \quad (7.46)$$

$$\equiv \frac{1}{2} A_\mu i (\Delta^{-1})^{\mu\nu} A_\nu. \quad (7.47)$$

Our discussion for the massive vector field shows the steps to follow which will result in the propagator:

$$\Delta(x_1, x_2)_{\mu\nu} = \int \frac{d^4p}{(2\pi)^4} \frac{-i}{p^2} \left(\eta_{\mu\nu} - (1 - \xi) \frac{p_\mu p_\nu}{p^2} \right) e^{-ip \cdot (x_1 - x_2)} \quad (7.48)$$

Remember, our constraint required $\partial^\mu A_\mu + \xi \lambda = 0$ which in the limit $\xi \rightarrow 0$ requires $\partial^\mu A_\mu = 0$. If we take $\xi \rightarrow 0$ in our propagator the part in parenthesis becomes:

$$\Pi_{T,\mu\nu} = \left(\eta_{\mu\nu} - \frac{p_\mu p_\nu}{p^2} \right), \quad (7.49)$$

$$\Pi_T^{\mu\nu} \Pi_{T,\nu\rho} = \left(\eta^{\mu\nu} - \frac{p^\mu p^\nu}{p^2} \right) \left(\eta_{\nu\rho} - \frac{p_\nu p_\rho}{p^2} \right), \quad (7.50)$$

$$= \Pi_{T,\rho}^\mu. \quad (7.51)$$

This is the transverse polarization projection operator. In the limit $\xi \rightarrow 0$ our gauge fixing has enforced that our propagator is transverse which physical photons are. A quick look at our Lagrangian after integrating out λ , Eq. 7.45 appears to present a problem with $\xi \rightarrow 0$. The $1/\xi$ term goes to infinity. While it appears problematic it is actually the same statement as $\xi \rightarrow 0$ enforces the constraint exactly: in this form of our Lagrangian $1/\xi$ is behaving like our Lagrange multiplier, making this quantity larger corresponds to making the restoring force that requires $\partial_\mu A^\mu \rightarrow 0$ stronger.

This discussion was nice, and is in fact correct. But as it stands if we're particularly careful, may leave us with some open questions: the treatment of λ as imposing a constraint is purely classical, how do we know that our results still hold when incorporating the quantum nature of these theories? Further, how do we know that λ as a nondynamical field does not become dynamical in a quantum system, thereby spoiling our argument that its introduction doesn't change the physics?

These two questions are quickly rectified with the “Fadeev-Poppov” procedure for gauge fixing. This more general approach will also allow us to treat more complicated gauge theories, such as QCD, as well as gauge theories with spontaneous symmetry breaking, such as the electroweak sector of the Standard Model.

7.4 Gauge Fixing II: Faddeev Popov procedure

With this last step we'll develop the last major tool we need to start looking at specific theories and their predictions. Some of those specific theories are QCD and the Standard Model, which like QED are gauge theories, but unlike QED their symmetries are described by nonabelian groups. These nonabelian groups lead to amazing phenomenology we'll want to study, but in order to be able to do so we'll need to be able to describe vector fields that transform under these symmetries. The approach we made in the last section doesn't extend naturally to these types of theories.

Let's start with normal integration. Consider some integral Z which is defined as an integral over dx , we'll rewrite it as a double integral by simply

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introducing a δ function:

$$Z = \int dx e^{iS(x)}, \quad (7.52)$$

$$= \int dx dy \delta(y) e^{iS(x)}. \quad (7.53)$$

Nothing has changed as there is no y dependence outside the δ function. We can take this further, since there is no y dependence we can shift the delta function by any function of x we want. The delta function tells us to replace all instances of y with that $f(x)$, but since there are no instances of y we still haven't changed the integral:

$$Z = \int dx dy \delta(y - f(x)) e^{iS(x)}. \quad (7.54)$$

If instead we had some $g(x, y)$ and $\delta(g(x, y))$, we have a rule for this too. We replace $\delta(g(x, y))$ by the δ function of the zero of g and divide by the derivative of g (evaluated at the zero),

$$\delta(g(x, y)) = \frac{\delta(y - f(x))}{\left| \frac{\partial g}{\partial y} \right|_{y=f(x)}}, \quad (7.55)$$

or in the case of multiple zeroes, we sum them. We can use this relation to further rewrite our integral Z as:

$$Z = \int dx dy \left| \frac{\partial g}{\partial y} \right| \delta(g(x, y)) \quad (7.56)$$

This trick is going to be the basis of the “Faddeev Popov” procedure. This delta function trick is basically the entire basis of the proof. We'll also rely on multiplying our path integral by some infinite constants, but we know that doesn't change the physics.

If we extend this to n dimensions, the derivative term becomes a determinant giving:

$$Z = \int d^n x d^n y \det \left(\frac{\partial G_i}{\partial y_j} \right) \Pi_i \delta(G_i) e^{iS(x_i)}. \quad (7.57)$$

We can use this same approach for our path integrals. Our analogue of x will be A_μ^a , and we'll take for y some $\theta^a(x)$ field which we'll shortly relate

to symmetries of the vector field. Notice that we've given both fields a superscript a , we may have more than one vector and scalar θ . Our path integral then becomes:

$$Z[J] = \int \mathcal{D}A_\mu^a e^{i \int d^4x \mathcal{L}_{\text{free}}}, \quad (7.58)$$

$$\rightarrow \int \mathcal{D}\theta^a \mathcal{D}A_\mu \det \left(\frac{\delta(G^a - \omega^a)}{\delta\theta^b} \right) \delta(G^a - \omega^a) e^{i \int d^4x \mathcal{L}_{\text{free}}}, \quad (7.59)$$

$$= \int \mathcal{D}\theta^a \mathcal{D}A_\mu \det \left(\frac{\delta G^a}{\delta\theta^b} \right) \delta(G^a - \omega^a) e^{i \int d^4x \mathcal{L}_{\text{free}}}, \quad (7.60)$$

$$(7.61)$$

In the second line we've just shifted G^a by some arbitrary direction ω^a that doesn't depend on θ . The third line uses that ω is independent of θ . We'll make use of that delta function to impose a constraint on our theory, much in the same way as the Lagrange multiplier did in our last section.

We have a functional determinant of G , and that functional determinant is in the numerator. Remember that for the Grassmann gaussian integral we obtained a functional determinant in the numerator. So we'll rewrite this functional derivative in terms of some Grassmann fields c and $\bar{c}$:

$$\det \left(\frac{\delta G^a(x)}{\delta\theta^b(y)} \right) \propto \int \mathcal{D}c \mathcal{D}\bar{c} e^{-i \int d^4x \bar{c}^a \left(\frac{\delta G^a}{\delta\theta^b} \right) c^b} \quad (7.62)$$

$$= \int \mathcal{D}c \mathcal{D}\bar{c} e^{i \int d^4x \mathcal{L}_{\text{ghost}}}, \quad (7.63)$$

We refer to $\bar{c}$ and c as ghost fields. They're an added cost of our gauge fixing procedure, but the benefits of this gauge fixing are usually worth it. For tree level calculations, it is sometimes preferable to use other gauge fixing that avoids ghosts. The “ $\propto$ ” indicates that we have dropped an infinite constant from our equality.

Notice that this really doesn't say anything about the representation of the Lorentz group the ghosts are in. We could choose anything we wanted, the simplest is to choose them to be scalars. So despite it being strange, the ghosts are anticommuting scalars. This is fine because the ghosts are unphysical, they will never be external lines of physical processes, they just enter into internal lines to ensure our physical vectors behave as they should. Having chosen them to be scalars we could say $\bar{c} \rightarrow c^\dagger$, but the standard

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notation is to refer keep the notation $\bar{c}^3$.

After all of this our path integral has taken the form:

$$Z[J] \propto \int \mathcal{D}\theta^a \mathcal{D}A_\mu^a \mathcal{D}c \mathcal{D}\bar{c} \delta(G^a - \omega^a) e^{i \int d^4x (\mathcal{L}_{\text{free}} + \mathcal{L}_{\text{ghost}})}, \quad (7.64)$$

Again, $\propto$ refers to an infinite constant that drops out of observables.

Our next step is to rewrite the $\delta(G)$. To do this we'll actually introduce another infinite constant of the form:

$$N = \int \mathcal{D}\omega \exp \left(-\frac{i}{2\xi} \int d^4x \omega^a \omega^a \right) \quad (7.65)$$

Absorbing this infinite constant into our definition of $Z[J]$ we have:

$$Z[J] \propto \int \mathcal{D}\theta^a \mathcal{D}A_\mu^a \mathcal{D}c \mathcal{D}\bar{c} \mathcal{D}\omega \delta(G^a - \omega^a) e^{i \int d^4x (\mathcal{L}_{\text{free}} + \mathcal{L}_{\text{ghost}} - \frac{i}{2\xi} \omega^a \omega^a)}, \quad (7.66)$$

$$= \int \mathcal{D}\theta^a \mathcal{D}A_\mu^a \mathcal{D}c \mathcal{D}\bar{c} e^{i \int d^4x (\mathcal{L}_{\text{free}} + \mathcal{L}_{\text{ghost}} - \frac{1}{2\xi} G^a G^a)}. \quad (7.67)$$

This is our final result. In terms of the individual Lagrangians we have:

$$\mathcal{L}_{\text{free}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + J^\mu A_\mu, \quad (7.68)$$

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi} G^a G^a, \quad (7.69)$$

$$\mathcal{L}_{\text{ghost}} = -\bar{c}^a \left(\frac{\delta G^a}{\delta \theta^b} \right) c^b. \quad (7.70)$$

Arriving here came at the cost of inserting a δ function in a clever manner, and a couple of infinite constants that won't impact any observables.

In the context of our massless vector from the last section, we had that our free Lagrangian was invariant under the transformation,

$$A_\mu \rightarrow A_\mu + \partial_\mu \theta. \quad (7.71)$$

We chose to restrict $\partial^\mu A_\mu$, so we'll similarly take $G = (\partial^\mu A_\mu)$. We can

³We could also choose them to be real, but we need a separate $\bar{c}$ and c , so using complex scalars is a bit cleaner presentation-wise and in terms of Feynman rules as we'll see later.

obtain δG from the chain rule:

$$\delta G = \frac{\delta G}{\delta A_\mu} \delta A_\mu, \quad (7.72)$$

$$= \frac{\delta G}{\delta \partial^\mu A_\mu} \partial^\mu (\delta A_\mu), \quad (7.73)$$

$$= \square \theta, \quad (7.74)$$

where we've chosen that our variation of A , δA is “along” the symmetry direction. That is we're choosing to only consider directions that retain our symmetry. Extending this to the $\delta G/\delta \theta$ we need we find:

$$\frac{\delta G(x)}{\delta \theta(y)} = \square \delta^4(x - y). \quad (7.75)$$

For our application we just need $\delta G(x)/\delta \theta(x)$ which allows us to drop the $\delta^4(x - y)$. We plug all of this into Eqs. 7.69 and 7.70, and we find for our Gauge fixing and Ghost Lagrangians:

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi} (\partial^\mu A_\mu)^2, \quad (7.76)$$

$$\mathcal{L}_{\text{ghost}} = -\bar{c}^a \square c^a. \quad (7.77)$$

Notice we've recovered our gauge-fixing term from the Lagrange multiplier approach. What's new is we have a ghost term, but for such a simple symmetry as we had for this massless vector they don't interact with anything. They have no effect on our calculations and can be neglected/integrated out of the theory.

We should stress at this point, that our treatment of gauge fixing has been rigorous this time. We did not simply add a nondynamic field to our Lagrangian and state it's a Lagrange multiplier. We followed functional methods to manipulate our path integral into a form we needed without changing the physics of the path integral.

For “local” $SU(N)$ gauge theory we will soon see that a vector field transforms instead as:

$$A_\mu^a(x) \rightarrow A_\mu^a(x) + (\partial_\mu \delta^{ac} + g f^{abc} A_\mu^b) \theta^b(x), \quad (7.78)$$

where f^{abc} are the structure constants of the Lie algebra and g is a coupling constant (analogous to e in electrodynamics). In this case we can again

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choose $G^a = \partial^\mu A_\mu^a$, but we'll find for our ghost Lagrangian:

$$\mathcal{L}_{\text{ghost}} = -\bar{c}^a \partial^\mu (\partial_\mu + g f^{abc} A_\mu^b) c^c, \quad (7.79)$$

and now we do have interactions between the ghosts and the vector fields.

Part III

Calculations in QFT

Chapter 8

The Feynman Rules

Having quantized our classical field theories for scalars, Dirac spinors, and vectors we can now start to discuss how we actually calculate observables in QFT. Path integrals are an elegant method for upgrading our relativistic field theories to QFTs, but in practice for calculations taking a repeated variations of the generating functional quickly becomes impractical (as we saw with our longer calculations). The way around this is the **Feynman rules**.

With just a few rules, which we'll infer from our previous path integral calculations, we can use diagrammatic methods to immediately write down time-ordered products in interacting theories. In the last chapters we used explicit calculations from the path integral perspective and re-expressed them as diagrams. Now we'll do the reverse: we draw the diagrams, and they tell us how to write down the expressions that we would have gotten from (a lot of) variations on the generating functional.

At the end of this section we're going to introduce a few tools that we won't have to derive. These include the LSZ formula, which relates time-ordered correlation functions to S-matrix elements, which should be familiar from scattering in QM, and some useful identities for on-shell Dirac fermions and vectors. These are related to the classical solutions to the EOM for these fields.

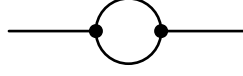


Figure 8.1: Bubble diagram representing an order g^2 correction to the scalar two-point function in ϕ^3 theory. Reproduced from the full set of (disconnected) diagrams in Fig. 4.5.

8.1 Feynman rules for scalars

In Eq. 4.73 we found an expression for the order g^2 correction to the two-point function for a scalar in ϕ^3 theory. Figure 4.5 shows the sum of the (disconnected diagrams). Of these the only two physical diagrams were found to be the tadpole diagram and the **bubble diagram** we reproduce in Fig. 8.1.

We're going to break down the details of the bubble diagram in order to infer the Feynman rules for scalar theory. We'll start with the expression from Eq. 4.73 which reads:

$$D_{\text{bubble}} = -\frac{g^2}{2} \int d^4 z_1 d^4 z_2 \Delta_{y_1, z_1} \Delta_{z_1, z_2}^2 \Delta_{z_2, y_2}, \quad (8.1)$$

$$\begin{aligned} &= -\frac{g^2}{2} \int d^4 z_1 d^4 z_2 \int \frac{d^4 p_1}{(2\pi)^4} \frac{ie^{ip_1 \cdot (y_1 - z_1)}}{p_1^2 - m^2} \int \frac{d^4 p_2}{(2\pi)^4} \frac{ie^{ip_2 \cdot (z_1 - z_2)}}{p_2^2 - m^2} \\ &\quad \times \int \frac{d^4 p_3}{(2\pi)^4} \frac{ie^{ip_3 \cdot (z_2 - z_1)}}{p_3^2 - m^2} \int \frac{d^4 p_4}{(2\pi)^4} \frac{ie^{ip_4 \cdot (z_2 - y_2)}}{p_4^2 - m^2}. \end{aligned} \quad (8.2)$$

Where we're referring to our diagram by D . We've also dropped the infinite constant N , which we saw drops out in calculations using our generating functional of connected diagrams. In the second line we've explicitly rewritten each propagator in terms of our expression from Eq. 4.21. Notice each propagator comes with its own momentum p_i , which is integrated over all momentum configurations.

Remember, this diagram is basically saying, propagate our scalar from our external point y_1 to the first interaction point z_1 , we then propagate two scalars separately from z_1 to the next interaction point z_2 where they “combine” back into a single scalar and propagate to the second external point y_2 ¹. In order to be general (and by construction from the path integral), we're *integrating over all possible interaction points z_1 and z_2* .

¹This, like most physical interpretations are very much heuristic. Remember, these

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These integrals over the interaction points actually turn out to be enormously helpful. We'll write:

$$\int d^4 z_1 e^{i z_1 \cdot (p_2 - p_1 - p_3)} = (2\pi)^4 \delta^4(p_2 - p_1 - p_3), \quad (8.3)$$

$$\int d^4 z_2 e^{i z_2 \cdot (p_3 - p_2 + p_4)} = (2\pi)^4 \delta^4(p_3 - p_2 + p_4). \quad (8.4)$$

Allowing us to rewrite our expression as:

$$D_{\text{bubble}} = -\frac{g^2}{2} \int \frac{d^4 p_1 d^4 p_2 d^4 p_3 d^4 p_4}{(2\pi)^{4 \cdot 4}} \frac{e^{i(p_1 \cdot y_1 - p_4 \cdot y_2)}}{(p_1^2 - m^2)(p_2^2 - m^2)(p_3^2 - m^2)(p_4^2 - m^2)} \\ \times (2\pi)^{2 \cdot 4} \delta^4(p_2 - p_1 - p_3) \delta^4(p_3 - p_2 + p_4). \quad (8.5)$$

This is very interesting: these delta functions are enforcing momentum conservation at each vertex. We're slowly replacing all the spacetime arguments with momenta and preserving momentum along the way.

We'll rename $p_3 = \ell$, for reasons we'll explain soon, and use the first delta function to rewrite this expression as:

$$D_{\text{bubble}} = \int \frac{d^4 p_1 d^4 p_4}{(2\pi)^{2 \cdot 4}} e^{i(p_1 \cdot y_1 - p_4 \cdot y_2)} \frac{-1}{(p_1^2 - m^2)(p_4^2 - m^2)} \\ \times \left(\frac{g^2}{2} \right) \int \frac{d^4 \ell}{2\pi^4} \frac{(2\pi)^4 \delta(p_4 - p_1)}{[(\ell + p_4)^2 - m^2](\ell^2 - m^2)}. \quad (8.6)$$

Notice that the first line of this equation corresponds to the external propagators, while the second line corresponds to the internal lines. This is on purpose: next we're going to use the LSZ formula to "amputate" external lines and relate them to particles.

We will not have time to derive the "Lehman-Symanzik-Zimmermann (LSZ) reduction formula," so we'll simply outline what it does here. The LSZ reduction formula relates the time-ordered products we've been working with so far to the S -matrix scattering amplitudes which are the observables we calculate in these notes. The LSZ reduction is not at all an assumption, but a rigorous and powerful result of canonical quantization² of QFTs.

are time ordered products, everything can be reversed if y_2 comes before y_1 , and further complicated by the ordering of z_1 and z_2 .

²As opposed to our path integral quantization. I am unaware of any path integral based derivations of LSZ.

LSZ tells us, in order to obtain a scattering amplitude from the time ordered product, we need to:

1. amputate external lines, i.e. remove propagators for external lines.
2. replace the corresponding fields with asymptotic on-shell one-particle states – that is, with states that don't interact and freely propagate forever (in reality, until they reach a particle detector).

Physically and very much qualitatively: we remove from the external legs this abstract and quantum concept of a field and replace them with particles. LSZ reduction bridges our ideas of scattering particles and the S -matrix and interacting fields and the corresponding correlation functions.

In the case of our two-point function, we'll refer to y_1 as the label for an incoming particle and y_2 for an outgoing particle giving:

$$\begin{aligned} \langle f | S | i \rangle = & \underbrace{\left(-i \int d^4 y_1 e^{-i p_i \cdot y_1} (p_i^2 - m^2) \right)}_{\text{incoming}} \underbrace{\left(-i \int d^4 y_2 e^{i p_f \cdot y_2} (p_f^2 - m^2) \right)}_{\text{outgoing}} \\ & \times \langle \Omega | T[\phi(y_1), \phi(y_2)] | \Omega \rangle \end{aligned} \quad (8.7)$$

This rather intuitive formulate relates the S -matrix element on the left to the time-ordered product on the right. The $-i(p^2 - m^2)$ removes the propagators and the exponentials are the asymptotic solutions of the Klein Gordon equation. Notice that the exponential has the opposite sign for incoming vs outgoing, this is going to be important. We've specifically labeled this S -matrix elements as $\langle f | S | i \rangle_D$ as it is incomplete. *We must sum all terms at a given order in g .* We also have the leading order term (the two-point function) which we're neglecting here.

Multiplying these new exponentials by our existing ones in Eq. 8.6 gives us some new delta functions and two propagators are removed. This leaves us with,

$$\langle f | S | i \rangle_D = \int \frac{d^4 p_1 d^4 p_4}{(2\pi)^{2 \cdot 4}} \frac{g^2}{2} \int \frac{d^4 \ell}{(2\pi)^4} \frac{(2\pi)^{3 \cdot 4} \delta^4(p_4 - p_1) \delta^4(p_i - p_1) \delta^4(p_f - p_4)}{[(\ell + p_4)^2 - m^2](\ell^2 - m^2)}, \quad (8.8)$$

$$= \frac{g^2}{2} \int \frac{d^4 \ell}{(2\pi)^4} \frac{(2\pi)^4 \delta^4(p_f - p_i)}{[(\ell + p_f)^2 - m^2](\ell^2 - m^2)}. \quad (8.9)$$

Notice that replacing external fields with the corresponding asymptotic states has removed all dependence on the spacetime arguments. This expression is

8. The Feynman Rules

entirely in terms of momenta, and momentum conservation was enforced in every step along the way.

Generally, we'll expand the S -matrix elements in terms of the identity matrix which represents no scattering, just free propagation, and the matrix elements that tell us about the scattering as,

$$\langle f|S|i\rangle = \langle f|i\rangle + (2\pi)^4 \delta^4(\sum_i p_i) i\mathcal{M}_{f,i}. \quad (8.10)$$

So we can associate our specific diagram with part of $i\mathcal{M}_{f,i}$. We'll drop the " f,i " subscript for simplicity and write:

$$i\mathcal{M}_D = +\frac{g^2}{2} \int \frac{d^4\ell}{(2\pi)^4} \frac{1}{[(p_i + \ell)^2 - m^2](\ell^2 - m^2)} \quad (8.11)$$

The $i\mathcal{M}$ are precisely the scattering amplitudes we will design our Feynman rules to extract. This will frequently allow us to jump over explicitly calculating time-ordered products and performing the LSZ reduction. At tree level, because we have "canonically normalized" our propagators/Lagrangians (i.e. fixing the pole and its residue appropriately) this is true. When we calculate processes involving loops we will generally need to fix the residue and the pole again *or* adjust the LSZ reduction formula for non-canonically normalized Lagrangians.

The bubble diagram's amplitude in Eq. 8.11 in its present form contains all the ingredients that will generalize into our Feynman rules. For scalar ϕ^3 theory, we will extract those Feynman rules in a way that allows us to directly obtain the scattering amplitude $i\mathcal{M}$ for any given diagram:

1. For all external lines we write 1. (This will change for Dirac fields and vectors, so we write it explicitly despite being trivial.)
2. All internal lines correspond to propagators *in momentum space*. That means we don't have the integral over position with an exponential anymore. These exponentials all enforced momentum conservation, so we label the propagator's momenta by strictly enforcing momentum conservation at each vertex. The scalar propagator (canonically normalized) is:

$$\Delta(p^2, m^2) = \frac{i}{p^2 - m^2}. \quad (8.12)$$

3. For any closed loop, we'll need to integrate over all possible momenta configuration as:

$$\text{loop} \Rightarrow \int \frac{d^4\ell}{(2\pi)^4}. \quad (8.13)$$

Rule	Origin in path integral and/or LSZ	Feynman rule
External lines	replacement of external field insertion via LSZ	1
Internal line	contraction of two variations via a (configuration space) propagator	$i/(p^2 - m^2)$
Momentum Conservation	integration over interaction z_i or external y_i from LSZ	–
Vertex	interaction term $\mathcal{L}_{\text{int}} = g/3!(-i\delta_{z_i})^3$	ig
Loop	Internal contraction resulting in unfixed momentum after imposing momentum conservation	$\int d^4\ell/(2\pi)^4$

Table 8.1: Summary of Feynman rules focusing on their origin from the Path integral formulation of QFT and LSZ reduction.

4. Each vertex comes with a factor of ig .
5. Multiply by some symmetry factor s corresponding to the number of different ways one can write the same diagram. E.g. if you can swap identical internal lines that connect the same vertices (as in our diagram in Fig. 8.1).

Table 8.1 outlines these Feynman rules and their origins from the path integral derivation of correlation functions and LSZ reduction. These Feynman rules are a rigorous approach to obtaining scattering amplitudes.

Let's explicitly apply these rules to calculating our bubble diagram and see that they work. We'll follow the Fig. 8.2 where we've labelled momenta for each line:

$$i\mathcal{M} = (1)^2(ig)^2s \int \frac{d^4\ell}{(2\pi)^4} \frac{i}{[(p_i + \ell)^2 - m^2]} \frac{i}{(\ell^2 - m^2)}, \quad (8.14)$$

$$= sg^2 \int \frac{d^4\ell}{(2\pi)^4} \frac{1}{[(p_i + \ell)^2 - m^2](\ell^2 - m^2)}. \quad (8.15)$$

Our labelling of momenta corresponds to matching our discussion above, but since the measure $d^4\ell$ is invariant under a shift, we could just as easily have chosen the momentum for the lower propagator of the loop to be $\ell + p_i/2$ and that for the upper propagator of the loop $\ell - p_i/2$. These types of reshuffling of loop momenta can sometimes greatly simplify calculations.

Comparing with Eq. 8.1 we can see that our symmetry factor should be $\frac{1}{2}$. Let's take a moment to try to understand these symmetry factors better.

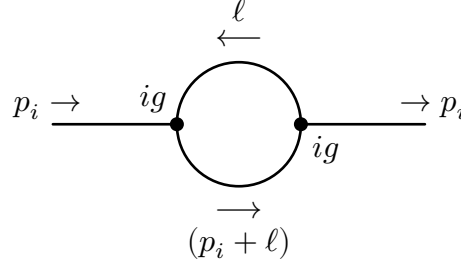


Figure 8.2: Bubble diagram representing an order g^2 correction to the scalar two-point function in ϕ^3 theory. Reproduced from the full set of (disconnected) diagrams in Fig. 4.5.

Symmetry factors

The combinatorics of symmetry factors are probably the least obvious part of using the Feynman rules and Feynman diagrams. In the case of our time-ordered products we had for the order g^2 corrections to the two-point function:

$$\langle \Omega | T[\phi(y_2)\phi(y_1) | \Omega \rangle = \frac{1}{2} \frac{1}{3!} \frac{1}{3!} \left[9 \text{ (diagram 1)} + 6 \text{ (diagram 2)} + 36 \text{ (diagram 3)} + 36 \text{ (diagram 4)} + 18 \text{ (diagram 5)} \right] \quad (8.16)$$

The path integral “counts” 9 different ways to produce the first diagram, 6 to produce the second, and so on. It explicitly takes into account the symmetry in the exchange of the two interaction vertices via the factor $1/2$ from the Taylor expansion of the interaction Lagrangian. That is, the symmetry under the exchange $z_1 \leftrightarrow z_2$ in:

$$\int d^4 z_1 d^4 z_2 \left(\frac{1}{i} \delta_{z_1} \right)^3 \left(\frac{1}{i} \delta_{z_2} \right)^3. \quad (8.17)$$

And we’ve explicitly put in a $1/3!$ for each vertex because we know ϕ^3 is symmetric in the three ϕ s. Dropping for a moment the $1/2$ from the interaction exchange and the $1/3!$ s we can explain the symmetry factor for the bubble diagram we’ve been considering as:

1. **2 options:** Assign the external legs to a vertex, Δ_{y_1, z_1} or Δ_{y_1, z_2} , one this fixed, Δ_{y_2, z_i} is fixed.

2. **3 options:** in connecting the external point y_1 to the chosen $\phi_1(z_i)\phi_2(z_i)\phi_3(z_i)$ any of the three ϕ s can be chosen. The label i on ϕ_i is strictly for counting purposes here.
3. **3 options:** in connecting the external point y_2 to $\phi^3(z_j)$ the same as above applies
4. **2 options:** finally, there are two remaining $\phi_1(z_i)\phi_2(z_i)$ at either vertex, these are contracted across the vertices for the bubble diagram. Choosing to pair $\phi_1(z_1)$ with $\phi_1(z_2)$ with a propagator fixes $\phi_2(z_1)$ to $\phi_2(z_2)$. Alternatively we can pair $\phi_1(z_1)$ with $\phi_2(z_2)$ fixing the other option.

This gives a total of $2 \times 3 \times 3 \times 2 = 36$ different ways to express the diagram, dividing by 2 for the symmetry in $z_1 \leftrightarrow z_2$, then further dividing by $3!$ twice for symmetry in the ϕ s appearing in the interaction Lagrangian gives an overall symmetry factor of $s = \frac{1}{2}$. This is precisely the factor of $\frac{1}{2}$ we have in Eq. 8.1 which was produced directly through applying variations to the generating functional in Eq. 4.73.

Often symmetry factors aren't even needed because so many fields are complex in QED, QCD, and the SM. Complex fields break the exchange symmetries that largely give rise to these combinatorial problems in real scalar theory.

The real problem is, when a symmetry factor pops up and you've forgotten it. At first glance, one might think this would cause a (hopefully) small numerical error in our calculation, but in gauge theories this can cause the calculated observable to be *arbitrary*. Only the sum of all diagrams (at a given order) will prove to be gauge invariant. Breaking gauge symmetry in the presence of our gauge fixing will cause this careful cancellation to fail. An example is (the accidental) explicit dependence of observables on the parameter ξ which, if missed, is a parameter that could arbitrarily change the amplitude which should just be a complex number.

For the combinatorically challenged, computer algebra systems that handle calculations in particle physics automate this by doing all the hard work analogous to our variations. Symmetry factors can be obtained immediately or checked using these programs.

Other interactions

So far we've only discussed $g\phi^3$ theory, but there are infinitely many operators leading to interactions we can consider. We'll discuss this after we introduce the Feynman rules for fermions and vectors which are relatively

8. The Feynman Rules

straightforward additions. We'll make use of our understanding of the generating functional of 1PI diagrams to directly obtain the Feynman rules so we don't have to infer them from known results derived from variations on $Z[J]$.

8.2 Feynman rules for Fermions

For a theory including fermions we'll need to add some additional rules. Some of these will be familiar from our discussion of path integrals for fermions, others we unfortunately don't have time to cover.

In particular we'll have "on-shell spinors" which correspond to the free solutions of the classical EOM for Dirac fields, the Dirac equation. One of these we'll call u_a , which corresponds to an *incoming Dirac particle* it's Dirac conjugate (barred particle) is $\bar{u}_a$ which corresponds to an *outgoing* Dirac particle. The subscript a is a Dirac index, just like those appearing in our propagators and gamma matrices. In contrast we call an *outgoing* Dirac antiparticle v_a and it's Dirac conjugate an *incoming* Dirac antiparticle.

The particle and antiparticle solutions come from the same field, these ideas of particle and antiparticles come directly from the classical solutions of the Dirac equation. For example the positron e^+ is the antiparticle of the electron e^- , and they simply correspond to two distinct solutions of the Dirac equation.

Since we *must* connect Dirac indices (otherwise we'll violate Lorentz invariance), these on-shell spinors for external lines/particles will be connected by propagators and/or interaction vertices. In our diagrammatic methods for the Dirac path integral we had arrows that went from barred to unbarred *field indices*. Now we'll have the arrows flow from unbarred spinors to barred spinors. That is, the arrow will be flowing away from u , v , and flowing toward $\bar{u}$ and $\bar{v}$. This is a convention we would have set if we rigorously derived the LSZ reduction formula.

Internal propagators don't correspond to particles though, they correspond to the time-ordered product of ψ and $\bar{\psi}$. Therefore the arrow in an internal propagator is taken to flow according to the direction fixed by external on-shell spinors. In the case of a closed internal fermion loop, the direction can be chosen at will, but we do not include both directions as separate diagrams.

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We can write the Feynman rules as follows:

1. For external lines we write:
Incoming: u_a for a particle, $\bar{v}_a$ for an antiparticle; arrow away from spinor.
Outgoing: $\bar{u}_a$ for a particle, v_a for an antiparticle; arrow toward spinor.
2. For internal lines we write a fermionic propagator:

$$\Delta_{ab}(p, m) = \frac{i(\gamma^\mu p_\mu + m)_{ab}}{p^2 - m^2} \quad (8.18)$$

3. If the propagator connects external lines, it's arrow must flow with the direction dictated by external lines.
4. If the propagator is part of a closed internal fermion loop, the arrows/flow of the loop is arbitrary. Different flow directions *do not* constitute different diagrams, they're effectively just a relabelling of the loop momentum.
5. The momentum of the propagator is assigned to flow with the arrow. The numerator of the propagator is linear in the momentum, so this is very important. This is not a problem with bosonic propagators which only depend on the momentum quadratically. We can assign momentum flow against the arrow, but in this case we need to take $p \rightarrow -p$ in the numerator of the propagator.
6. If two *identical* external fermion lines are “crossed” (i.e. two particle lines or two antiparticle lines), we introduce a minus sign. This is the minus sign we found for the time-ordered product of four fermion fields in Eq. 6.74. See Fig. 8.3. For n identical external lines, we can (nonphysically) label each leg and calculate the sign by considering which exchanges correspond to *swapping the fermions themselves, rather than just relabeling the vertices*. Relabeling the vertices would correspond to overcounting the number of diagrams. This rule is straightforward to apply in theories where all operators involve at most one bilinear in fermions ($\psi\psi$), theories which involve more bilinears require more care, but we will not encounter them in these notes.
7. If we have a *closed loop of Dirac indices* we introduce a minus sign. This is the minus sign we found for the vacuum bubble diagrams in Eq. 6.46. Recall that the number of minus signs corresponded to the number of loops of closed Dirac indices, *not the number of loops formed of Dirac fermions*. See Fig. 8.4 as well as the discussion surrounding Eq. 6.46.

Many of these rules will become much clearer as we do calculations in theories involving Dirac fields. A clarification of the origin of these rules includes: Spinors for the external lines come from the LSZ reduction formula for fermions, which we discuss briefly at the end of this chapter. The propagator in momentum space results from using the spatial integral and expo-

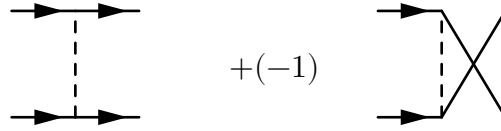


Figure 8.3: Crossed lines for identical Dirac fermions result in an extra factor of -1 . In the right diagram we drop the arrows on two lines, they are implied by the lines to which they are connected. In this diagram no interaction occurs at the crossed fermion lines.

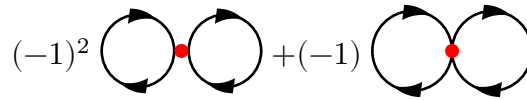


Figure 8.4: Closed loops of Dirac indices contribute a sign (-1) . The left diagram has two closed loops of Dirac fields occurring at the same interaction point (red dot), in the right diagram the Dirac indices *do not close* at the interaction point, instead they only close once over the two loop diagram.

nential for momentum conserving delta functions. The flow of the arrows for fermions follows from careful consideration of the flow coming from the path integral *and* for external fermions setting a convention when introducing the LSZ reduction. The flow of the momentum with the propagator arrow is fixed by the convention for arrows in the presence of LSZ as well as the fact that *order matters* in the Dirac propagator as discussed in Chapter 6. Finally, the special sign rules follow directly from the Grassmann nature of fermionic fields and were discussed at length in Chapter 6.

Common on-shell spinor identities

Now we'll outline without proof some important relations among on-shell spinors we'll need. We'll also start using the Feynman slash notation. For any four-vector p_μ (i.e. not just momentum) we write: $p_\mu \gamma^\mu = \not{p}$.

The spinors are solutions to the Dirac equation, which can be written con-

8. The Feynman Rules

cisely as:

$$(\not{p} - m)_{ab} u_b(p, m) = 0, \quad (8.19)$$

$$\bar{u}(p, m)(\not{p} - m) = 0, \quad (8.20)$$

$$(\not{p} + m)v(p, m) = 0, \quad (8.21)$$

$$\bar{v}(p, m)(\not{p} + m) = 0. \quad (8.22)$$

In the first equivalence we included the Dirac indices for clarity, the rest of the equations can similarly be written in terms of Dirac indices. We'll frequently get terms like $(\not{p}u)$ from putting together our Feynman rules, simplifying the expression using the above on-shell relations will greatly simplify our lives, especially in the Electroweak Standard Model where fermion masses are almost always negligibly small.

Implicit in these on-shell spinors is that they are solutions for spin $\frac{1}{2}$ particles, which have two spin projections (e.g. $s_z = \pm\frac{1}{2}$). Summing over those spin projections will prove *extremely useful and particularly common* in our calculations. These spin sums are given by:

$$\sum_s u_a^{(s)}(p, m) \bar{u}_b^{(s)}(p, m) = (\not{p} + m)_{ab}, \quad (8.23)$$

$$\sum_s v_a^{(s)}(p, m) \bar{v}_b^{(s)}(p, m) = (\not{p} - m)_{ab}. \quad (8.24)$$

When we square amplitudes for calculating cross sections and decay rates we'll end up precisely with the square of these spin sums which we'll write as:

$$(\not{p} + m)_{ab} (\not{p} + m)_{ba} = \text{Tr} [(\not{p} + m)(\not{p} + m)], \quad (8.25)$$

where the trace follows from recalling the trace of a matrix can be written as: $\text{Tr}[M_{ab}] = M_{aa}$. Generally, though, we'll be dealing with many more products of Dirac matrices in the traces. When we do these calculations we'll figure out how to evaluate these traces.

8.3 Feynman Rules for Vectors

As vector fields are bosonic, their rules are basically the same as scalars. Under LSZ reduction, external vector fields (propagators) are replaced with (on-shell) “polarization vectors.” This is analogous to the factor of 1 for scalars and the on-shell spinors for Dirac fermions. We’ll have to choose a gauge for massless vectors which may result in ghost interactions. Without specifying our symmetries we can’t really specify the gauge fixing procedure and resulting ghosts. We’ll generally use R_ξ gauge, as was introduced in Chapter 7, so we’ll specify the corresponding propagator here.

The Feynman rules for vector fields are:

1. For external lines we write:
 incoming: $\epsilon_\mu^a(p)$
 outgoing: $\epsilon_\mu^{a,*}(p)$
 where a labels a vector field (A_μ^a) in the case that there are more than one. If there is only one vector field we can drop this symbol. It is typically dropped in QFT discussions.
2. In the R_ξ gauge our propagator for a massless vector takes the form:

$$\Delta(p^2)_{\mu\nu}^{ab} = \frac{-i\delta^{ab}}{p^2} \left[\eta_{\mu\nu} - (1 - \xi) \frac{p_\mu p_\nu}{p^2} \right]. \quad (8.26)$$

Again, the a, b label the vectors. If there is only one vector this symbol can be dropped.

3. For massive vectors we don’t need to fix a gauge, the propagator is given by:

$$\Delta(p^2, m^2)_{\mu\nu}^{ab} = \frac{-i\delta^{ab}}{p^2 - m^2} \left(\eta_{\mu\nu} - \frac{p_\mu p_\nu}{m^2} \right) \quad (8.27)$$

For one vector, δ^{ab} can be dropped.

We should take a moment to note that when we discuss symmetries of vector fields (Yang Mills theory) we may separate vectors based on which symmetries they transform under. For example in the SM we’ll have one B_μ , three W_μ^i , and eight G_μ^a corresponding to $U(1)_{\text{hypercharge}}$, $SU(2)_{\text{Left}}$, and $SU(3)_{\text{color}}$ respectively.

It’s also important to keep in mind that the form of these propagators assumes no kinetic mixing (e.g. $A_\mu \partial^\mu \partial^\nu B_\nu$) and no mass mixing (e.g. $m_{AB} A^\mu B_\mu$) in the Lagrangian. This is represented by the δ^{ab} which ensures if a vector a “comes in”, it’s the vector a that goes out. It is possible to define propagators in the presence of mixing but we won’t discuss that in these notes.

In these notes, we won't deal with the massive vector propagator in the context presented in Chapter 7. We will see massive vectors in the SM, but they'll gain their mass through "spontaneous symmetry breaking" or the "Higgs mechanism." We will recover the same massive propagator only by taking $\xi \rightarrow \infty$, which is referred to as the "unitary gauge." This will be clearer when we discuss the SM.

Various choices of ξ can be chosen, the most standard are:

- **Feynman gauge**, $\xi = 1$: greatly simplifies the Lorentz part of the propagator.
- **Landau gauge**, $\xi = 0$: the propagator is transverse.
- **'t Hooft gauge**, keep ξ arbitrary: can check explicitly that observables are explicitly ξ independent (as they must be).
- **Unitary gauge**, $\xi \rightarrow \infty$: only works in the presence of a vector mass.

Useful polarization vector identities

Just as we will need some identities for simplifying expressions involving on-shell spinors, we'll need some identities to help us work with polarization vectors. Again, we will not prove them. They are identities that are enforced classically by solving the classical field equations. Again, we'll write them with labels a, b in case there is more than one vector, but they are usually written without this index in the literature.

Polarization vectors are orthogonal:

$$\epsilon_\mu^{a*}(p, \lambda) \epsilon^{b, \mu}(p, \lambda') = -\delta_{\lambda\lambda'} \delta^{ab} \quad (8.28)$$

where λ and λ' label the possible polarizations. The minus sign is convention that comes from the mostly minus metric $\eta_{\mu\nu}$. This orthogonality applies to massless and massive vectors.

In most observables, we'll sum over the vector's possible polarizations. For massless vectors this gives³:

$$\sum_\lambda \epsilon_\mu^{a*}(p, \lambda) \epsilon_\nu^b(p, \lambda) = -\eta_{\mu\nu} \delta^{ab}, \quad (8.29)$$

³More generally we can include unphysical terms resulting from gauge fixing. This is great for checking your calculations' gauge invariance, but we won't work with this more careful polarization sum in these notes.

and for massive vectors we have,

$$\sum_{\lambda} \epsilon_{\mu}^{a*}(p, \lambda) \epsilon_{\nu}(p, \lambda) = \left(-\eta_{\mu\nu} + \frac{p_{\mu} p_{\nu}}{m^2} \right) \delta^{ab}, \quad (8.30)$$

Massless vectors are transverse, giving rise to the identity:

$$p^{\mu} \epsilon_{\mu}^a(p, \lambda) = 0, \quad (8.31)$$

this identity holds generally only for massless vector fields. It is often used in the literature for massive fields in very specific situations, such as the rest frame of the massive vector. Be careful: this identity *does not* generally apply to massive vectors in a moving frame.

8.4 Feynman rules for interactions

In all of this discussion we've only mentioned the interaction term $g\phi^3$. For what follows we'll need more interesting interactions, and to effectively know how to obtain the Feynman rule for any interaction. This is usually sketched in textbooks and rarely presented in a concise approach.

To make this concept concrete, we'll invoke our knowledge of the 1PI generating functional (effective action) from Chapter II. This generating function only gives us 1PI diagrams, for example when we calculated the time ordered product of three scalar fields at order g in Chapter 4 we found:

$$\langle \Omega | T[\phi(y_3)\phi(y_2)\phi(y_1)] | \Omega \rangle = \text{diagram 1} + \text{diagram 2} + \text{diagram 3} + \text{diagram 4} \quad (8.32)$$

The last of these four diagrams is the only 1PI diagram. If we had generated this time-ordered product from Γ instead of $Z[J]$ we would only have obtained the last diagram, which is precisely (up to external propagators) the Feynman rule for the $g\phi^3$ vertex.

Let's repeat that because it's important: to order g , the effective action gives us the Feynman rule for our interaction (the tree-level "vertex function"). So for the purpose of obtaining these tree-level n -point vertex functions we'll simply make the exchange $i\Gamma \rightarrow iS = i \int d^4x \mathcal{L}$.

Let's try this for $g\phi^3$:

$$i\Gamma = i \int d^4x \frac{g}{3!} \phi(x)^3, \quad (8.33)$$

$$\frac{\delta^3 i\Gamma}{\delta\phi(y_1)\delta\phi(y_2)\delta\phi(y_3)} = ig \int d^4x \delta^4(x - y_1)\delta^4(x - y_2)\delta^4(x - y_3) \quad (8.34)$$

This isn't quite our Feynman rule ig , and that's because our Feynman rules are in momentum space. We also may have operators that depend on derivatives, so moving to momentum space is a must⁵. Our next few steps will be to get $i\Gamma (= iS)$ into a convenient form, changing to momentum space, and then figuring out how to extract Feynman rules from this.

⁴Really, what we've observed here is that at tree level the effective action is the classical action.

⁵Technically not a must, but working in configuration space can be tricky (difficult).

Now recall that n variations on Γ gave the 1PI “component” of a connected diagram,

$$i \frac{\delta^n \Gamma}{\delta \phi^n} \bigg|_{\bar{\phi}=0} \equiv i\Gamma^{(n)}. \quad (8.35)$$

That i is actually precisely the i in the Feynman rule for $(g/3!)\phi^3$: ig . We’ve implicitly defined this n -variation vertex as $i\Gamma^{(n)}$ which we’ll use from now on.

With this in mind, we expand the effective action Γ in terms of the 1PI vertices it contains:

$$i\Gamma = \sum_{n=0}^{\infty} \frac{1}{n!} \int d^4x \, i\Gamma^{(n)}(x) \phi^n(x), \quad (8.36)$$

$$= \sum_{n=0}^{\infty} \frac{1}{n!} \int d^4x_1 \cdots d^4x_n \, i\Gamma^n(x_1, \dots, x_n) \left[\prod_{i=1}^n \delta^4(x_i - x_1) \right] \phi(x_1) \cdots \phi(x_n). \quad (8.37)$$

In the first line there are three things to keep in mind: $\Gamma^{(n)}$ is a nontrivial function of derivatives which in general act on different $\phi(x)$ within the ϕ^n . The second is technically for the tadpole and two-point function in this equations we’ve messed up the normalizations we defined in Chapter 5.2. We’re only using this for interactions so the last point won’t matter, except for a quick example below where we rederive the propagator. Finally, the $n!$ in the denominator is only there as a representative of the counting for the vertex. For a $(\phi^\dagger)^m \phi^n$ we would have $m!n!$ in the denominator instead, similarly $\partial\phi$ is treated as distinct from ϕ itself.

In the second line, we’ve inserted a bunch of δ -functions so that we can evaluate each scalar at a different spacetime point. The equivalence between the first and second line follows directly from evaluating all of the delta functions.

We’ll make the change to momentum space by expressing the fields as,

$$\phi(x_i) = \int \frac{d^4p}{(2\pi)^4} \phi(p) e^{-ip \cdot x}, \quad (8.38)$$

where the convention in the exponent of $-ip \cdot x$ fixes that in the vertex function, $i\Gamma^{(n)}$, *all momenta are incoming*.

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As a quick example, consider the $n = 3$ term:

$$\begin{aligned}
& \frac{1}{3!} \int d^4x_1 d^4x_2 d^4x_3 \delta^4(x_2 - x_1) \delta^4(x_3 - x_1) \\
& \times \int \frac{d^4p_1 d^4p_2 d^4p_3}{(2\pi)^{3 \cdot 4}} i\Gamma^{(n)}(x_1, x_2, x_3) e^{-i(p_1 \cdot x_1 + p_2 \cdot x_2 + p_3 \cdot x_3)} \phi(p_1) \phi(p_2) \phi(p_3) \\
& = \frac{1}{3!} \int \frac{d^4p_1 d^4p_2 d^4p_3}{(2\pi)^{3 \cdot 4}} d^4x_1 e^{-i(p_1 + p_2 + p_3) \cdot x_1} i\Gamma^{(n)}(p_1, p_2, p_3) \phi(p_1) \phi(p_2) \phi(p_3)
\end{aligned} \tag{8.39}$$

$$= \frac{1}{3!} \int \frac{d^4p_1 d^4p_2 d^4p_3}{(2\pi)^{3 \cdot 4}} (2\pi)^4 \delta^4(p_1 + p_2 + p_3) i\Gamma^{(n)}(p_i) \phi(p_1) \phi(p_2) \phi(p_3) \tag{8.40}$$

In the second line a lot happened:

First, and it's important it's first, the derivatives in $\Gamma^{(n)}$ acted on the $\exp(-ip_i \cdot x_i)$ so that they pull down the relevant momentum (this will come back shortly and with explicit examples when we get some actual Feynman rules.). Without performing this action first $\Gamma^{(n)}$ has implicit dependence on the x_i .

Then, once all dependence on x_i is in the exponential, we can integrate over the $x_{i \neq 1}$ and get the exponential that only depends on x_1 . Finally in the last line we integrate over x_1 obtaining a total momentum conserving delta function.

Performing this task infinitely many times in Eq. 8.37 gives:

$$i\Gamma = \sum_{n=0}^{\infty} \frac{1}{n!} \int \frac{d^4p_1}{(2\pi)^4} \cdots \frac{d^4p_n}{(2\pi)^4} (2\pi)^4 \delta^4(p_1 + \cdots + p_n) i\Gamma^{(n)}(p_1, \cdots, p_n) \phi(p_1) \cdots \phi(p_n). \tag{8.41}$$

Notice that momentum conservation is enforced at our vertex. Based on our convention of all momenta incoming, we must either label all momenta as incoming when we write down a Feynman diagram or introduce a minus sign for each outgoing momentum.

From here we can obtain the 1PI functions *in momentum space*. This is exactly what we need for our Feynman rules which are designed specifically to give us $i\mathcal{M}$ in momentum space.

The ϕ^3 vertex, revisited

For the ϕ^3 vertex let's try this again, now that we have the momentum space details worked out. We'll write:

$$i\Gamma = i \int d^4x \frac{g}{3!} \phi^3(x) \quad (8.42)$$

$$= \frac{ig}{3!} \int d^4x_1 d^4x_2 d^4x_3 \delta^4(x_2 - x_1) \delta^4(x_3 - x_1) \phi(x_1) \phi(x_2) \phi(x_3) \quad (8.43)$$

$$= \frac{ig}{3!} \int d^4x_1 d^4x_2 d^4x_3 \delta^4(x_2 - x_1) \delta^4(x_3 - x_1) \\ \times \int \frac{d^4p_1 d^4p_2 d^4p_3}{(2\pi)^{3 \cdot 4}} \phi(p_1) \phi(p_2) \phi(p_3) e^{-i(p_1 \cdot x_1 + p_2 \cdot x_2 + p_3 \cdot x_3)} \quad (8.44)$$

$$(8.45)$$

This is the point at which, if we had derivatives, we would stop to apply them to the relevant exponentials and pull down momentum dependence.

Lacking any derivatives we can move on to the next step, which is setting all the $x_i = x_1$ with the first delta function, then getting an overall momentum conserving delta function:

$$i\Gamma = \frac{ig}{3!} \int \frac{d^4p_1 d^4p_2 d^4p_3}{(2\pi)^{3 \cdot 4}} (2\pi)^4 \delta^4(p_1 + p_2 + p_3) \phi(p_1) \phi(p_2) \phi(p_3). \quad (8.46)$$

Having agreed that our Feynman rule is $i\Gamma^{(3)}$, we now need to take three variations of this effective action to obtain the Feynman rule. We'll agree our variations are with respect to the momenta k_i and separately that we can write $\delta^4(p_i - k_j) = \delta_{ij}$. This is because we're going to have, unsurprisingly, $3! = 6$ terms to fit on the page.

Taking the variations we have:

$$\frac{\delta^3 i\Gamma}{\delta\phi(k_1) \delta\phi(k_2) \delta\phi(k_3)} = \frac{ig}{3!} \int \frac{d^4p_1 d^4p_2 d^4p_3}{(2\pi)^{(3-1) \cdot 4}} \delta^4(p_1 + p_2 + p_3) \left[\delta_{11} \delta_{22} \delta_{33} \right. \\ \left. + \delta_{11} \delta_{23} \delta_{32} + \delta_{12} \delta_{21} \delta_{33} + \delta_{12} \delta_{23} \delta_{31} \right. \\ \left. + \delta_{13} \delta_{22} \delta_{31} + \delta_{13} \delta_{21} \delta_{32} \right], \quad (8.47)$$

$$= \frac{ig}{(3\pi)^{3 \cdot 4}} (2\pi)^4 \delta^4(k_1 + k_2 + k_3). \quad (8.48)$$

8. The Feynman Rules

All six of the momentum delta functions coming from the variations trivially gives us the delta function in momenta. If we had considered an operator depending on momenta we would have had momentum dependence and this could have been *much more complicated*.

Now we just need to relate this to $i\Gamma^{(3)}(p_1, p_2, p_3)$ in Eq. 8.41. We'll take the same three variations which will then allow us to extract the vertex function, i.e. the Feynman rule for the vertex:

$$\left. \frac{\delta^3 i\Gamma}{\delta\phi(k_1)\delta\phi(k_2)\delta\phi(k_3)} \right|_{\phi=0} = \frac{1}{3!} \int \frac{d^4 p_1 d^4 p_2 d^4 p_3}{(2\pi)^{3 \cdot 4}} (2\pi)^4 \delta^4(p_1 + p_2 + p_3) \\ \times i\Gamma^{(3)}(p_1, p_2, p_3) \left[\delta_{11} \delta_{22} \delta_{33} + \dots \right] \quad (8.49)$$

$$= \frac{i}{(2\pi)^{3 \cdot 4}} (2\pi)^4 \delta^4(k_1 + k_2 + k_3) \Gamma^{(3)}(k_1, k_2, k_3). \quad (8.50)$$

The δ s in the square brackets are precisely the ones from above, so we get a factor of $3!$ that cancels the denominator. The vertex function is then precisely the Feynman rule we're looking for:

$$i\Gamma^{(3)} = ig. \quad (8.51)$$

This is an important, and opportune, time to mention that there is no law that states our interaction Lagrangian should have the $3!$:

$$\mathcal{L} = \frac{g}{3!} \phi^3. \quad (8.52)$$

If we dropped the $3!$ we would have for our Feynman rule, $i\Gamma^{(3)} = 6ig$. This does affect our discussion of the symmetry factors in Sec. 8.1. The path integral doesn't know about the symmetry of the vertex in the exchange of the three fields, we've just artificially built that in. This gives us very nice symmetry factors as shown in Eq. 8.16.

If we dropped the $3!$, nothing changes about the counting for the symmetry factors we just no longer *normalize to the symmetry in the vertex* under exchange of fields. So our symmetry factors would instead be:

$$\langle \Omega | T[\phi(y_2)\phi(y_1)] | \Omega \rangle = \left[\frac{9}{2} \text{---} \text{---} \text{---} + 3 \text{---} \text{---} \text{---} + 18 \text{---} \text{---} \text{---} + 18 \text{---} \text{---} \text{---} + 9 \text{---} \text{---} \text{---} \right] \quad (8.53)$$

We should immediately stop and ask ourselves: does this change the physics? The answer is no. If we renamed the electron charge $e \rightarrow e/2$, the fine structure constant we actually extract from experiment ($\alpha = e^2/4/\pi$) would just be $e^2/16/\pi$ instead. We'll discuss this *a lot* between “renormalization schemes” when we cover QED and QCD, and “input parameter” schemes when we cover the SM.

In terms of sheer convenience, using $1/3!$ as a normalization wins every time. Remember, our perturbative expansion is an expansion in small g . If we have terms running around like $18g$ we might have a hard time gauging “small”. When the vertex is simply g the idea of small is a bit clearer. We'll discuss this more as well when we talk about renormalization. There we can get an idea of what is “small” by comparing g and g^2 contributions to the same process. The general (but also loose) rule turns out to be, assuming a normalization of $1/3!$: $g < 4\pi$. Any rational particle physicist insists instead: $g \ll 4\pi$.

The scalar propagator, *again*

Let's see how this with the easiest possible example for including momenta. We can reproduce our momentum space propagator in a straightforward manner. As we mentioned when we expanded $i\Gamma$ in terms of the various $\Gamma^{(n)}$, Eq. 8.36, our normalization chosen there corresponded to interaction vertices. For the propagator we need to relate the time ordered product of two fields to variations of the effective action, from Eqs. 5.10, 5.24, and 5.26 we find:

$$\Delta = \frac{\delta^2 Z[J]}{Z[J]} \Big|_{J=0} = i\delta^2 W[J] \Big|_{J=0} = i(\delta^2 \Gamma)^{-1} \Big|_{\bar{\phi}=0}. \quad (8.54)$$

$$(8.55)$$

We've suppressed the arguments of the variations, but they follow directly from Chapter 5.2. We've also set the tadpoles δZ to zero: for our free Lagrangian they are zero.

Let's evaluate this for the free Lagrangian for a real scalar field. To complicate our lives, and to better demonstrate the task at hand for more general cases, we'll write the free Lagrangian as:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2 \phi^2. \quad (8.56)$$

That is, we didn't use IBP to put both derivatives on one field.

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We start by rewriting the action in momentum space:

$$\begin{aligned} \int d^4x \mathcal{L} = & \int d^4x_1 \left[\frac{1}{2} [\partial_{x_1, \mu} \phi(x_1)] \int d^4x_2 \delta^4(x_2 - x_1) [\partial_{x_2}^\mu \phi(x_2)] \right. \\ & \left. - \int d^4x_2 \delta^4(x_2 - x_1) \frac{1}{2} m^2 \phi(x_1) \phi(x_2) \right]. \end{aligned} \quad (8.57)$$

Notice the way we directly made the “change of variables” to x_2 *after the remaining x_1 dependence* for the first term. We can see from this perspective how the derivative changes from argument x_1 to x_2 . With this more careful understanding, we will always just jump to the following expression:

$$\int d^4x \mathcal{L} = \int d^4x_1 d^4x_2 \delta^4(x_2 - x_1) \left[\frac{1}{2} [\partial_{x_1, \mu} \phi(x)] [\partial_{x_2}^\mu \phi(x_2)] - \frac{1}{2} m^2 \phi(x_1) \phi(x_2) \right], \quad (8.58)$$

$$= \int d^4x_1 d^4x_2 \delta^4(x_2 - x_1) \int \frac{d^4p_1 d^4p_2}{(2\pi)^{2 \cdot 4}} \quad (8.59)$$

$$\times \left[\frac{1}{2} \underbrace{[\partial_{x_1, \mu} e^{-ip_1 \cdot x_1} \phi(p_1)]}_{=-ip_{1, \mu} e^{-ip_1 \cdot x_1}} [\partial_{x_2}^\mu e^{-ip_2 \cdot x_2} \phi(p_2)] - \frac{1}{2} m^2 e^{-ip_1 \cdot x_1} \phi(p_1) e^{-ip_2 \cdot x_2} \phi(p_2) \right]. \quad (8.60)$$

Notice we’ve explicitly labelled the derivatives with which argument (x_i) they act on. In momentum space, the derivatives only act on the exponentials, pulling down factors of momenta. Since there are “just numbers,” i.e. not operators like derivatives, we can freely move them around. Applying all the derivatives “uses them up,” then we are free to use the spacetime delta function then rewrite the exponentials in terms of a delta function:

$$\int d^4x \mathcal{L} = \int \frac{d^4p_1 d^4p_2}{(2\pi)^{2 \cdot 4}} (2\pi)^4 \delta^4(p_1 + p_2) \frac{1}{2} (-p_1 \cdot p_2 - m^2) \phi(p_1) \phi(p_2), \quad (8.61)$$

$$= \int \frac{d^4p_1 d^4p_2}{(2\pi)^{2 \cdot 4}} (2\pi)^4 \delta^4(p_1 + p_2) \frac{1}{2} (p_1^2 - m^2) \phi(p_1) \phi(p_2). \quad (8.62)$$

In the second line we’ve just rewritten $p_2 = -p_1$ using the delta function.

Following Eq. 8.54 we’ll need to take two variations of this effective action

and our general effective action:

$$\frac{\delta^2 S}{\delta\phi(k_1)\delta\phi(k_2)} = \frac{1}{(2\pi)^{2\cdot 4}} (2\pi)^4 \delta^4(k_1 + k_2) (k_1^2 - m^2) \quad (8.63)$$

$$\frac{\delta^2 \Gamma}{\delta\phi(k_1)\delta\phi(k_2)} = \frac{1}{(2\pi)^{2\cdot 4}} (2\pi)^4 \delta^4(k_1 + k_2) \Gamma^{(2)}(k_1, k_2) \quad (8.64)$$

We have $\Gamma^{(2)} = k_1^2 - m^2$. In configuration space, we had $\Delta = i(\delta^2 \Gamma)^{-1}$. In momentum space we instead have:

$$\Delta(k^2, m^2) = i(\Gamma^{(2)})^{-1} = \frac{i}{k^2 - m^2}. \quad (8.65)$$

If it's not clear why this is the case in momentum space, rewrite the relation,

$$\Delta(x - y) = i \left(\frac{\delta^2 \Gamma}{\delta\phi(x_1)\delta\phi(x_2)} \right)^{-1}, \quad (8.66)$$

by transforming to momentum space. It's the same procedure we followed for writing the general form of Γ in momentum space.

A momentum dependent interaction

Let's consider a momentum dependent interaction. This will be directly analogous to the propagator example, but with one extra field. It serves as one last example. We'll go through this quite quickly compared to the last two examples, deriving Feynman rules isn't an overly tedious task once we have the understanding we developed in the last two sections.

Consider the interaction:

$$\mathcal{L} = c\phi(\partial_\mu\phi)(\partial^\mu\phi). \quad (8.67)$$

We rewrite this interaction Lagrangian in momentum space following the

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same steps as before:

$$iS = i \int d^4x_1 d^4x_2 d^4x_3 \delta^4(x_2 - x_1) \delta^4(x_3 - x_1) c\phi(x_1) [\partial_{x_2,\mu} \phi(x_2)] [\partial_{x_3}^\mu \phi(x_3)] , \quad (8.68)$$

$$= i \int d^4x_1 d^4x_2 d^4x_3 \delta^4(x_2 - x_1) \delta^4(x_3 - x_1) \quad (8.69)$$

$$\times c \int \frac{d^4p_1 d^4p_2 d^4p_3}{(2\pi)^{3\cdot 4}} e^{-ip_1 \cdot x_1} \phi(p_1) (-ip_{2,\mu}) e^{-ip_2 \cdot x_2} \phi(p_2) (-ip_3^\mu) e^{-ip_3 \cdot x_3} \phi(p_3) \quad (8.70)$$

$$= ic \int \frac{d^4p_1 d^4p_2 d^4p_3}{(2\pi)^{3\cdot 4}} (2\pi)^4 \delta^4(p_1 + p_2 + p_3) (-p_2 \cdot p_3) \phi(p_1) \phi(p_2) \phi(p_3) \quad (8.71)$$

We aren't done yet, now we need to take variations with respect to $\phi(k_1)$, $\phi(k_2)$, and $\phi(k_3)$. This will give us a result from which we can directly extract $i\Gamma^{(3)}$. These variations give the same 6 combinations of delta functions as in the ϕ^3 vertex case (see Eq. 8.47). This time, they aren't trivial because we have momentum dependence in our expression! This is going to effectively take into account that the path integral, or ourselves in the case of using the Feynman rules, doesn't know which ϕ in the vertex is connected to which other ϕ s via propagators.

$$\begin{aligned} & \frac{\delta^3 iS}{\delta\phi(k_1) \delta\phi(k_2) \delta\phi(k_3)} \\ &= ic \int \frac{d^4p_1 d^4p_2 d^4p_3}{(2\pi)^{3\cdot 4}} (2\pi)^4 \delta^4(p_1 + p_2 + p_3) (-p_2 \cdot p_3) \\ & \times \left[\delta_{11} \delta_{22} \delta_{33} + \delta_{11} \delta_{23} \delta_{32} + \delta_{12} \delta_{21} \delta_{33} \right. \end{aligned} \quad (8.72)$$

$$\begin{aligned} & \left. + \delta_{12} \delta_{23} \delta_{31} + \delta_{13} \delta_{21} \delta_{32} + \delta_{13} \delta_{22} \delta_{31} \right] , \\ &= -\frac{ic}{(2\pi)^{3\cdot 4}} (2\pi)^4 \delta^4(k_1 + k_2 + k_3) \times 2 [k_1 \cdot k_2 + k_1 \cdot k_3 + k_2 \cdot k_3] . \end{aligned} \quad (8.73)$$

Notice that the variations effectively symmetrized our expression in terms of the momenta.

We already have the triple variations applied to the effective action from our ϕ^3 example. We found:

$$\left. \frac{\delta^3 i\Gamma}{\delta\phi(k_1)\delta\phi(k_2)\delta\phi(k_3)} \right|_{\phi=0} = \frac{i}{(2\pi)^{3\cdot 4}} (2\pi)^4 \delta^4(k_1 + k_2 + k_3) \Gamma^{(3)}(k_1, k_2, k_3) \quad (8.74)$$

And so we've found that our vertex Feynman rule for the interaction in Eq. 8.67 is given by:

$$\Gamma^{(3)}(k_1, k_2, k_3) = -2ic [k_1 \cdot k_2 + k_1 \cdot k_3 + k_2 \cdot k_3] \quad (8.75)$$

Including Grassmann variables

Remember: order matters for Grassmann variables. We'll take a very simple approach to this and always pair our variations as:

$$\frac{\delta^2}{\delta\psi(k_2)\delta\bar{\psi}(k_1)} . \quad (8.76)$$

This choice avoids signs as $\delta\bar{\psi}$ will act directly on the $\bar{\psi}$ which is always to the left of ψ . For four- or more fermion operators, the variations will *commute through any bilinears* $\bar{\psi}\psi$, as they belong to the bosonic subalgebra.

Consider the interaction Lagrangian:

$$\mathcal{L}_I = c(\bar{\psi}_a \Gamma_{ab}^{(1)} \psi_b)(\bar{\psi}_c \Gamma_{cd}^{(2)} \psi_d) . \quad (8.77)$$

Where $\Gamma^{(1)}$ and $\Gamma^{(2)}$ are any elements of either of the bases of gamma matrices we defined in Tab. 2.1. The combination of the two *must be Lorentz invariant*, so if $\Gamma^{(1)}$ has open Lorentz indices, then $\Gamma^{(2)}$ has the same open Lorentz indices.

By our convention in Eq. 8.76, we obtain the Feynman rule by associating the interaction Lagrangian with our effective action:

$$\begin{aligned} & \frac{\delta^4 iS}{\delta\psi_{a'}(k_1)\delta\bar{\psi}_{b'}(k_2)\delta\psi_{c'}(k_3)\delta\bar{\psi}_{d'}(k_4)} \\ &= \frac{\delta^4}{\delta\psi_{a'}(k_1)\delta\bar{\psi}_{b'}(k_2)\delta\psi_{c'}(k_3)\delta\bar{\psi}_{d'}(k_4)} \int \frac{d^4 p_1 d^4 p_2 d^4 p_3 d^4 p_4}{(2\pi)^{4\cdot 4}} \\ & \quad \times (2\pi)^4 \delta^4(p_1 + p_2 + p_3 + p_4) c[\bar{\psi}(p_1)_a \Gamma_{ab}^{(1)} \psi_b(p_2)][\bar{\psi}_c(p_3) \Gamma_{cd}^{(2)} \psi_d(p_4)] \end{aligned} \quad (8.78)$$

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Applying the variations we find:

$$\begin{aligned} & \frac{\delta^4 iS}{\delta\psi_{a'}(k_1)\delta\bar{\psi}_{b'}(k_2)\delta\psi_{c'}(k_3)\delta\bar{\psi}_{d'}(k_4)} \\ &= \frac{1}{(2\pi)^{4\cdot 4}} (2\pi)^4 \delta^4(k_1 + k_2 + k_3 + k_4) \end{aligned} \quad (8.79)$$

$$\times ic \left[\Gamma_{a'b'}^{(1)} \Gamma_{c'd'}^{(2)} - \Gamma_{a'd'}^{(1)} \Gamma_{c'b'}^{(2)} - \Gamma_{c'b'}^{(1)} \Gamma_{a'd'}^{(2)} + \Gamma_{c'd'}^{(1)} \Gamma_{a'b'}^{(2)} \right] \quad (8.80)$$

And we find that our Feynman rule is:

$$i\Gamma_{abcd}^{(4)} = ic \left[\Gamma_{ab}^{(1)} \Gamma_{cd}^{(2)} - \Gamma_{ad}^{(1)} \Gamma_{cb}^{(2)} - \Gamma_{cb}^{(1)} \Gamma_{ad}^{(2)} + \Gamma_{cd}^{(1)} \Gamma_{ab}^{(2)} \right] \quad (8.81)$$

As we develop models like QED and QCD, we'll revisit this to get the Feynman rules for the interactions. QED has only one interaction vertex, QCD has three, the Standard Model has dozens. We'll only derive Feynman rules we need, but now we have the tools to derive any Feynman rule for any theory we can think up. In practice many physicists use computer based tools to derive Feynman rules.

8.5 A closer look at LSZ

In this section we will take a quick look at the LSZ reduction formula. We'll start by motivating the classical asymptotic one-particle states we used for the Feynman rules for external lines then we'll state the LSZ reduction for scalars, Dirac fermions, and vectors. To do this with any level of rigor is beyond these notes due to time constraints, but technically straightforward. We would just need to solve some differential equations.

Recall from our discussions of propagators for scalars, Dirac fermions, and vectors that the “canonically” normalized Lagrangians (i.e. Lagrangians resulting in propagators with physical poles and residues) were given by:

$$\mathcal{L}_\phi = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2 \phi^2, \quad (8.82)$$

$$\mathcal{L}_\psi = i\bar{\psi}\gamma^\mu \partial_\mu \psi - m\bar{\psi}\psi, \quad (8.83)$$

$$\mathcal{L}_A = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}. \quad (8.84)$$

The equations of motion for these three fields follow from the Euler Lagrange equation we derived in Eq. 4.27. They take the form:

$$(\square + m^2)\phi = 0, \quad (8.85)$$

$$(i\gamma^\mu \partial_\mu - m)\psi = 0, \quad (8.86)$$

$$\partial^\mu F_{\mu\nu} = (\square \delta_\nu^\mu + \partial^\mu \partial_\nu)A_\mu = 0. \quad (8.87)$$

Equation 8.85 is the Klein-Gordon equation, which admits plane wave solutions:

$$\phi(x) \sim 1 \times e^{-ip \cdot x}, \quad 1 \times e^{+ip \cdot x}. \quad (8.88)$$

Those “1”s we’ve written explicitly are the same “1” we assign for each external line in our Feynman rules. They just indicate that the scalar has no internal structure. We’ll upgrade them to objects with components for our fields with components; i.e. to on-shell spinors and polarization vectors. The exponential factors are the plane-wave terms that appear in the LSZ reduction formula we wrote in Eq. 8.7. By convention for incoming particles we choose the exponential with a minus sign, and for outgoing the positive sign. Altogether these solutions are our “asymptotic single particle states.”

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Next we can take a look at the “Dirac equation” in Eq. 8.86. If we multiply the Dirac equation by $(i\gamma^\nu\partial_\nu + m)$ we again obtain the Klein-Gordon equation:

$$0 = (i\gamma^\nu\partial_\nu + m)(i\gamma^\mu\partial_\mu - m)\psi, \quad (8.89)$$

$$= (-\gamma^\nu\gamma^\mu\partial_\nu\partial_\mu - m^2)\psi, \quad (8.90)$$

$$= -(\square + m^2)\psi \quad (8.91)$$

The third line follows from using that derivatives commute, which allows us to symmetrized in μ, ν without changing the expression, which then allows us to apply the Clifford Algebra:

$$\gamma^\nu\gamma^\mu\partial_\nu\partial_\mu = \frac{1}{2}\{\gamma^\nu, \gamma^\mu\}\partial_\nu\partial_\mu = \eta^{\nu\mu}\partial_\nu\partial_\mu = \square \quad (8.92)$$

Since the Dirac equation satisfies the Klein-Gordon equation, we know it also admits plane-wave solutions. The internal (spin) structure of the Dirac equation is encoded in the “on-shell spinors” we introduced in Chapter 8.2. The Dirac field ψ has four components, these are encoded in pairs in the spinor for particles u and the spinor for antiparticles v .

Then we can write solutions to the classical Dirac equation (roughly) as:

$$\psi(x) = u(p)e^{-ip \cdot x}, \quad v(p)e^{ip \cdot x} \quad (8.93)$$

The signs of the exponentials are not convention, they’re fixed by solving the Dirac equation. Comparing with the interpretation of the exponentials for the scalars explains why our Feynman rule for an *incoming particle* uses a spinor u and for an *outgoing antiparticle* a spinor v . The Dirac conjugate, $\psi^\dagger\beta$, of this field solution then tells us an incoming antiparticle requires with a $\bar{v}$, while for an outgoing particle we write $\bar{u}$.

Finally for massless vectors, if we assume a plane wave solution,

$$A_\mu \sim \epsilon_\mu(p)e^{-ip \cdot x}, \quad (8.94)$$

where $\epsilon_\mu(p)$ is now our polarization vector. Plugging this solution into Eq. 8.87 we find:

$$0 = (\square\delta_\nu^\mu + \partial^\mu\partial_\nu)A_\mu, \quad (8.95)$$

$$= -p^2\epsilon_\nu - (p \cdot \epsilon)p_\nu \quad (8.96)$$

For physical massless vectors we have $p^2 = 0$ and the transverse property of massless vectors gives $p \cdot \epsilon = 0$. So the plane wave solution in Eq. 8.94 works and we conclude we have the general solution:

$$A_\mu \sim \epsilon_\mu(p) e^{-ip \cdot x}, \quad \epsilon_\mu^*(p) e^{ip \cdot x}. \quad (8.97)$$

Following the discussion above we conclude that ϵ_μ is appended to external incoming lines, while ϵ_μ^* is appended for outgoing vectors.

For massive vectors the discussion is similar, and we append the same polarization vectors. In the massive case the polarization vector has an extra longitudinal component in addition to the two transverse components for the massless vector.

With these concepts established we can now take a closer look at the general LSZ reduction for scalars, Dirac fermions, and vectors. Let's start by defining an operator that amputates the external propagator from our time ordered product and adjoins an asymptotic single particle state. For an incoming scalar particle we'll write this as:

$$\mathcal{O}_{\text{in},i}^\phi = -i \int d^4 y_i e^{-ip_i \cdot y_i} (p_i^2 - m^2). \quad (8.98)$$

This is precisely what we had for the incoming particle in Eq. 8.7. The $-i(p_i^2 - m^2)$ is the inverse propagator (in momentum space), so it removes the propagator as promised. The integral over the exponential is the solution to the Klein-Gordon equation we presented in Eq. 8.88. We write “ y_i ,” but notice from Eqs. 8.7 and 8.6 that this is just shorthand for “take y_i to be the spacetime point corresponding to the external field we're amputating in the time-ordered product.”

For an outgoing scalar particle we have instead:

$$\mathcal{O}_{\text{out},i}^\phi = -i \int d^4 y_i e^{ip_i \cdot y_i} (p_i^2 - m^2). \quad (8.99)$$

The only thing that's changed, in this case, is the sign of the exponential.

Fermions are analogous to the scalars, except now we have the spinors to account for. So we can write for a particle:

$$\mathcal{O}_{\text{in},i}^u = -i \int d^4 y_i u(p_i, m) (\gamma^\mu p_{\mu,i} - m) e^{-ip_i \cdot y_i}, \quad (8.100)$$

$$\mathcal{O}_{\text{out},i}^{\bar{u}} = -i \int d^4 y_i \overline{(\gamma^\mu p_{\mu,i} - m)} \bar{u}(p_i, m) e^{ip_i \cdot y_i}. \quad (8.101)$$

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Notice that for incoming particles the operator acts to the right, removing the propagator and appending the spinor solution. For outgoing it acts to the left and appends a $\bar{u}$ on the right. The ordering is important as the γ matrices do not commute.

And separately for antiparticles:

$$\mathcal{O}_{\text{in},i}^{\bar{v}} = -i \int d^4 y_i \overleftarrow{(\gamma^\mu p_{\mu,i} + m)} \bar{v}(p_i, m) e^{-ip_i \cdot y_i}, \quad (8.102)$$

$$\mathcal{O}_{\text{out},i}^v = -i \int d^4 y_i v(p_i, m) (\gamma^\mu p_{\mu,i} + m) e^{ip_i \cdot y_i}. \quad (8.103)$$

The $+$ sign here isn't a mistake – it comes from the way the v spinor is defined over positive energy states.

And finally for vectors, where we'll include a mass that can simply be taken to zero for a massless vector, we have:

$$\mathcal{O}_{\text{in},i}^\epsilon = -i \int d^4 y_i [(p_i^2 - m^2) \eta^{\mu\nu} - p_i^\mu p_i^\nu] \epsilon_\mu(p) e^{-ip_i \cdot y_i}, \quad (8.104)$$

$$\mathcal{O}_{\text{out},i}^{\epsilon^*} = -i \int d^4 y_i [(p_i^2 - m^2) \eta^{\mu\nu} - p_i^\mu p_i^\nu] \epsilon_\mu^*(p) e^{ip_i \cdot y_i}. \quad (8.105)$$

All that remains is to put it all together for a general process. We write for n incoming and $\bar{n}$ outgoing scalars, m incoming and $\bar{m}$ outgoing Dirac particles, o incoming and $\bar{o}$ outgoing Dirac antiparticles, and p incoming and $\bar{p}$ outgoing vector fields:

$$\begin{aligned} \langle f | S | i \rangle &= \left(\prod_{i=1}^n \mathcal{O}_{\text{in},i}^\phi \right) \left(\prod_{\bar{i}=1}^{\bar{n}} \mathcal{O}_{\text{out},\bar{i}}^\phi \right) \left(\prod_{j=1}^m \mathcal{O}_{\text{in},j}^u \right) \left(\prod_{\bar{j}=1}^{\bar{m}} \mathcal{O}_{\text{out},\bar{j}}^{\bar{u}} \right) \\ &\times \left(\prod_{k=1}^o \mathcal{O}_{\text{in},k}^{\bar{v}} \right) \left(\prod_{\bar{k}=1}^{\bar{o}} \mathcal{O}_{\text{out},\bar{k}}^v \right) \left(\prod_{l=1}^p \mathcal{O}_{\text{in},l}^\epsilon \right) \left(\prod_{\bar{l}=1}^{\bar{p}} \mathcal{O}_{\text{out},\bar{l}}^{\epsilon^*} \right) \\ &\times \langle \Omega | T[\phi^{n+\bar{n}}, \psi^{m+o}, \bar{\psi}^{\bar{m}+\bar{o}}, A^{p+\bar{p}}] | \Omega \rangle. \end{aligned} \quad (8.106)$$

The time ordered product at the end, the correlation function for a bunch of fields, is written very compactly. We need to, naturally, introduce separate external spacetime points at which each field is evaluated. Which field in the time-ordered product corresponds to in and out states are fixed by our choices in the $\mathcal{O}$.

This sketch hopefully elucidates how we use the LSZ reduction to obtain matrix elements. The LSZ reduction takes us from time ordered products of fields (QFT) to the scattering amplitude for single particle states. It can be, and generally needs to be, extended for more than one type of scalar, Dirac fermion, or vector.

For example in the QED there's one vector and one Dirac fermion, but in QCD there are eight vectors (one 8 dimension irrep of $SU(3)_{\text{color}}$) and (up to) 6 Dirac fermions. In the SM the scalar boson is a complex field, so this needs to be extended to complex scalars. All of this follows directly from what we've already seen.

At the end of Chapter 4.1 we mentioned that we didn't actually have to work with canonically normalized fields. Physical scattering properties only need to be imposed on the observables, not on our time-ordered products. By modifying LSZ for non-canonically normalized fields we can still relate time-ordered products back to physical processes.

Consider the non-canonically normalized scalar free Lagrangian,

$$\mathcal{L} = Z_\phi \frac{1}{2} (\partial_\mu \phi)(\partial^\mu \phi) - Z_m \frac{m^2}{2} \phi^2, \quad (8.107)$$

Where we now have some “Z-factors” that shift the kinetic and mass terms. Our propagator in this case looks like:

$$\Delta(p^2, m^2) = \frac{i}{Z_\phi p^2 - Z_m m^2}. \quad (8.108)$$

In Eq. 4.4 this corresponds to $a_1 = -Z_\phi/2$ and $a_2 = -Z_m/2$, so we can check this propagator against Eq. 4.11.

Immediately we can see that our LSZ reduction formula is going to require some readjustment. Our operators defined in Eqs. 8.98 and 8.99 become:

$$\mathcal{O}_{\text{in},i}^\phi = -i \int d^4 y_i e^{-ip_i \cdot y_i} (Z_\phi p_i^2 - Z_m m^2), \quad (8.109)$$

$$\mathcal{O}_{\text{out},i}^\phi = -i \int d^4 y_i e^{ip_i \cdot y_i} (Z_\phi p_i^2 - Z_m m^2). \quad (8.110)$$

The factors $-i(Z_\phi p_i^2 - Z_m m^2)$ exactly cancel the propagator for the external field, just as in the canonically normalized case.

With this small adjustment, we can now perform calculations in our QFT using non-canonically normalized kinetic terms. This becomes relevant when

we discuss renormalizability. When we cancel infinite terms coming from loop diagrams we'll get corrections to these Z -factors even if we started with canonical form for our fields. It won't always make sense to go through the effort of redefining everything so we retain a canonical form. In those cases we'll want this more general approach to LSZ reduction.

For Dirac fermions we write,

$$\mathcal{L} = iZ_\psi \bar{\psi} \gamma^\mu \partial_\mu \psi - Z_m \bar{\psi} \psi, \quad (8.111)$$

resulting in the following changes to our LSZ operators:

$$\mathcal{O}_{\text{in},i}^u = -i \int d^4 y_i u(p_i, m_P) (Z_\psi \gamma^\mu p_{\mu,i} - Z_m m) e^{-ip_i \cdot y_i}, \quad (8.112)$$

$$\mathcal{O}_{\text{out},i}^{\bar{u}} = -i \int d^4 y_i \overline{(Z_\psi \gamma^\mu p_{\mu,i} - Z_m m)} \bar{u}(p_i, m_P) e^{ip_i \cdot y_i}, \quad (8.113)$$

$$\mathcal{O}_{\text{in},i}^{\bar{v}} = -i \int d^4 y_i \overline{(Z_\psi \gamma^\mu p_{\mu,i} + Z_m m)} \bar{v}(p_i, m_P) e^{-ip_i \cdot y_i}, \quad (8.114)$$

$$\mathcal{O}_{\text{out},i}^v = -i \int d^4 y_i v(p_i, m_P) (Z_\psi \gamma^\mu p_{\mu,i} + Z_m m) e^{ip_i \cdot y_i}. \quad (8.115)$$

Notice in the on-shell spinors we've changed $m \rightarrow m_P$. We're distinguishing the pole mass (m_P) from the “mass” appearing in the (inverse) propagator! The external spinor lines are physical spinors and so they are functions of the physical pole mass. We replace the unphysical (non-canonically normalized) external propagators with the physical on-shell one-particle states.

This is really just bookkeeping and not as substantial as it seems. We'll discuss it more when we do renormalization, in particular when we discuss “renormalization schemes.”

For vectors more care is needed due to subtleties relating to gauge fixing. It turns out in general we need to renormalize the gauge fixing parameter ξ as well. This is beyond these notes so we'll simply state for a massive vector field our LSZ reduction operators look like:

$$\mathcal{O}_{\text{in},i}^\epsilon = -i \int d^4 y_i [(Z_A p_i^2 - Z_m m^2) \eta^{\mu\nu} - Z_A p_i^\mu p_i^\nu] \epsilon_\mu(p) e^{-ip_i \cdot y_i}, \quad (8.116)$$

$$\mathcal{O}_{\text{out},i}^{\epsilon^*} = -i \int d^4 y_i [(Z_A p_i^2 - Z_m m^2) \eta^{\mu\nu} - Z_A p_i^\mu p_i^\nu] \epsilon_\mu^*(p) e^{ip_i \cdot y_i}. \quad (8.117)$$

Chapter 9

Common Formulae

9.1 Gaussian Integrals

Not many integrals can be solved exactly, the most important integral for us will be the Gaussian,

$$\int dp \exp\left(-\frac{1}{2}ap^2 + Jp\right) = \sqrt{\frac{2\pi}{a}} \exp\left(\frac{J^2}{2a}\right). \quad (9.1)$$

When an integral is too hard to solve, we'll just approximate it as a Gaussian with a small perturbation, and solve the integral perturbatively.

Element	Bilinear	Bilinear in terms of LR	mixes chirality?
$\mathbb{1}$	$\bar{\psi}\chi$	$\bar{\psi}_L\chi_R + \bar{\psi}_R\chi_L$	yes
γ^μ	$\bar{\psi}\gamma^\mu\chi$	$\bar{\psi}_L\gamma^\mu\chi_L + \bar{\psi}_R\gamma^\mu\chi_R$	no
$\sigma^{\mu\nu}$	$\bar{\psi}\sigma^{\mu\nu}\chi$	$\bar{\psi}_L\sigma^{\mu\nu}\chi_R + \bar{\psi}_R\sigma^{\mu\nu}\chi_L$	yes
γ_5	$\bar{\psi}\gamma_5\chi$	$\bar{\psi}_L\gamma_5\chi_R + \bar{\psi}_R\gamma_5\chi_L$ $= +\bar{\psi}_L\chi_R - \bar{\psi}_R\chi_L$	yes
$\gamma_5\gamma^\mu$	$\bar{\psi}\gamma_5\gamma^\mu\chi$	$\bar{\psi}_R\gamma_5\gamma^\mu\chi_R + \bar{\psi}_L\gamma_5\gamma^\mu\chi_L$ $= -\bar{\psi}_R\gamma^\mu\chi_R + \bar{\psi}_L\gamma^\mu\chi_L$	no
P_L	$\bar{\psi}P_L\chi$	$\bar{\psi}_R\chi_L$	yes
P_R	$\bar{\psi}P_R\chi$	$\bar{\psi}_L\chi_R$	yes
$\gamma^\mu P_L$	$\bar{\psi}\gamma^\mu P_L\chi$	$\bar{\psi}_L\gamma^\mu\psi_L$	no
$\gamma^\mu P_R$	$\bar{\psi}\gamma^\mu P_R\chi$	$\bar{\psi}_R\gamma^\mu\psi_R$	no
$\sigma^{\mu\nu}$	$\bar{\psi}\sigma^{\mu\nu}\chi$	$\bar{\psi}_L\sigma^{\mu\nu}\chi_R + \bar{\psi}_R\sigma^{\mu\nu}\chi_L$	yes

Table 9.1: Projections onto right- and left-handed Dirac spinors of the V - A and chiral bases of gamma matrices. The first column is the element of either basis under consideration. The V - A basis elements are above the double line while the chiral basis elements are below. In the case of the terms involving γ_5 , we can use γ_5 to project out the chirality eigenvalue as per Eqs. 2.92 and 2.93. The last column indicates whether the term mixes left- and right-handed spinors.

9.2 Clifford Algebra & Gamma Matrices

The Clifford Algebra is given by:

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \quad (9.2)$$

The table containing projections of various fermion bilinears onto chiralities can be found in Tab. 2.1, reproduced here as Tab. 9.1

9.3 Common Lagrangians

Real scalar field

The free (canonically normalized) Lagrangian for a real scalar is given by:

$$\mathcal{L}_{\text{free}} = -\frac{1}{2}\phi\Box\phi - \frac{m^2}{2}\phi^2 \quad (9.3)$$

we also considered the ϕ^3 interaction:

$$\mathcal{L}_1 = \frac{g}{3!}\phi^3 \quad (9.4)$$

Complex scalar field

The free (canonically normalized) Lagrangian for a complex scalar is given by:

$$\mathcal{L}_{\text{free}} = -\phi^\dagger\Box\phi - m^2\phi^\dagger\phi \quad (9.5)$$

The kinetic term is hermitian up to a term related by IBP, more frequently it is written manifestly hermitian:

$$\mathcal{L}_{\text{free}} = \partial_\mu\phi^\dagger\partial^\mu\phi - m^2\phi^\dagger\phi \quad (9.6)$$

Dirac spinor/fermion

The free (canonically normalized) Lagrangian for a Dirac spinor is given by:

$$\mathcal{L}_{\text{free}} = i\bar{\psi}\not{\partial}\psi - m\bar{\psi}\psi. \quad (9.7)$$

While the kinetic term doesn't appear to be manifestly hermitian, it is up to a total derivative:

$$[i\bar{\psi}\not{\partial}\psi]^\dagger = -i\partial_\mu\psi^\dagger(\gamma^\mu)^\dagger\gamma_0^\dagger\psi \quad (9.8)$$

$$= -i\partial_\mu\psi^\dagger\gamma_0^2(\gamma^\mu)^\dagger\gamma_0\psi \quad (9.9)$$

$$= -i\partial_\mu\bar{\psi}\gamma^\mu\psi \quad (9.10)$$

$$= i\bar{\psi}\not{\partial}\psi + \partial_\mu(\bar{\psi}\gamma^\mu\psi) \quad (9.11)$$

Remember this is a Lagrangian level manipulation of operators, so our Hermitian conjugate ignores the Grassmann nature of the Dirac spinors (see discussion below Eq. 6.22). In the second line we used $\gamma_0^2 = \mathbb{1}$, in the third we used $\gamma_0 (\gamma^\mu)^T \gamma_0 = \gamma^\mu$, and in the last line we used IBP.

Massive vector

The free (canonically normalized) Lagrangian for a massive vector is given by:

$$\mathcal{L}_{\text{free}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{m^2}{2}A_\mu A^\mu, \quad (9.12)$$

where $F_{\mu\nu}$ is the usual field strength tensor from electrodynamics:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (9.13)$$

Massless vector, R_ξ gauge ($U(1)$ gauge theory gauge)

The free (canonically normalized) Lagrangian for a single massless vector can be written:

$$\mathcal{L}_{\text{free}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2\xi}(\partial_\mu A^\mu)^2, \quad (9.14)$$

where the second term corresponds to R_ξ gauge fixing. Other gauge fixing options result in different Lagrangians. The F^2 term is invariant under the local symmetry,

$$A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu \theta(x), \quad (9.15)$$

while the gauge fixing term is only invariant under this symmetry subject to the further constraint $\partial^2 \theta(x) = 0$.

General massless vector, Faddeev Popov gauge fixing

The free (canonically normalized) Lagrangian for a vector with multiple components $A_\mu^a(x)$ can be written:

$$\mathcal{L}_{\text{free}} = -\frac{1}{4}F_{\mu\nu}^a F^{a,\mu\nu} - \frac{1}{2\xi}G^a G^a + \bar{c}^a \left(\frac{\delta G^a}{\delta \theta^b} \right) c^b, \quad (9.16)$$

where the last two terms come from the Faddeev Popov procedure. The second term is the gauge fixing term and the third term is the ghost term.

These are fully expressed in terms of some G^a which can be chosen to be a function of A_μ^a , and is chosen such that the gauge-fixing term allows us to invert the propagator of the massless vector and the ghost Lagrangian balances the cost of breaking part of the gauge symmetry.

The field strength tensor in the case of an $SU(N)$ symmetry can be written:

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a - gf^{abc} A_\mu^b A_\nu^c, \quad (9.17)$$

where g is some coupling constant analogous to e in electrodynamics and f^{abc} are the structure constants of the $SU(N)$ Lie Algebra.

Glossary

V-A basis: basis of gamma matrices,

$$\Gamma_{VA} = \left\{ \mathbb{1}, \gamma^\mu, \sigma^{\mu\nu} = \frac{i}{2} [\gamma^\mu, \gamma^\nu], \gamma_5, \gamma_5 \gamma^\mu \right\}, \quad (9.18)$$

frequently employed in Nuclear physics .

See page(s): 49

(Euler Lagrange) Equations of Motion (EOM): derived from:

$$\frac{\partial \mathcal{L}}{\partial \phi(x)} - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \phi(x)} = 0, \quad (9.19)$$

where in general ϕ can be any field, one needs only adjust the variations appropriately to maintain covariance with the associated symmetry .

See page(s): 83

Adjoint (representation): the (N^2-1) -dimensional representation of some group. This is the representation used for gauge fields .

See page(s): 21

bubble diagram: a one-loop Feynman diagram consisting of two propagators forming a closed loop, connecting two vertices which in turn connect the bubble to external lines or the rest of the diagram..

See page(s): 146

Chiral basis: basis of gamma matrices,

$$\Gamma_{\text{chiral}} = \left\{ P_L, P_R, P_L \gamma^\mu, P_R \gamma^\mu, \sigma^{\mu\nu} = \frac{i}{2} [\gamma^\mu, \gamma^\nu] \right\}, \quad (9.20)$$

frequently employed when dealing with the electroweak standard model and other chiral theories .

See page(s): 50

Chirality: an analogue of **helicity** that is frame independent. Left- (Right-) handed Weyl spinors have by definition left- (right-) handed chirality, as this handedness corresponds to the spinor's irrep of the Lorentz group it is reference frame independent. For massless particles Helicity and Chirality are the same .
See page(s): 52, 188

Clifford algebra: the *associative algebra* of the gamma matrices, $[\gamma^\mu, \gamma^\nu] = 2\eta^{\mu\nu}$.
See page(s): 47, 48

connected diagram: a diagram where all lines and vertices are connected to each other. Contrast with **disconnected diagrams**.
See page(s): 90, 96, 186

contravariant (of transformations): we say a Lorentz vector ∂_μ transforms contravariantly when it transforms as,

$$\partial_\mu \rightarrow \Lambda_\mu^\nu \partial_\nu . \quad (9.21)$$

In contrast, the spacetime point x^μ transforms **covariantly** .
See page(s): 34, 186

covariant (of transformations): we say a Lorentz vector x^μ transforms covariantly when it transforms as,

$$x^\mu \rightarrow \Lambda^\mu_\nu x^\nu . \quad (9.22)$$

In contrast, the derivative of x^μ transforms **contravariantly** .
See page(s): 34, 186

Dirac spinor: a spinor defined as a combination of the (2,1) and (1,2) irreps of the . If a theory is invariant under the full Lorentz group Dirac spinors replace Weyl spinors as the irreps as Weyl spinors are not invariant under Parity .
See page(s): 28, 46

disconnected diagram: a diagram where separate subdiagrams exist with no connection via propagators or interaction vertices. Contrast with **connected diagrams**.
See page(s): 90, 96, 186

Doublet (representation): a two-dimensional representation acted on by some group action, frequently used with respect to $SU(2)_L$ in the SM

See page(s): 21

dual space: the space of linear maps V^* , which are vectors in their own right, mapping the vector space V onto a Field (in the mathematical sense) .

See page(s): 41

Feynman diagram: a diagrammatic representation of a term of a certain order in perturbation theory in the calculation of a **time-ordered product** of fields .

See page(s): 85, 187

Feynman rules: a set of rules which allows convert a **Feynman diagrams** into the corresponding calculation in perturbation theory directly (without having to resort to using variational methods for every calculation).

See page(s): 145

functional: a map of the space of functions, e.g. $x(t)$ to the real numbers. Analogous to $f(x)$ being a map of the variable x to the real numbers.

See page(s): 66

Fundamental (representation): the N -dimensional representation of some group. This is the representation used for example for quarks under $SU(3)_c$. The anti-fundamental is used for antiquarks .

See page(s): 21

generating functional: a functional of an external source that encodes informations about correlation functions (e.g. **time-ordered products**) of the theory. In these notes we consider the generating functional for disconnected diagrams, connected diagrams, and **one-particle irreducible diagrams (1PI)** .

See page(s): 101

Grassmann variables: variables which are characterized by the fact they anticommute. This leads to a unique algebra and integration calculus, they are used in the context of QFTs to define fermions consistent with Fermi statistics .

See page(s): 113

Helicity: for a particle with spin, left- (right-) handed helicity corresponds to the spin of the particle being anti-aligned (aligned) with its momentum. For massive particles a Lorentz boost can change the direction of momentum therefore helicity is a frame dependent quantity. The frame independent analogue of Helicity is **Chirality** .
See page(s): 52, 186

integrate out: to over a field in the path integral. In our example from gauge fixing in Chapter 7.3, we integrate over all of the field's configuration completely removing it. In EFTs like those discussed in these notes we integrate over all of the field's configurations, but subject to the assumption that M , the fields mass, is large. In other fields of particle physics other approaches are taken, such as only integrating out the high energy (small wavelength) modes and leaving the “part” of the fields configurations..
See page(s): 135

integration by parts (IBP): the statement that Lagrangian terms are redundant up to total derivatives. For example,

$$\mathcal{L} = \partial_\mu (\text{fields and derivatives})^\mu = 0, \quad (9.23)$$

or a specific example is:

$$\mathcal{L} = \partial_\mu \phi \partial^\mu \phi = -\phi \square \phi - \underbrace{\partial_\mu (\phi \partial^\mu \phi)}_{=0, \text{ by IBP}}, \quad (9.24)$$

$$= -\phi \square \phi. \quad (9.25)$$

Proof of the ability to use IBP identities requires Gauss's theorem and the assumption fields fall off sufficiently quickly at infinity.

See page(s): 35

invariant symbol: a tensor which after applying transformations returns itself. For example:

$$\eta_{\rho\sigma} = \eta_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma, \quad (9.26)$$

$$\epsilon_{ab} = L_a{}^c L_b{}^d \epsilon_{cd}. \quad (9.27)$$

Similar invariances apply to $\delta^a{}_c$, $\epsilon^{\dot{a}\dot{b}}$, $\sigma^\mu_{a\dot{a}}$, and $\epsilon^{\mu\nu\rho\sigma}$.
See page(s): 39

irreducible representation: an irreducible representation, or “irrep” is a vector space V whose only subrepresentations are V itself and the trivial subspace. For the spin addition example, the four states $|\pm\pm\rangle$ and $|\pm\mp\rangle$ were not an irrep, while the $s = 1$ and $s = 0$ representations were.

See page(s): 12, 20

Klein-Gordon equation: A Lorentz invariant differential equation,

$$(\square - m^2) \phi = 0, \quad (9.28)$$

that generalizes the energy-momentum relation $E^2 = \vec{p}^2 + m^2$ into a field equation. Its solutions take the form of plane waves .

See page(s): 83

Lagrangian density: the abstraction of the usual Lagrangian which is related to the action by,

$$S = \int dt L, \quad (9.29)$$

to:

$$S = \int d^4x \mathcal{L}. \quad (9.30)$$

Where $\mathcal{L}$ is the Lagrangian density .

See page(s): 31

Lie Algebra: an Algebra in the formal math sense, it is unique in addition to addition it has the operation of the “Lie Bracket” which is antisymmetric and obeys the Jacobi identity.

See page(s): 16, 189

Lie Group: a group in the formal math sense, it is intimately connected to the **Lie Algebra**. They form a smooth, continuous space with a smooth operation allowing us to perform calculus on the group.

See page(s): 17

loop: a (part of) a diagram with an unspecified momentum we must integrate over. This frequently leads to ultraviolet divergences which must be renormalized away..

See page(s): 90

one-particle irreducible diagram: (also 1PI) a diagram that remains connected after cutting a single internal line (i.e. propagator) .

See page(s): 97, 187

Orthogonal Group $O(N)$: a group with the defining proper $O^T O = \mathbb{1}$.
See page(s): 18

perturbation theory: solving hard integrals by Taylor expansion in small parameters.
See page(s): 86

Projection operators: Projection operators satisfy $\sum_i P_i = \mathbb{1}$, $P_i P_{i \neq j} = 0$ and $P_i^2 = P_i$, they project onto specific states. The chiral projection operators,

$$P_{\pm} = \frac{1}{2} (\mathbb{1} + \gamma_5) , \quad (9.31)$$

decompose a Dirac spinor into its left- and right-handed components,

$$P_L \psi = \psi_L \quad P_R \psi = \psi_R . \quad (9.32)$$

.
See page(s): 49

propagator: the inverse of the kinetic term in the Lagrangian, describes the propagation of a field from one spacetime point to another. At higher orders in perturbation theory, the propagator can be also be thought of the inverse of the sum of all two-point functions, although this is not always well defined due to gauge symmetry.
See page(s): 81

Proper Orthochronous Lorentz group: the subgroup $SO^+(1, 3)$ of the full Lorentz Group. Our fields are typically defined as irreps of this subgroup. This subgroup is related to the full Lorentz group by including the discrete transformations Parity and Time reversal .
See page(s): 27

quantum electrodynamics: (QED) the $U(1)$ gauge theory describing the interaction of electrically charged particles with the photon.
See page(s): 54

representation: effectively a mapping of the abstract concept of a group onto matrices. For our use we'll frequently abuse the terminology and, what we really mean is: a field transforming according to a representation.
See page(s): 20, 21

scalar field: a field that transforms under the trivial $(1, 1)$ representation of the Lorentz group .

See page(s): 25

spinor: general term for a field in one of the spin $\frac{1}{2}$ irreps of the Lorentz group, including the Dirac field .

See page(s): 37

subrepresentation: a subspace of the vector space a group acts on which is itself invariant under the group action.

See page(s): 20

symmetry factor: the number, S , of distinct permutations of identical vertices and internal lines leaving a diagram topologically unchanged. The symmetry factor needs to be included in the Feynman rules to cancel overcounting in the Feynman rules approach as a factor $1/S$ times the relevant diagram..

See page(s): 90

tadpole diagram: a diagram where the external line loops back on itself, the external line is only connected to itself, see also **vacuum expectation value**.

See page(s): 90, 96

tensor: an N^{th} rank tensor is a quantity whose components transform with one transformation matrix for each component. A scalar is a 0^{th} rank tensor, a vector a 1^{st} rank tensor.

See page(s): 6

time-ordered product: an operation $T\{\}$ which rearranges a product of e.g. fields into a product which is ordered by their time argument. For example, $T\{\phi(x_1), \phi(x_2), \phi(x_3)\}$.

See page(s): 72, 187

Triplet (representation): a three-dimensional representation acted on by some group action .

See page(s): 21

Unitary Group $U(N)$: a group with the defining property $U^\dagger U = \mathbb{1}$.

See page(s): 18

vacuum bubble diagram: closed diagrams with no external legs at all..

See page(s): 97

vacuum expectation value: (also “vev”) the expectation of a single field over the vacuum, $\langle \Omega | \phi(x) | \Omega \rangle$. For most theories it is something we renormalize away so we perturb around zero vacuum expectation value. For Higgs fields we expand about some nonzero expectation value and can calculate corrections to the vev in perturbation theory.
See page(s): 92, 191

variation: (or variational derivative), the functional analogue of $dx_i/dx_j = \delta_{ij}$:

$$\frac{\delta}{\delta f(y)} f(x) = \delta(x - y) . \quad (9.33)$$

Importantly for a special-relativistic theory, delta function ensures that we maintain locality. Due to its linearity, all of the usual “rules” from calculus carry over, such as the power rule, product rule, and chain rule.

See page(s): 67

vector: a quantity whose components transform in a specific way under rotations.

See page(s): 4

Weyl spinor: spinors in the $(1, 2)$ (left-handed Weyl spinor) or $(2, 1)$ irrep (right-handed Weyl spinor) of the Lorentz group. .

See page(s): 37

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